ExxonMobil Connection Evaluation Program

Second Edition

June 2020

Revisions

Edition	**Date**	**Revisions**	**Technical Editors/Reviewers**
First	March 2001	Initial Release	Ismail Ceyhan David Khemakhem David Coe Jim Myers Jim Powers
Second	June 2020	• Finite Element Analysis workflow added using Dassault Systemès SIMULIA Abaqus ® plug-ins • Physical testing protocol changed from EMCEP 1st Edition to CAL IV of API RP 5C5, 4th Edition • Added Testing Variants	Adam Aylor Ajinkya Patil Anirban Manna Brian O'Donnell Jim Myers Jithin Devan

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Executive Summary

Overview

This document describes the latest program ExxonMobil uses to evaluate performance of premium connections. The program is stewarded by ExxonMobil Upstream Integrated Solutions (UIS). The intent of this document is to be transparent and complete on the process involved in evaluation; to allow a proactive Connection Owner to complete testing and Finite Element Analysis (FEA) without ExxonMobil involvement in a manner that would still be acceptable to ExxonMobil. In some instances it may allow tests performed as per CAL IV of API RP 5C5, 4th Edition, prior to the release of EMCEP 2nd Edition, to be considered compliant. This document supersedes EMCEP, 1st Edition. The latest version of this document is available on the *ExxonMobil Wells Technical Quality extranet site.* ExxonMobil representatives may be contacted to gain access to this extranet site.

The connection evaluation program consists of performing FEA and physical testing on the connection. Performance envelopes are defined through the results of FEA. Physical testing is performed on a limited number of specimens to verify the FEA results and to explore performance parameters that cannot be studied through FEA (e.g., galling resistance).

A connection also requires approved manufacturing procedures to ensure that the connections delivered for field use are identical to the evaluated connections. The manufacturing and inspection procedures will be reviewed and approved by ExxonMobil UIS-Wells Technical.

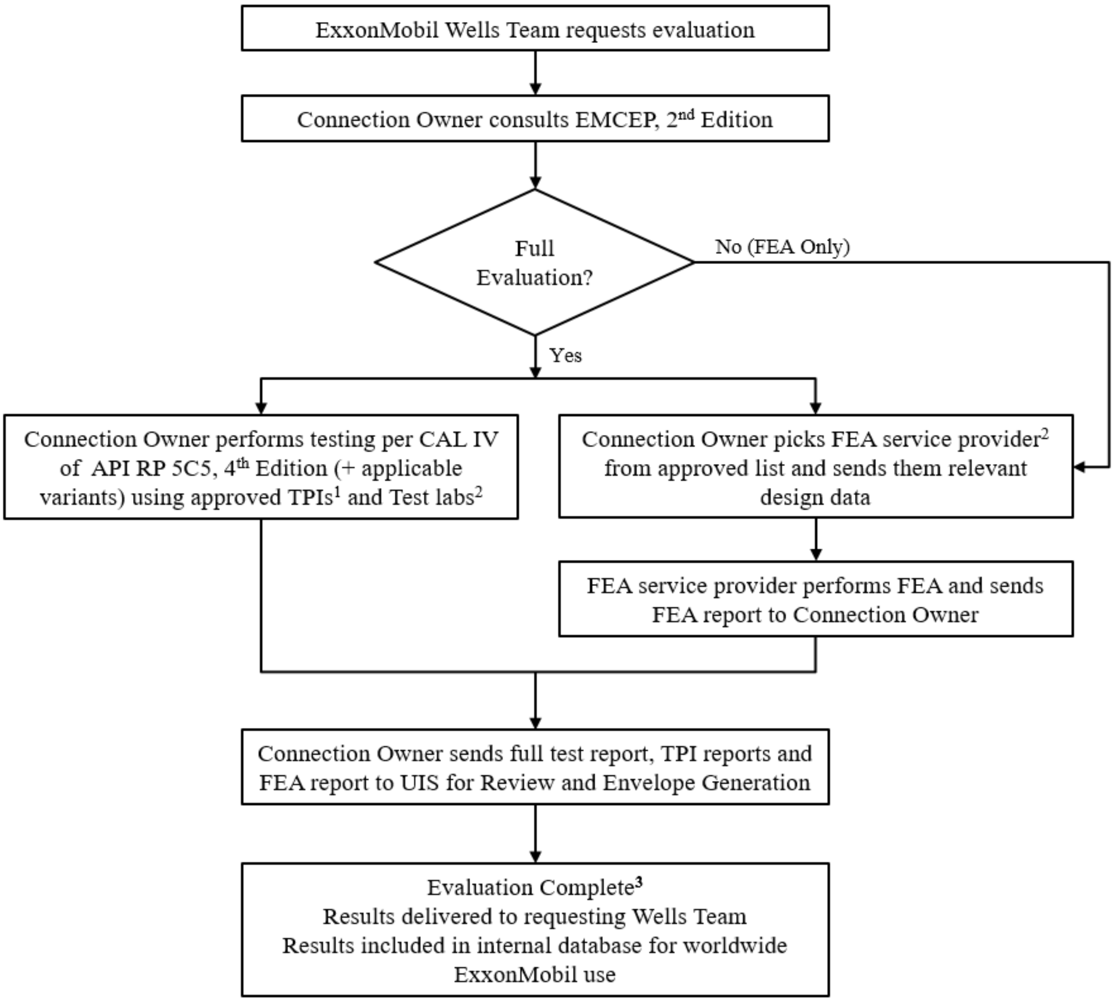
Workflow

The Connection Owner performs physical testing per **Connection Assessment Level IV (CAL IV) of API RP 5C5, 4th Edition**. If any modifications/additions are requested by UIS or the Connection Owner, they have to be agreed upon by both parties. The Connection Owner is free to choose any ExxonMobil approved Third Party Inspector (TPI) services and connection testing facilities.

The FEA shall be performed as per the ExxonMobil FEA procedure in Part I of this document by any ExxonMobil approved FEA service provider. The FEA service provider may be an independent third party or the Connection Owner themselves.

Connection Owner then sends the consolidated report that includes FEA, physical testing and TPI reports to UIS for review and performance envelope generation.

Figure 1 shows a flowchart representation of the ExxonMobil Connection Evaluation Program (EMCEP) 2nd Edition workflow.



Notes:

1. List of approved TPI's available on the ExxonMobil Wells Technical Quality extranet site
2. A process is available to evaluate and approve Test labs/FEA service providers
3. Any questions/discussion between UIS and Connection Owner is embedded in this step

Figure 1 Flowchart of EMCEP 2nd Edition

Part I – Finite Element Analysis

Finite Element Analysis (FEA) is the first of the two components of the ExxonMobil Connection Evaluation Program. FEA is used to study structural and sealability performance of the connection design.

Finite Element Analysis Overview

The FEA model is used to evaluate performance of both metal-to-metal seals and the structural integrity of the connection, and considers maximum thread and minimum seal interferences (100% and 0% of the interference range calculated based on the manufacturing tolerances respectively). Results are summarized in the form of detailed metal-to-metal seal contact stress profiles to assess pressure sealability in the form of stress, strain, or displacement contour plots to examine potential failure modes and locations. Performance envelopes for each design are derived from the FEA results.

Connection Analysis Data Requirements

The following information is required to perform the premium connection Finite Element Analysis.

1- Connection name, size, weight, grade, drawing number and revision level

2- Data Requirements:
- a) Connection data sheet (specification sheet)
- b) Electronic copy of the CAD drawings (e.g., IGES format)
- c) Connection Engineering Drawing(s) in PDF format
- d) Pin and Box critical cross section areas
- e) Material Properties:
  - Young's Modulus
  - Poisson's ratio
  - Stress – Strain data
  - Anisotropic properties, if any

3- CAD Drawing Requirements:
- a) Seal interference (internal and external): Minimum
- b) Thread interference: Maximum
- c) Torque shoulder interference: Minimum make-up torque*
- d) All other dimensions: Nominal

* Torque shoulder interference should reflect minimum make-up torque of the connection. Torque shoulder interference of 0.001-in is also acceptable (measured at the mid-point of the pin torque shoulder when the interference at the closest seal is adjusted to zero).
Dimensions and interferences are calculated based on the tolerances specified in connection engineering drawing(s).

Discussion of Methodology

Metal-to-metal seal integrity is assessed from the distribution of contact stress between the seal faces. The characteristics of the seal contact stress distribution (Figure 2), such as length, height, enclosed area, and general shape, provide the major contributions to affecting a seal. The effects of surface finish and thread compound have importance to galling sensitivity and are evaluated by the physical testing program.

FEA provides significantly more insight into connection sealability performance than physical testing. FEA can be used to quantitatively evaluate the seal quality, whereas the physical test results are binary (i.e., "leak" or "no leak"). For example, although the FEA may indicate marginal seal performance, it is possible that a leak is not observed during testing due to the limited test duration. Also, FEA allows the separate assessment of seal quality for liquid or gas service.

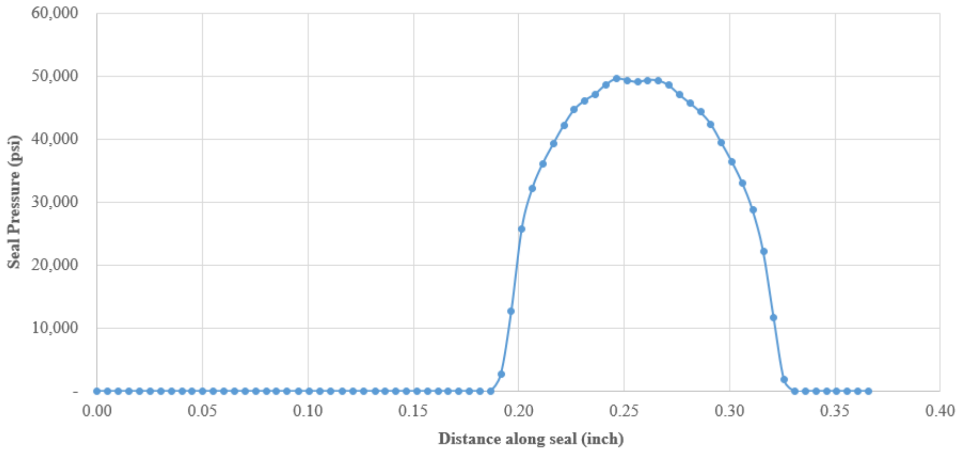


Figure 2 Contact Stress Distribution Example

The FEA methodology considers only the metal-to-metal seal as sealing element, i.e., threads and torque shoulders are treated as non-sealing elements. It also considers phenomena that affect pressure sealability in a connection under applied axial load and pressure, including "penetration" of the metal-to-metal seal by pressure. For instance, if a portion of the seal separates, internal pressure can "penetrate" past that portion up to a point at which the contact stress is sufficient to prevent further penetration. This "pressure penetration" phenomenon is fundamental to the mechanics of the seal. A marginal seal condition can be mistakenly identified as a seal if "pressure penetration" is not modeled.

Structural integrity of the connection is assessed by evaluating the plastic strain distribution throughout the connection. Through-wall or through-shoulder equivalent plastic strain above 2% indicates structural weakness.

The FEA Methodology and instructions to the FEA analyst are described in the following six sections:

1. Geometry
2. Assembly
3. Mesh
4. Analysis
5. Result Verification
6. Result Extraction

Geometry

This section describes the connection geometry configuration required for FEA.

The connection geometry:

a) Shall be as per connection engineering drawing's latest revision.

b) Shall be in maximum thread and minimum seal interference configuration (Max/Min configuration). Max/Min configuration is defined as a connection assembly which has 100% of the thread diameter interference range and 0% of the seal diameter interference range calculated based on the manufacturing tolerances. i.e., the thread and seal diameters in the geometry shall be as per Max/Min configuration.

c) Should have the rest of the dimensions specified in the engineering drawing as per their nominal values.

d) Should reflect worst case sealability configuration.

The analyst shall ensure the pin and the box lengths are sufficient to avoid end effects in the analysis.

The analyst shall contact the Connection Owner and request for updated CAD files if there is any difference(s) between the dimensions in the CAD, and dimensions in the engineering drawings.

Assembly

This section describes the connection assembly required for FEA. In a fully assembled condition, the connection geometry shall reflect the worst case sealability configuration.

The axial position of the pin with respect to the box shall be to reflect the minimum make-up torque on the connection. For a connection with torque shoulder, axial position of pin with 0.001-in torque shoulder interference is also acceptable. See Appendix I for details on adjusting torque shoulder interference.

Mesh

This section describes the steps to generate a quality mesh. ExxonMobil requires the following mesh controls and element sizes for the various connection regions as follows:

a) Connection geometries may be partitioned to generate a smooth, quality mesh (see Figure 3).

b) Element type shall be 4-noded bilinear axisymmetric quadrilateral with full integration (e.g., CAX4 in Abaqus). No triangular elements are allowed.

c) Element size at seal, threads and shoulder edges shall be ≤ 0.001-inches (see Figure 4).

d) Element size at outer diameter (OD) and inner diameter (ID) edges away from the seal, threads and torque shoulder shall be ≤ 0.05-inches (see Figure 5).

e) Thread profile radii edges shall have 4 or more elements (see Figure 6).

f) Pin and Box shall have 6 or more elements across the wall thickness from OD to ID.

g) Mesh shall not contain more than 1% distorted elements. Distorted element shape is defined as:

    i. Quad-Face Corner Angle less than 30°

ii. Quad-Face Corner Angle greater than 150º

iii. Aspect ratio greater than 5.0

These distorted elements shall not occur at critical regions like seal, thread and torque shoulder.

h) Verify boundaries, there shall not be any cracks (unequivalenced edges, mis-matched meshes) in the model.

i) Verify that all element normals are in the same direction (e.g., the positive z-direction in Abaqus Standard).

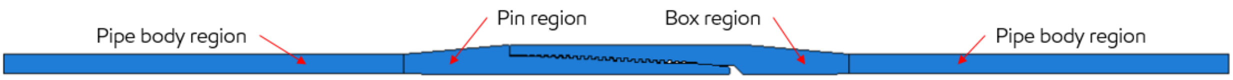


Figure 3 Pin and Box Geometry Partitioning – Example

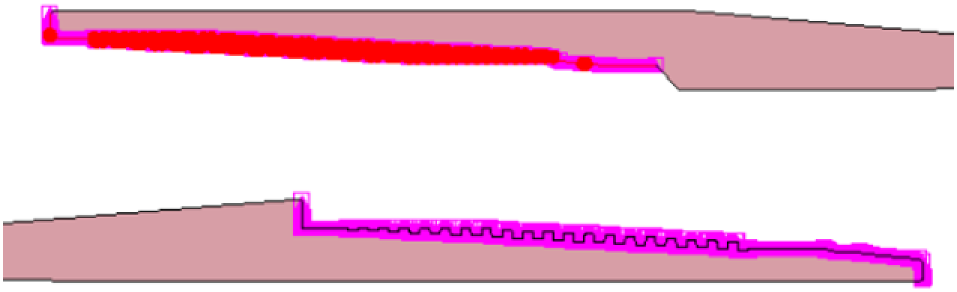


Figure 4 Local Seeds – Seal, Threads and Shoulder edges

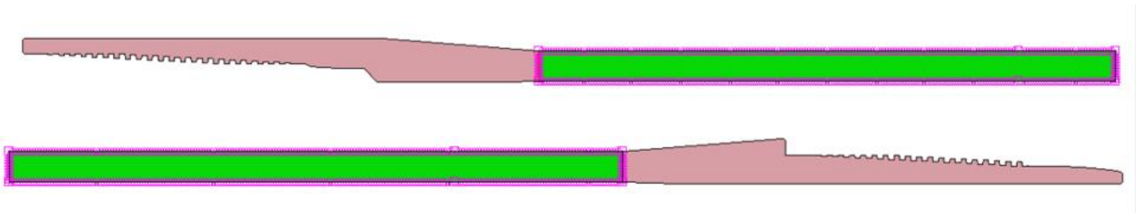


Figure 5 Local Seeds – OD and ID edges away from Seal, Threads and Shoulder

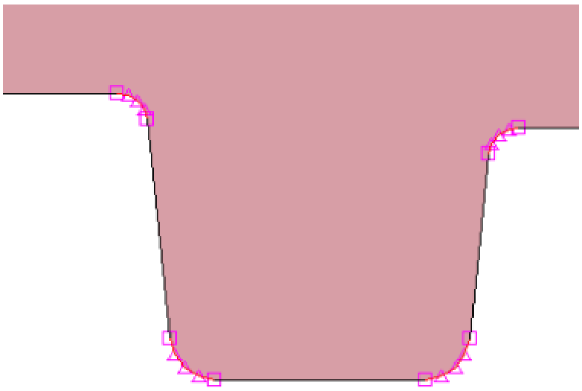


Figure 6 Local Seeds - Thread profile radii edges

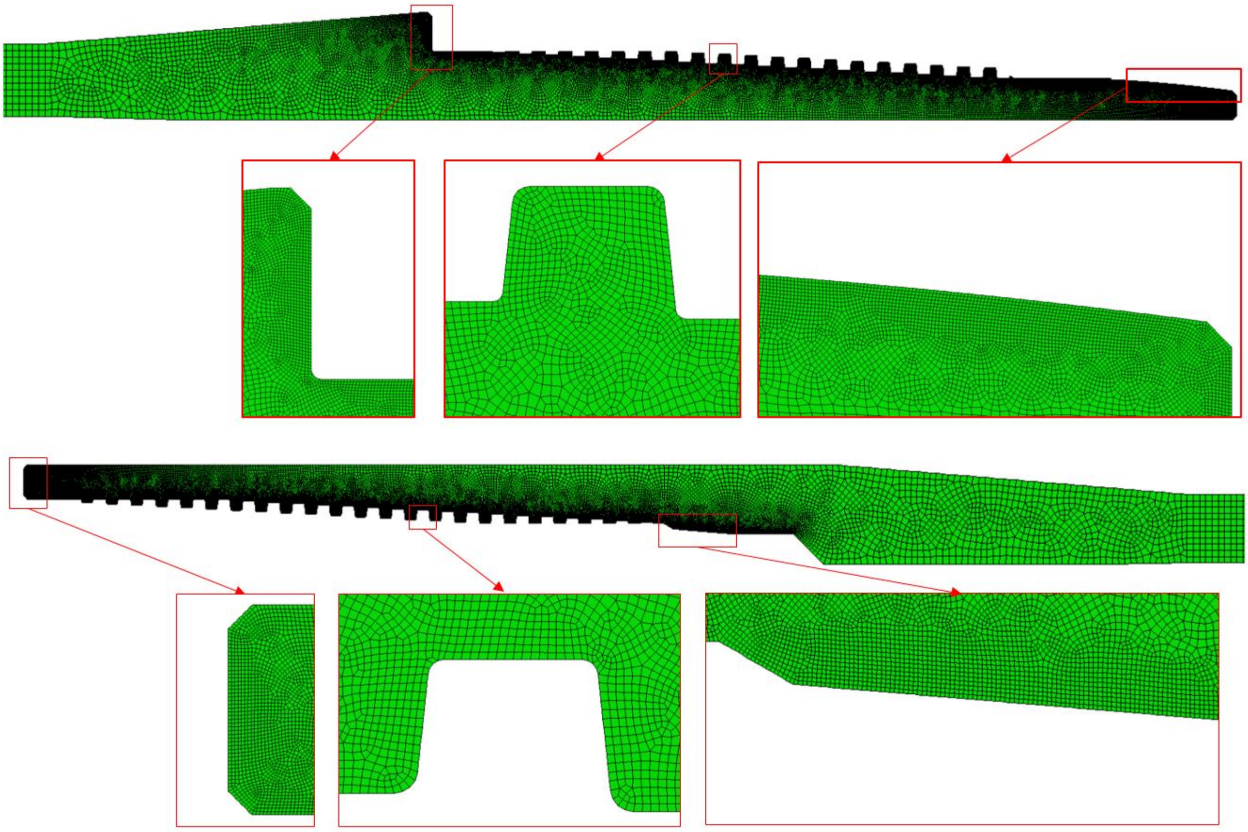


Figure 7 Connection FEA Mesh - Example

Analysis

This section describes the steps to generate FEA input files. ExxonMobil has developed an Abaqus plug-in to automate a substantial portion of the steps explained in this section, Appendix II describes the instructions to use this plug-in. The analyst may use this plug-in or follow the instructions described in the sub-sections below to generate FEA input files.

Material Properties

UIS recommends the following material properties for API grades:

- Young's Modulus : $30 \times 10^6\ psi$
- Poisson's ratio : 0.30
- Plastic behavior : Bi-linear Stress – Strain curve (see Figure 8)

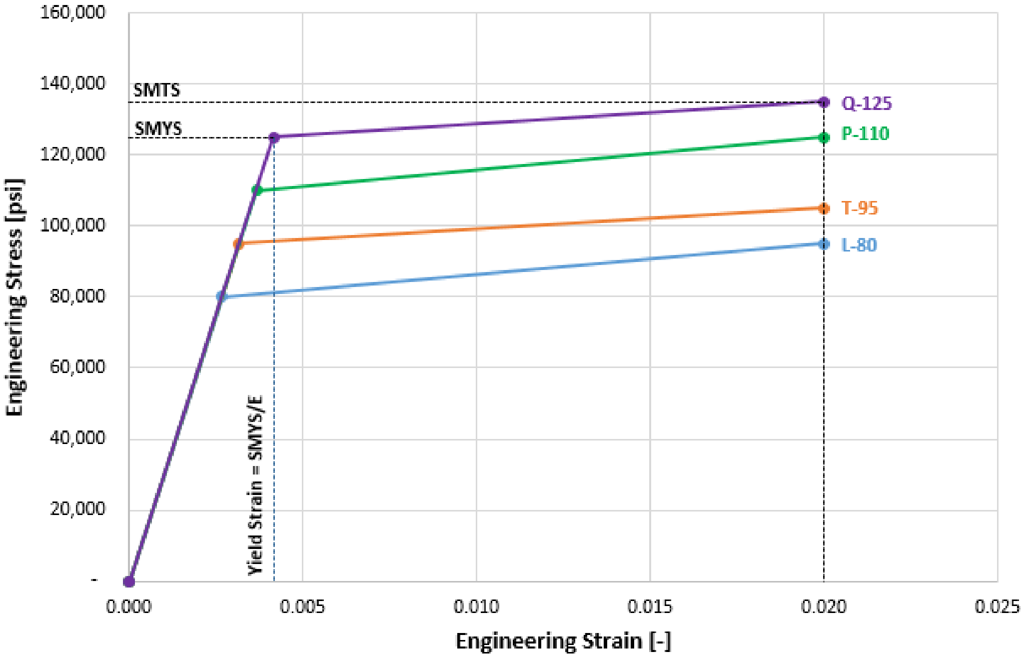


Figure 8 Engineering Stress-Strain curve for API grades

The points on the bi-linear curve should be calculated using specified minimum yield stress (SMYS) and specified minimum tensile strength (SMTS) tabulated in Table E.5 of API 5CT.

For proprietary grades, the specified minimum yield stress (SMYS) and specified minimum tensile strength (SMTS) shall be as defined by the connection owner. The elastic behavior should be only up to SMYS and strain at SMTS should be 2%.

Hardening model should be "Isotropic" or equivalent.

Contact Interactions

To define contact interactions in the FEA Model, the analyst shall define contact surface pairs on seal(s), torque shoulder(s) and threads using the guidelines described below. An example is shown in Figure 9.

- All contact pair interactions shall be defined with "**Node to Surface"** or "**Surface to Surface**" discretization method. Sliding formulation shall be "**Finite Sliding**".
- Tangential behavior shall be defined with "**Penalty**" friction formulation or similar and **friction coefficient = 0**

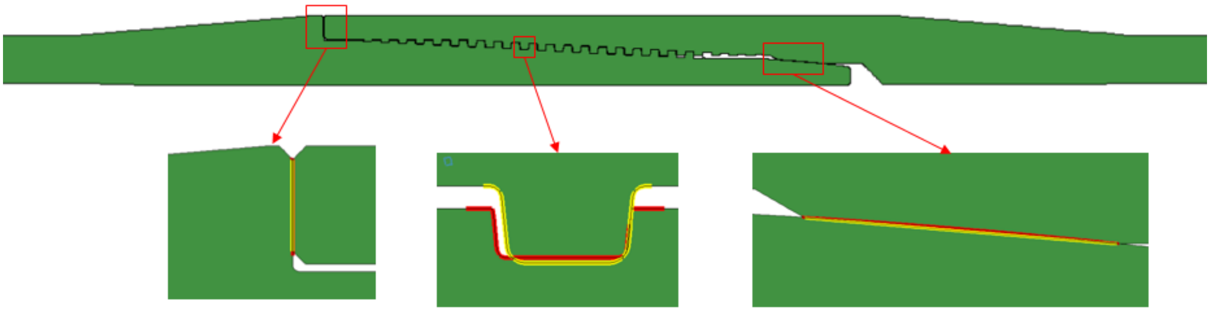


Figure 9 Contact surface pairs on Pin and Box - Example

Make-up simulation

Connection make-up (pre-load) shall be simulated as a non-linear static interference analysis. In this analysis the analyst shall resolve thread, seal(s) and torque shoulder(s) interferences with box end axially constrained. UIS preferred method is to complete the simulation in 4 incremental steps. The incremental steps are:

1. Resolution of thread contact pairs with pin and box ends axially constrained
2. Resolution of seal contact pair(s) with pin and box ends axially constrained
3. Resolution of torque shoulder contact pair(s) with pin and box ends axially constrained
4. Resolution of contact with **only the box** end axially constrained

All the steps of make-up simulation shall allow radial displacement. An example of surfaces where the axially constrained boundary condition shall be applied is shown in Figure 10.

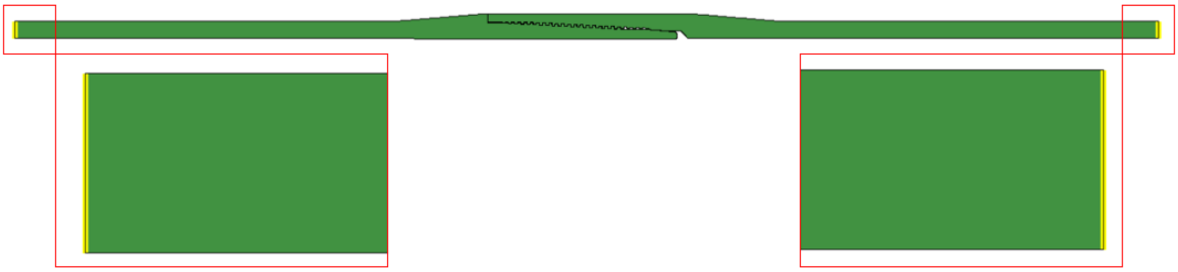


Figure 10 Axial Constraint surfaces on Pin and Box – Example

Output variables shall be defined as per the Output Variables section and Model convergence criteria shall be defined as per the Model Convergence Criteria section.

The analyst shall verify the Make-up simulation results using the guidelines described in Result Verification section.

Load schedule simulation

33 load schedules, as listed in Table 1 shall be simulated separately. These load schedules have different combinations of axial and pressure loads. Loads for each of the load schedules listed in Table 1 shall be calculated using connection data sheet and instructions described in Appendix III.

Load schedule shall be simulated as a non-linear static analysis with pressure penetration. In this analysis the analyst shall apply the axial and pressure loads as per the load schedule on the appropriate surfaces of the make-up simulated model with box end axially constrained. All the steps of load schedule simulation shall allow radial displacement.

Table 1 Load Schedules and Descriptions

No.	Load schedule	Type	Description
1	c25epmxmn	Standard	25% Compression Rating with External Pressure
2	c25pmxmn	Standard	25% Compression Rating with Internal Pressure
3	c50epmxmn	Standard	50% Compression Rating with External Pressure
4	c50pmxmn	Standard	50% Compression Rating with Internal Pressure
5	c75epmxmn	Standard	75% Compression Rating with External Pressure
6	c75pmxmn	Standard	75% Compression Rating with Internal Pressure
7	c90epmxmn	Standard	90% Compression Rating with External Pressure
8	c90pmxmn	Standard	90% Compression Rating with Internal Pressure
9	cepmxmn	Standard	Compression with External Pressure
10	cmxmn	Standard	Compression Only
11	cpmxmn	Standard	Compression with Internal Pressure
12	epmxmn	Standard	External Pressure Only
13	ruppmxmn	Standard	Tension with Internal Pressure on Rupture curve
14	mtrpmxmn	Standard	100% Tension Rating with Internal Pressure
15	pmxmn	Standard	Internal Pressure Only
16	t100pmxmn	Standard	100% Pipebody Tension Rating with Internal Pressure
17	t25epmxmn	Standard	25% Pipebody Tension Rating with External Pressure
18	t25pmxmn	Standard	25% Pipebody Tension Rating with Internal Pressure
19	t50epmxmn	Standard	50% Pipebody Tension Rating with External Pressure
20	t50pmxmn	Standard	50% Pipebody Tension Rating with Internal Pressure
21	t75epmxmn	Standard	75% Pipebody Tension Rating with External Pressure
22	t75pmxmn	Standard	75% Pipebody Tension Rating with Internal Pressure
23	tepmxmn	Standard	Tension with External Pressure
24	tmxmn	Standard	Tension only
25	tpmxmn	Standard	Tension with Internal Pressure
26	vmpmxmn	Standard	Tension with Internal Pressure on VonMises curve
27	vmcpmxmn	Standard	Compression with Internal Pressure on VonMises curve
28	vmepmxmn	Standard	Tension with External Pressure on VonMises curve
29	vmcepmxmn	Standard	Compression with External Pressure on VonMises curve
30	sp3mxmn	Specimen	Axial Load with Internal Pressure as per EMCEP 1st Ed. SP3
31	sp5mxmn	Specimen	Axial Load with Internal Pressure as per EMCEP 1st Ed. SP5
32	apiseramxmn	Specimen	Axial Load with Internal Pressure as per API 5C5 Series-A
33	apiserbmxmn	Specimen	Axial Load with Internal Pressure as per API 5C5 Series-B

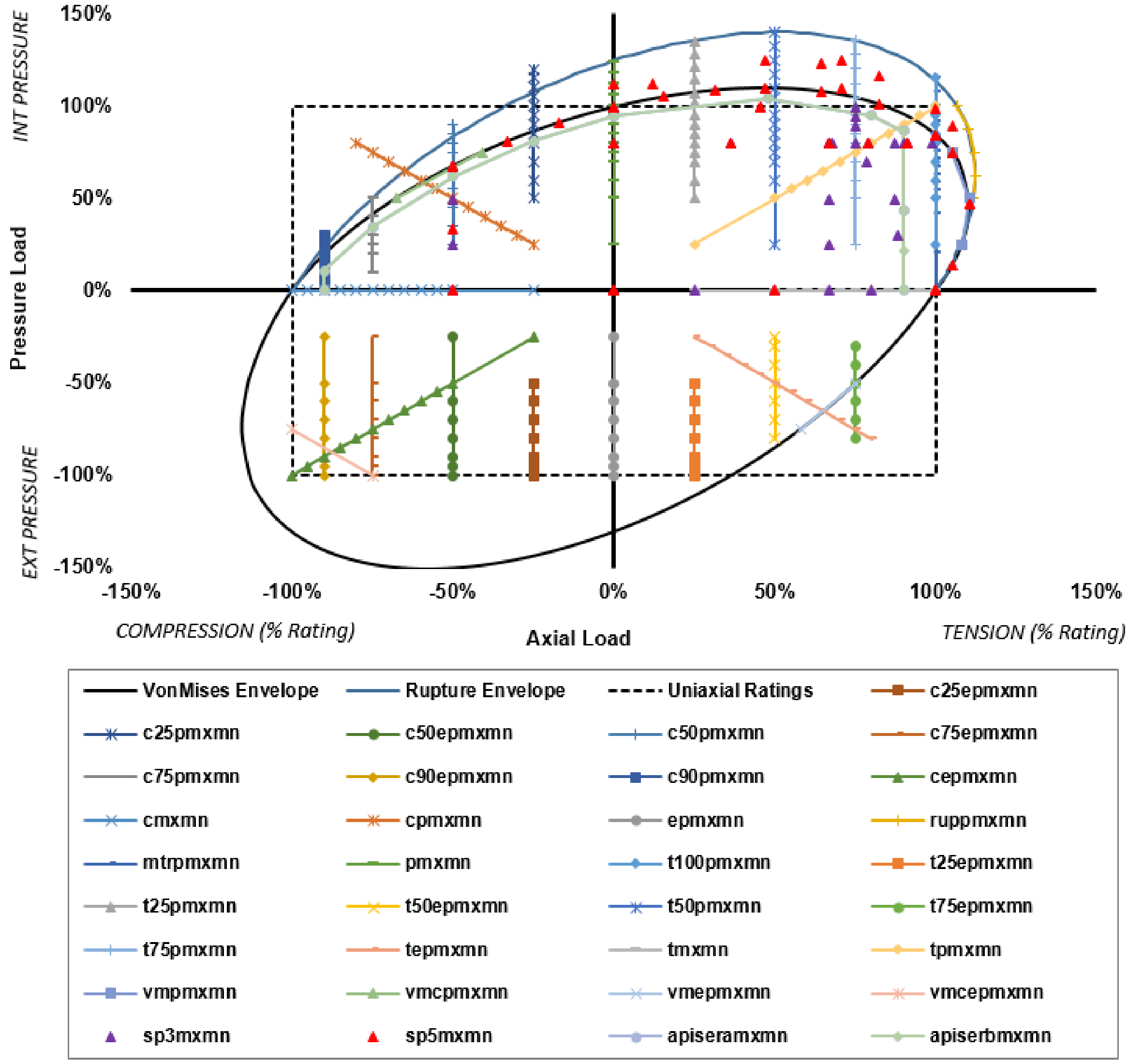


Figure 11 FEA Load Schedules

Load application – Axial Load

The analyst shall define the axial load application surface at the pin end. An example of surfaces where the axial load shall be applied is shown in Figure 12. Axial load shall be applied as distributed pressure.

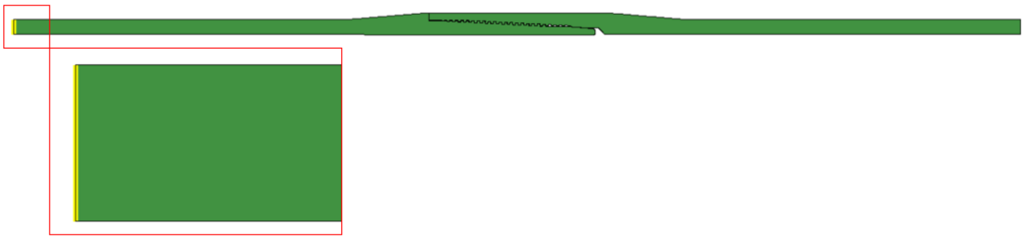


Figure 12 Axial load application surface – Example

Load application – Pressure Load

Only a metal-to-metal seal is considered as sealing element, i.e., threads and torque shoulders are treated as non-sealing elements. The analyst shall define the external pressure and internal pressure surfaces for both the pin and the box as per the guidelines below:

1. External pressure application surface on the Box shall start from the outer diameter node at the Box end, and end on the Box seal node which is exposed to external pressure.
2. External pressure application surface on the Pin shall start from the outer diameter node at the Pin end, and end on the Pin seal node which is exposed to external pressure.
3. Internal pressure application surface on the Box shall start from the inner diameter node at the Box end, and end on the Box seal node which is exposed to internal pressure.
4. Internal pressure application surface on the Pin shall start from the inner diameter node at the Pin end, and end on the Pin seal node which is exposed to internal pressure.
5. If there are two seals, the external pressure ends on the external seal and the internal pressure ends on the internal seal for "Standard Analysis". Both the external pressure and the internal pressure end on the primary seal for "Bleed-through Analysis".

An example of surfaces where the external pressure and internal pressure shall be applied for a 1-sealed connection is shown in Figure 13.

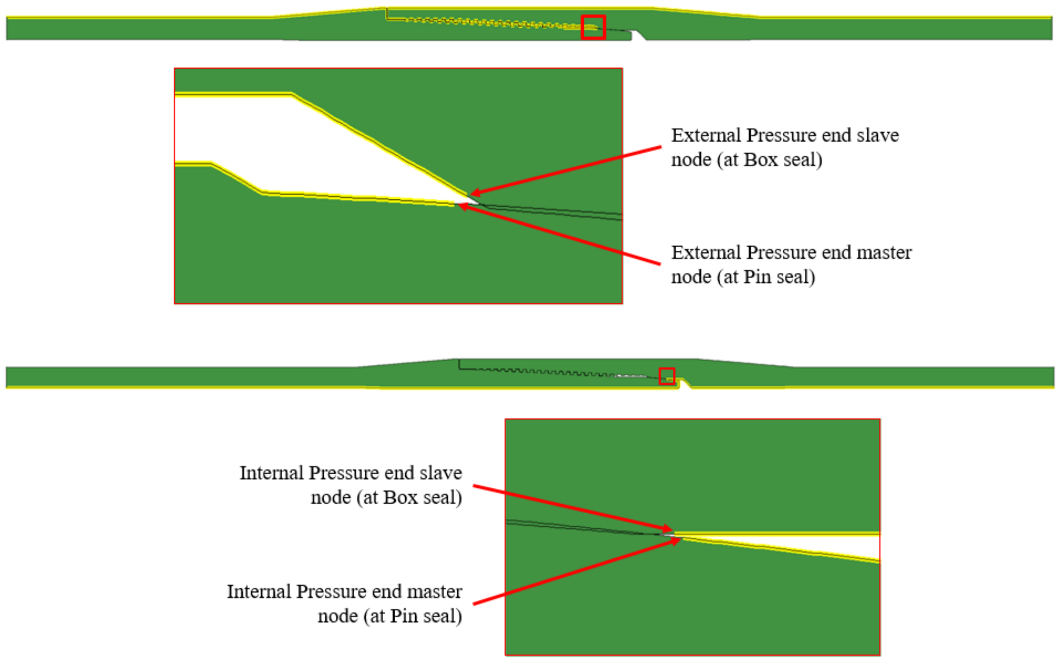


Figure 13 Pressure application surfaces – Example

Pressure Penetration

Pressure Penetration shall be defined for all the load steps which has internal or external pressure. To define pressure penetration, the pressure exposure must be determined. If there is one seal, then the seal will be exposed to both internal and external pressure.

If there are two seals, two sets of load schedule simulation shall be done:

1. "Standard Analysis" involves 2 Pressure Penetration Sets, referencing both the Internal and the External Seals. In this, the outer seal will be exposed only to external pressure and the inner seal exposed only to internal pressure.
2. "Bleed-through Analysis" only involves 1 Pressure Penetration set, using only the Internal Seal. This simulation is referred to as "Bleed-through" because it assumes the External Seal will fail and the External pressure will communicate past the threads to the Internal Seal.

The analysts shall use the seal surface contact pair(s) and applied pressure load to define "**Pressure Penetration**". Following guidelines shall be used while defining the Pressure Penetration:

Master surface :	Seal master surface
Slave surface :	Seal slave surface
Critical Contact stress :	Applied Pressure as per the load schedule
Fluid pressure :	Applied Pressure as per the load schedule
Start node on Master surface :	Node exposed to applied pressure on Seal master surface
Start node on Slave surface :	Node exposed to applied pressure on Seal slave surface

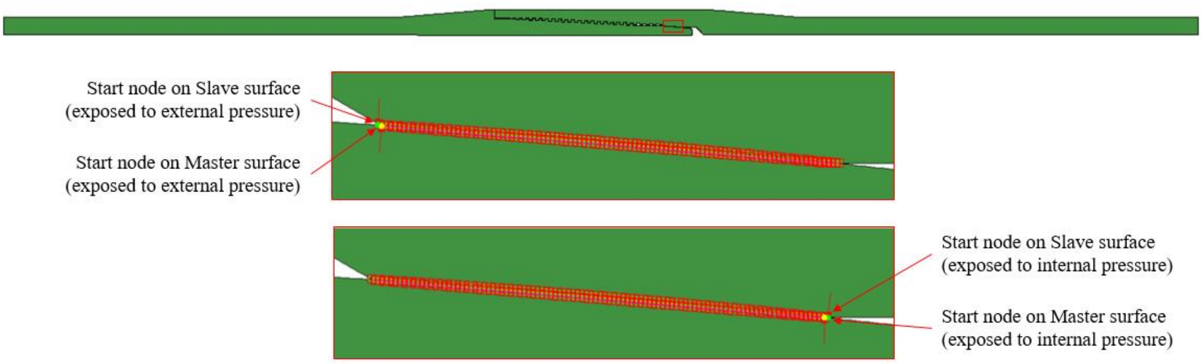


Figure 14 Start Nodes for Pressure Penetration

Output variables shall be defined as per the Output Variables section and Model convergence criteria shall be defined as per the Model Convergence Criteria section.

The analyst shall verify the load schedule simulation results using the guidelines described in Result Verification section.

Model Convergence Criteria

In commercial FEA software, the default convergence criteria settings/parameters shall be used to obtain a converged solution. The analyst may modify the convergence criteria only if there is a convergence issue after implementing stabilization* and contact pair double-up**. If convergence criteria is modified from the default, then the analyst shall document the modification and inform UIS. The acceptance of convergence criteria modified FEA results will be decided after UIS review.

*Stabilization may be used to enhance model convergence. If stabilization is used then the analyst shall ensure the ratio of stabilization energy to internal energy does not exceed 5%.

**Doubling-up contact pairs is allowed at the thread and torque shoulder(s) contact pairs to enhance model convergence. Doubling-up contact pairs at the seal(s) is not allowed.

Output Variables

Following output variables shall be defined in the make-up and load schedule simulations:

1. Reaction Forces
2. Displacements
3. Stress Distribution
4. Plastic Strain Distribution
5. Contact Stress
6. Contact Displacement
7. Fluid pressure for pressure penetration analysis or similar
8. Model Internal Energy
9. Model Stabilization Energy if stabilization is used

Result Verification

This section describes the result verification process. Result verification should be broken down into two distinct levels. The first should be a gross detailing of which load schedule input files ran to completion, and if not, when did they end. The second level should be to determine the fine aspects of each load schedules' run including body wall yielding, energy ratio checks, and reaction force checks.

Gross assessment of simulations

All the specimen type load schedules specified in Table 1 shall be fully converged, if not fully converged then convergence enhancements may be used. Notes shall be taken as per the instructions given in the FEA Documentation section to capture if any of the standard type load schedules has not fully converged.

Detailed assessment of simulations

A detailed analysis of the FEA results shall be performed by loading the FEA output files. This involves animation, plotting and visual techniques to determine plastic yielding, energy ratio checks and reaction force checks.

The following checks should be carried out for the Make-up simulation:

a) Axial stress should be close to zero at the end of the pin and the box.

b) Sum of reaction forces in axial direction should be close to zero.

c) Hoop stress should be positive on the box and negative on the pin. Exception will be for dovetail and hook threads.

The following checks shall be carried out for both Make-up simulation and Load schedule simulation:

a) No penetration of the master nodes into the slave surface.

b) If stabilization is used, then the maximum ratio of stabilization energy to internal energy shall not exceed 5%.

c) Sum of reaction forces in axial direction should be close to the applied axial load for **cmxmn** and **tmxmn** load schedules.

The analyst shall modify and re-run the FEA model if the model could not satisfy the checks mentioned above.

Result Extraction

This section describes the result extraction process. Result verifications shall be completed prior to result extraction process. As part of the result extraction, the analyst shall extract the Equivalent Plastic Strain Contours and Contact stress Profile at Seal(s) for all the steps of all the load schedules.

<u>Plastic Strain Contours</u>

The analyst shall plot the equivalent plastic strain contours for all the steps of all the load schedules listed in Table 1 and identify the load schedule steps in which the through-wall or through-shoulder equivalent plastic strain exceeds 2%.

The analyst shall document the Plastic strain contours as per the guidelines described in FEA Documentation section.

<u>Contact Stress Profile at Seal(s)</u>

The analyst shall extract the contact stress profile at Seal master surface(s) for the final step of make-up simulation and all the steps of all the load schedules listed in Table 1. The analyst shall capture following parameters from the contact stress profile:

1. Length-A as shown in Figure 15
2. Length-B as shown in Figure 15
3. Maximum Contact Stress (P_{max})
4. Contact Length (b-a)
5. Contact Energy (area under the curve)

Contact energy may be calculated using any numerical integration techniques such as Trapezoidal method, Simpson's method, Gauss Quadrature etc.

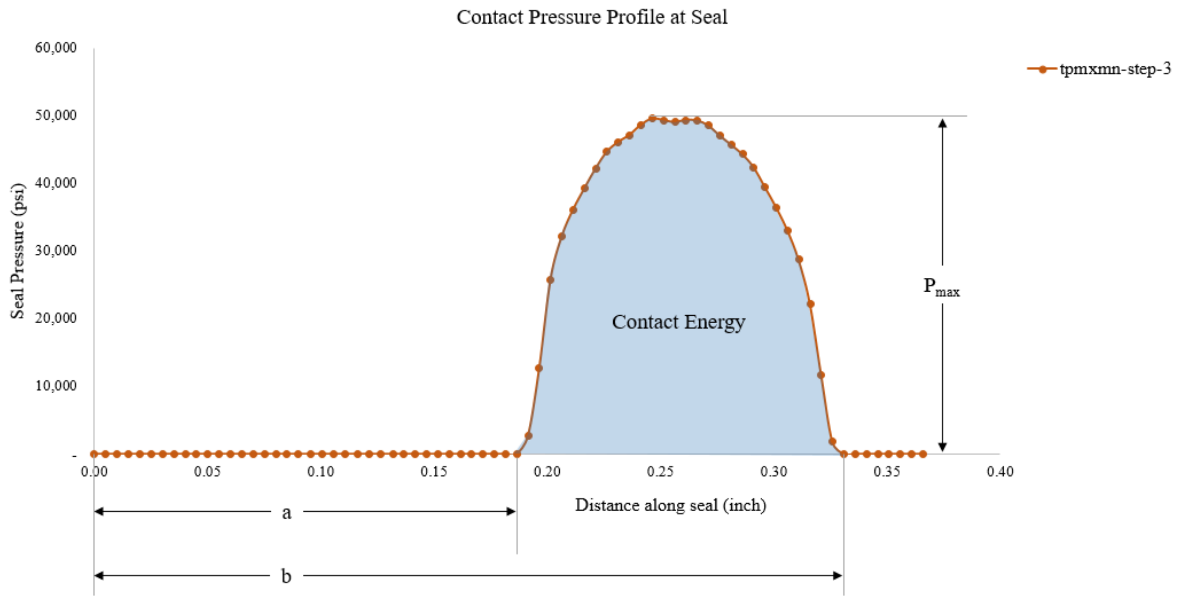


Figure 15 Contact Stress Profile at Seal – Example

ExxonMobil has developed an Abaqus plug-in to automate the contact stress profile extraction process, Appendix IV describes the instructions to use this plug-in. The analyst may use this plug-in along with the define connection plug-in or follow the instructions described in FEA Documentation to extract the contact stress profiles at seal(s).

FEA Service Provider

FEA shall be performed by any ExxonMobil approved FEA Service Provider. The FEA Service Provider may be an independent service provider or the Connection Owner's in-house FEA team.

The below flowchart (Figure 16) outlines the FEA Service Provider qualification process:

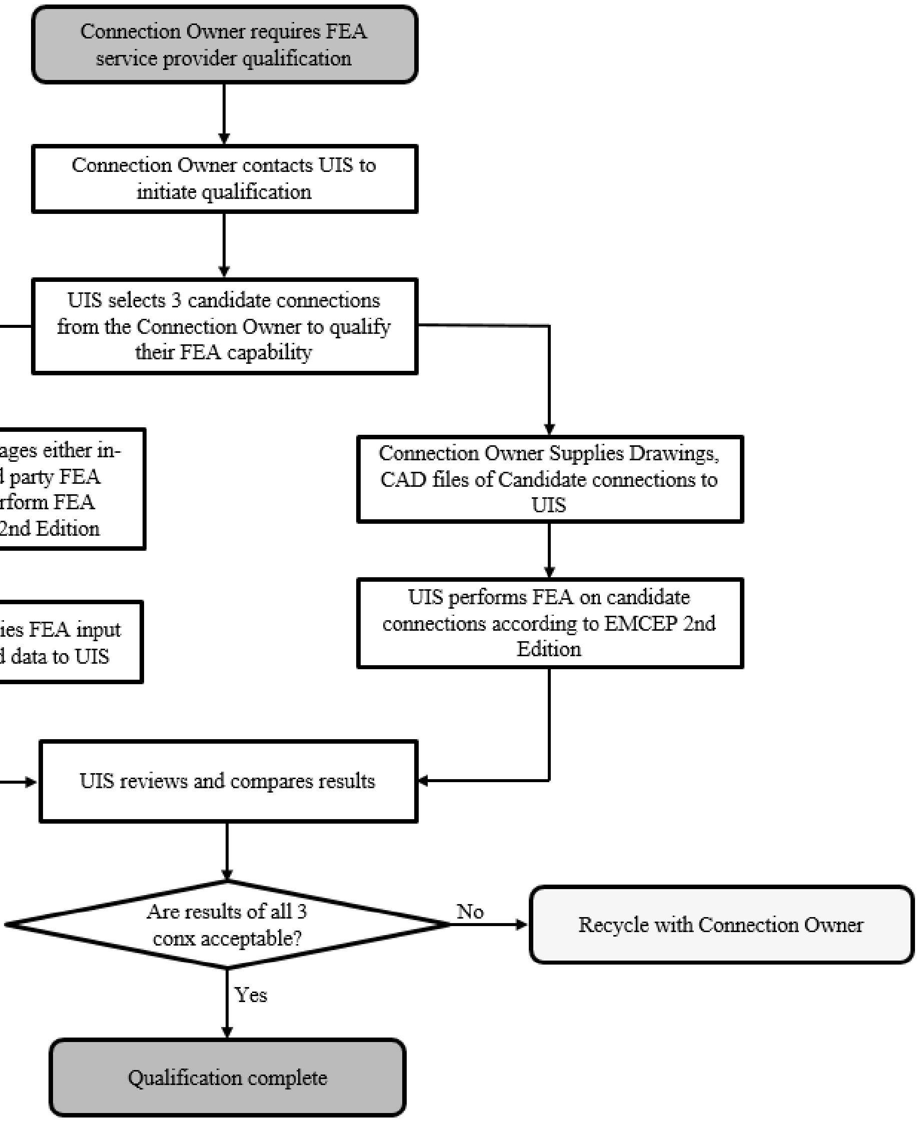


Figure 16 FEA Service Provider Qualification process flowchart

Part II – Physical Testing

Overview

Physical testing is the second of the two components of the ExxonMobil Connection Evaluation Program. Physical testing is used to verify Finite Element Analysis (FEA) results. In addition, physical testing is used to evaluate performance parameters that cannot be studied conveniently through FEA (e.g., galling resistance, performance sensitivity to thread compound type and amount).

Testing Protocol

Physical testing, from inception to test summary, shall be done as per **API RP 5C5, 4th Edition**. The assessment shall be at the **Connection Assessment Level IV (CAL IV)**. Note that ISO 13679:2019 is also an acceptable equivalent.

Sample Generation

Samples shall be generated by mill standard process. With the exception of the targeted tolerances required by the Standard, QA/QC processes for the manufacturing and acceptance of the connections shall remain as true to mill standard production processes as practical.

Safety

All the safety standards and considerations as per API RP 5C5, 4th Edition have to be met prior to the test. Any deviations from the safety standards needs to be discussed with all parties in attendance.

Lab personnel, third party inspector (TPI) and connection owner representatives are empowered to halt operations at any time if they believe it is unsafe to proceed. When agreed that the safety issue is addressed satisfactorily, operations can continue.

THE TESTING DESCRIBED IN API RP 5C5, 4TH EDITION PROGRAM INCLUDES PRESSURE AND TEMPERATURE TESTING AND CAN POSE A RISK OF PERSONAL INJURY OR PROPERTY DAMAGE IF APPROPRIATE TESTING AND SAFETY PROCEDURES ARE NOT FOLLOWED. EXXONMOBIL CORPORATION AND ANY OF ITS AFFILIATES OR EMPLOYEES SHALL NOT BE LIABLE FOR PERSONAL INJURY OR PROPERTY DAMAGE TO ANY PERSON RESULTING FROM THE TESTING DESCRIBED IN API RP 5C5, 4TH EDITION.

THE PERSON(S) CONDUCTING THIS TESTING SHOULD ENSURE THAT APPROPRIATE SAFETY PROCEDURES AND PRACTICES AS MENTIONED IN API RP 5C5, 4TH EDITION ARE IN PLACE PRIOR TO UNDERTAKING ANY TESTING UNDER THIS PROGRAM.

Third Party Witness

An independent, ExxonMobil approved third-party witness shall be used unless otherwise directed by ExxonMobil. All third party daily reports shall be documented and made available as an appendix in the test summary report. ExxonMobil shall identify suitably experienced and knowledgeable TPI's and provide the list of approved TPI's on the *ExxonMobil Wells Technical Quality extranet site.*

The third party inspectors shall record all deviations from CAL IV of API RP 5C5, 4th Edition test protocol in physical testing summary form.

TPI Instructions are given in Appendix V of this document.

Testing Facilities

Testing shall be conducted at any ExxonMobil approved connection testing facility. The connection testing facility may be an independent lab or the Connection Owner's in-house facility. A list of ExxonMobil Approved Independent Connection Testing Facilities is available on the *ExxonMobil Wells Technical Quality extranet site.*

The evaluation consists of a pre-screening questionnaire that helps decide the scope of subsequent stages of evaluation, which may involve an on-site audit by UIS Representatives.

Facilities reviewed and approved by UIS before 2020 remain approved provided the equipment remain substantially the same, the procedures remain substantially the same, there aren't events suggesting insufficient process controls and physical address and ownership remain the same. Any changes in equipment or procedures shall be reported to UIS.

Connection testing facilities can contact UIS for the latest qualification status of their lab.

Part III – Documentation

Overview

This part describes the format and the information which needs to be documented and reported to UIS.

The documentation consists of two parts:

- Finite Element Analysis Report
- Physical Testing Report as per Section 9 of API RP 5C5, 4th Edition

FEA Documentation

This section describes the format in which the FEA report shall be prepared. The report shall be prepared by the organization responsible for the FEA.

The FEA report shall be supplied in electronic format as a single document with Plastic Strain contour plots, Contact Stress Profiles and Connection data sheet as attachments. The test report shall contain the following sections:

1. Connection FEA Checklist
2. Summary
3. Connection Data Sheet
4. Geometry
5. Mesh
6. Material
7. Assessment of Simulations
8. Results
9. Analysis Deviations

An example of FEA report is available on the *ExxonMobil Wells Technical Quality extranet site.*

The instructions to prepare each of the FEA report sections are described below:

Connection FEA Checklist

Connection FEA checklist shall be prepared by the analyst and signed by analyst, reviewer and approver. The FEA checklist shall answer the questions listed in Table 2. The answers shall be in Yes/No format.

Table 2 Connection FEA Checklist

Yes	No	Summary
☐	☐	Does the CONNECTION SKETCH accurately depict the correct connection geometry?
☐	☐	Are Connection MANUFACTURER, CONNECTION type, SIZE, WEIGHT and GRADE are consistent with Connection Data Sheet?
☐	☐	Do PREPARED BY, REVIEWED BY and APPROVED BY boxes all have signatures and dates supplied?
		Geometry & Assembly
☐	☐	Are the CAD files as per the connection engineering drawing's latest revision?
☐	☐	Are Seal(s) and Thread diameters in the Model as per Max/Min Configuration?
☐	☐	Is the Assembly reflect minimum make-up torque configuration?

Yes	No	Mesh
☐	☐	Does the mesh contain only quad elements?
☐	☐	Does the mesh contain less than 1% distorted elements?
		Material
☐	☐	Does the Material have elastic behavior only up to specified minimum yield stress (SMYS)?
		Make-up Simulation
☐	☐	Are the thread, seal(s) and torque shoulder(s) interferences completely resolved at the end of the make-up simulation?
		Load Schedule Simulations
☐	☐	Is the maximum ratio of stabilization energy to internal energy ≤ 5% for all the load schedules for which stabilization is used?
☐	☐	Is the FEA status table contain remarks for the load schedules which didn't fully converge?
		Result Extraction
☐	☐	Are the Plastic Strain contour plots extracted with max. limit **0.02** and min. limit **0.00**?
☐	☐	Are the load steps with through-wall/shoulder plasticity identified and tabulated?
☐	☐	Are the contour plot PDF files generated accurately and referenced in result section?
☐	☐	Are the Contact Stress Profile parameters tabulated accurately? (N/A for FEA done w/ Post Process connector)
☐	☐	Are the XOM files generated accurately and referenced in result section? (N/A for FEA done w/o Post Process connector)

PREPARED BY	REVIEWED BY	APPROVED BY
____________________	____________________	____________________

Summary

Summary section of the FEA report shall contain following information:

1. Connection details (Manufacturer, Connection Name, Connection size, weight and grade)
2. Connection drawing number(s) used
3. Connection sketch
4. Critical cross section area of pin and box, in $inch^2$
5. Software(s) used with version

Connection Data Sheet

Connection Data Sheet section of the FEA report shall contain the latest version of the Connection Data Sheet supplied by the connection owner.

Geometry

Geometry section of the FEA report should contain Geometry Comparison table. The analyst should compare the dimensions in engineering drawing and that in CAD files, and document the differences in this table.

Mesh

Mesh section of the FEA report shall contain a screenshot of Element quality check with parameters described in the Discussion of Methodology - Mesh section. This section shall also contain an Image of Mesh with critical areas zoomed-in.

Material

Material section of the FEA report shall contain ALL the material properties specified in the FEA Model.

Assessment of Simulations

Assessment of Simulations section of the FEA report shall have two sub-sections:

1. Gross assessment
2. Detailed assessment

<u>Gross assessment</u>

This sub-section shall contain a table detailing the status and remarks of the load schedules. Remarks shall be written for the load schedules which did not fully converge. The remark should contain step number and stress/strain/contact behavior of the model at last converged step.

<u>Detailed assessment</u>

This sub-section shall contain Stress Contour, Reaction Loads and Model Energy assessments.

Stress Contour Assessment shall contain following stress contour plots:

a) Radial Stress contour plot at make-up simulation final step with clear close up view of seal(s), threads and torque shoulder(s).
b) Axial Stress contour plot at make-up simulation final step with clear view of Pin and Box ends and with following limits:
   - Maximum : 1.0 Minimum : -1.0
c) Hoop Stress contour plot at make-up simulation final step with clear view of Pin and Box ends and with following limits:
   - Maximum : 1.0 Minimum : -1.0

Reaction Loads Assessment shall contain a comparison table of reaction force at the Box end and applied axial load for final step of **cmxmn** and **tmxmn** load schedules.

Model Energy Assessment shall contain model energy plots with internal energy and stabilization energy for ALL the load schedules in which stabilization has been used.

Results

This section of the FEA report shall have two sub-sections; Plastic Strain Contours and Contact Stress Profiles. Instruction to prepare these two sub-sections are described below:

Plastic Strain Contours

The analyst shall capture the equivalent plastic strain contours in the four different views described in Table 3 for all the steps of all the load schedules listed in Table 1. The contour plots shall be captured with **maximum limit of 0.02 and minimum limit 0.00**

Table 3 Plastic Strain Contour Plot Views Description

View	Description
User-1	Global view of the connection
User-2	View of seal, threads & torque shoulder
User-3	Close-up view of primary seal
User-4	Close-up view of torque shoulder/ secondary seal, if applicable

Figure 17 shows an example of different views in which the equivalent plastic strain contour plots shall be extracted.

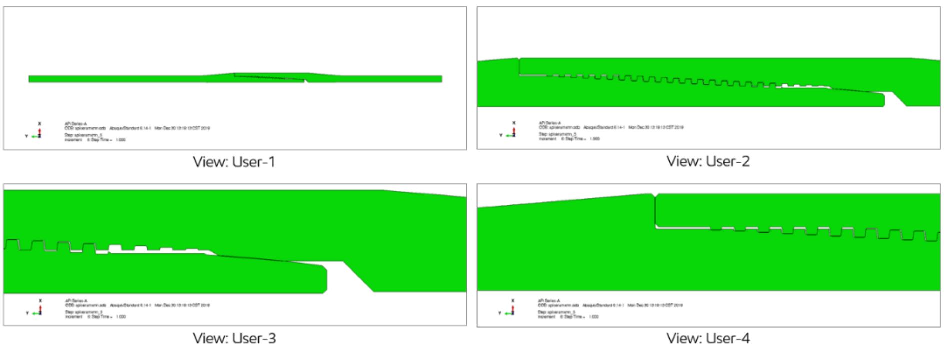


Figure 17 Plastic Strain Contour Plot Views - Example

First page of these PDF files shall have the list of load schedule steps in which the through-wall or through-shoulder equivalent plastic strain exceeds 2%. Examples of through-wall and through-shoulder plastic strain >2% are shown in Figure 18 and Figure 19 respectively.

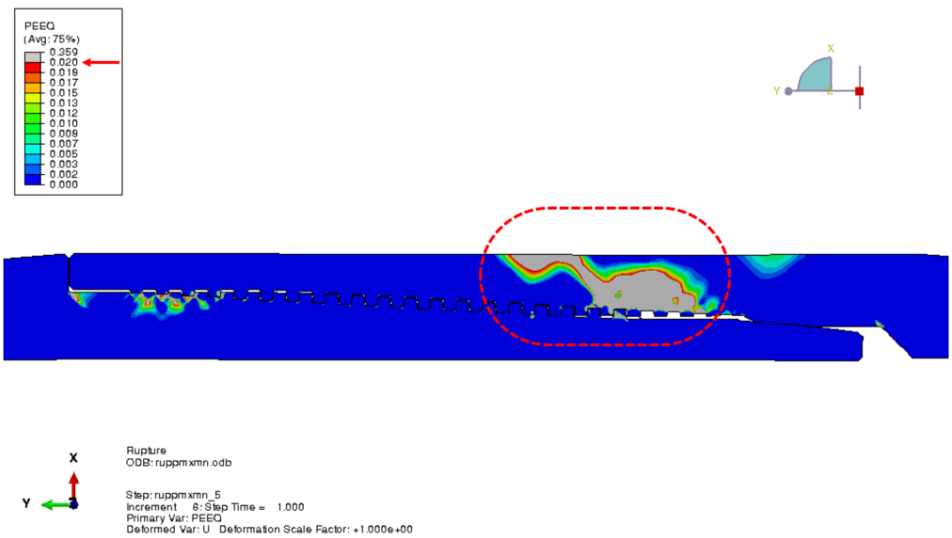


Figure 18 Through-wall plastic strain > 2% - Example

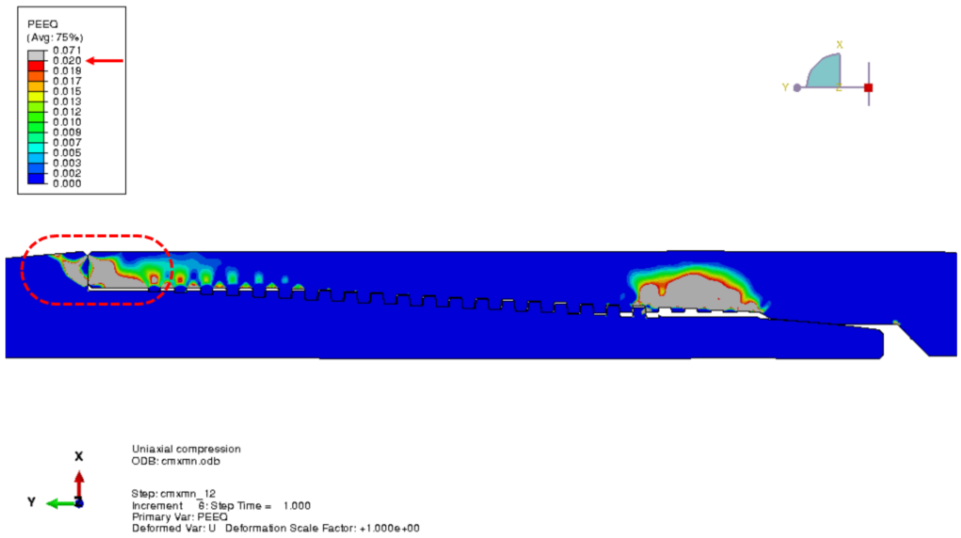


Figure 19 Through-shoulder plastic strain > 2% - Example

Upon extraction of the contour plots, the analyst shall group the images by views and combine the images to four PDF files, each corresponds to four views described in Table 3.

Names of these PDF files shall be referenced in the Plastic Strain Contours sub-section of the FEA report and the PDF files shall be attached with the FEA report.

In the case of two-sealed connections plastic strain contour files for "Standard Analysis" and "Bleed-through Analysis" shall be prepared separately.

Contact Stress Profiles

The analyst shall capture the contact stress parameters for the final step of make-up simulation and all the steps of all the load schedules listed in Table 1.

The contact stress profile parameters shall be captured at the seal(s) master surface with the following guidelines:

1. Contact pressure profile parameters shall be captured in Tabular form as shown in Table 4.
2. Columns for the Table shall be as per the order and units described below:
   Column-1 : Load Schedule Name
   Column-2 : Step Number
   Column-3 : Pressure in psi, (Internal Pressure positive and External Pressure negative)
   Column-4 : Axial Load in kips, (Tension loads positive and compression loads negative)
   Column-5 : Length "a" in inches (see Figure 15)
   Column-6 : Length "b" in inches (see Figure 15)
   Column-7 : Maximum contact stress in ksi
   Column-8 : Contact Length in inches
   Column-9 : Contact Energy in ksi-in
3. First row shall be the contact stress parameters of make-up final step, followed by that of load schedule steps.
4. Load schedules shall be **in the same order as they are tabulated in Table 1**.

5. Contact Stress Profile parameters shall be specified as **"<No_Data>"** for the Load steps in which a converged solution is not available. Applied Pressure and Axial Load for these load steps shall be captured accordingly.
6. Contact pressure profile Table shall have **361 rows** once fully filled.

Table 4 Contact Stress Profile Table - Template

Load Schedule	Step Number	Applied Pressure (psi)	Applied Axial Load (kips)	a (in)	b (in)	P_{max} (ksi)	b-a (in)	Energy (ksi-in)
Make-up	final	0	0					
c25epmxmn	1							
c25epmxmn	2							

In the case of two-sealed connections:

- For "Standard Analysis", Contact Stress Profile Tables shall be captured for internal seal and external seal separately.
- For "Bleed-through Analysis", Contact Stress Profile Table shall be captured for primary seal.

ExxonMobil has developed an Abaqus plug-in to automate the contact stress profile extraction process, Appendix IV describes the instructions to use this plug-in. If the analyst choose to use the plug-in then the names of plug-in generated XOM files shall be referenced in the Contact Stress Profiles sub-section and the XOM files shall be attached with the FEA report.

Analysis Deviations

In this section of the FEA report, the analyst shall capture **ALL** the deviations from the "Part I – Finite Element Analysis" section occurred during the analysis.

Physical Test Report

Physical test report shall be prepared in accordance to Section 9 of API RP 5C5, 4th Edition. The test report shall also contain:

- TPI reports
- Root cause failure analysis reports (if any).
- Information related to testing of any failed specimens or failed attempts during the test program.

The test report shall be endorsed by the parties/individuals involved in this test program with signatures by Connection Owner's representatives, Lab representative, third party witness and also representative of any other parties involved.

All documents generated out of physical testing activities which needs to be witnessed shall be endorsed by TPI.

A Physical test summary form shall be prepared as per the instructions shown below and shall be included as the first page of the test report. This form describes the progress and results of the test program at a glance. A sample of this form is shown in Figure 20.

Instructions

- Enter the tubular size, wall thickness, weight and grade on the first line under the form title.
- Enter the Connection Owner and connection name on the second line under the form title.
- Gray out cells for tests that were not required.
- For other cells:
    - For tests that were successfully completed, enter "PASSED" in green on the first line of the cell.
    - For tests that were not successfully completed, enter "FAILED" in red on the first line of the cell.
    - If a test was required but not completed (e.g., the specimen failed during a previous test), leave the cell blank.
    - Enter the date of test on the second line of the cell. For a test conducted over multiple days, enter the date the test was completed.
    - Use the third and subsequent lines for explanations:
        - For limit load tests describe the failure mode, location and mode (as percent of the appropriate connection owner's rating).
        - For failed tests, describe the failure (e.g., load step during which a leak was observed).
        - For any deviations from the test protocol.
- Use separate forms for each batch of specimens (i.e., specimens 1-5 on the first form, specimens 1R1-5R1 on the second form, etc.)
- The form must be signed and dated by
    - Connection Owner's representative
    - Lab representative
    - Third-party witness

PHYSICAL TESTING SUMMARY FORM

9-5/8-inch OD, 0.545 wall (53.50 lb/ft), P-110 Grade

ABC XYZ Connection

Specimen	Make & Break Test	Test Series B	Test Series C	Test Series A (90%)	Test Series A (95%)	Limit Load Test
Specimen 1	PASSED 12 Feb 2020	PASSED 15 Feb 2020	PASSED 20 Feb 2020	FAILED 25 Feb 2020 Leak during load step 133 (LP14a90); test suspended		
Specimen 1R1	PASSED 2 Mar 2020	PASSED 10 Mar 2020	PASSED 15 Mar 2020	PASSED 20 Mar 2020	PASSED 25 Mar 2020	PASSED 30 Mar 2020 Pipe-body failure, A-side, at 133% CEE
Specimen 2	PASSED 12 Feb 2020	PASSED 15 Feb 2020	PASSED 20 Feb 2020	PASSED 22 Feb 2020	PASSED 25 Feb 2020	PASSED 3 Mar 2020 at 125% of CEE, no leak was observed & testing was stopped
Specimen 3	PASSED 12 Feb 2020 With make and breaks performed on Side B as well	PASSED 15 Feb 2020	PASSED 20 Feb 2020	PASSED 22 Feb 2020	PASSED 25 Feb 2020	PASSED 7 Mar 2020 Pipe-body fracture, A-side, 120% MTR
Specimen 4	PASSED 12 Feb 2020	PASSED 15 Feb 2020	PASSED 20 Feb 2020	PASSED 22 Feb 2020	PASSED 25 Feb 2020	PASSED 3 Mar 2020 at 120% of CEE, no leak was observed & testing was stopped
Specimen 5	PASSED 12 Feb 2020					PASSED 16 Feb 2020 Pipe-body fracture, B-side, 115% MTR

Connection Owner's Representative

Third Party Witness

Lab Representative

Figure 20 Physical Testing Summary Form (Sample)

Part IV – Testing Variants

Overview

This section contains additions and modifications to the Connection Evaluation Program to account for variation from the standard test protocol.

Typical variations have been detailed in the following sub-sections:

1. Variant 1: Incremental Tests for Tubing Evaluation of a Previously Evaluated Casing
2. Variant 2: Incremental Evaluation of Connections Previously Evaluated with Other Alloy Contents
3. Variant 3: Modified Thread Compound, Surface Finish, and Recommended Make-Up Torque: Incremental Tests for Previously Evaluated Connections
4. Variant 4: Testing for High Torque
5. Variant 5: Gas Sealability Testing for Large Diameter Connections
6. Variant 6: Modified Drawing or Revision Number: Re-Evaluation of Previously Evaluated Connections
7. Variant 7: Special Clearance and Special Drift Connections

The application of Variants shall be within the same grade band. ExxonMobil has designated three material grade bands based on API minimum yield strength:

1. Less than 75 ksi
2. 75 to 95 ksi (inclusive)
3. Greater than 95 ksi

The terminologies used in the testing Variants' flowcharts are as per API RP 5C5, 4th Edition.

In the instances where multiple Variants are applicable refer to Table 5 for testing guidance. Variant 4 shall be treated and tested independently if it is applicable along with other Variants. Variant 7 is only required when no other Variants are to be performed.

Table 5 Testing Guideline if Multiple Variants are Applicable

Applicable Variants			Test to be done as per
Variant 1	Variant 2	Variant 3	
✓	✓	✓	Variant 2
✓	✓		Variant 2
	✓	✓	Variant 2
✓		✓	Variant 3

Variant 1: Incremental Tests for Tubing Evaluation of a Previously Evaluated Casing

This section describes the incremental test program needed to evaluate for tubing use, a connection that was previously evaluated as per EMCEP, 1st Edition or later for casing use. The previous evaluation must have been performed using the same connection design with the same diameter, weight and grade band.

Specimens 3X and 5X as per CAL IV of API RP 5C5, 4th Edition tubing application shall be tested i.e., 9 make and breaks shall be performed. The incremental test program for both threaded-and-coupled and integral joint connections is shown in Figure 21 (Note that the terms and references used in the figure are taken from API RP 5C5, 4th Edition).

The make and breaks for specimen 3X shall be performed on the B end of the connection. Series C shall not be performed on Specimen 3X.

A connection that is evaluated for tubing use is available for casing use without any further evaluation.

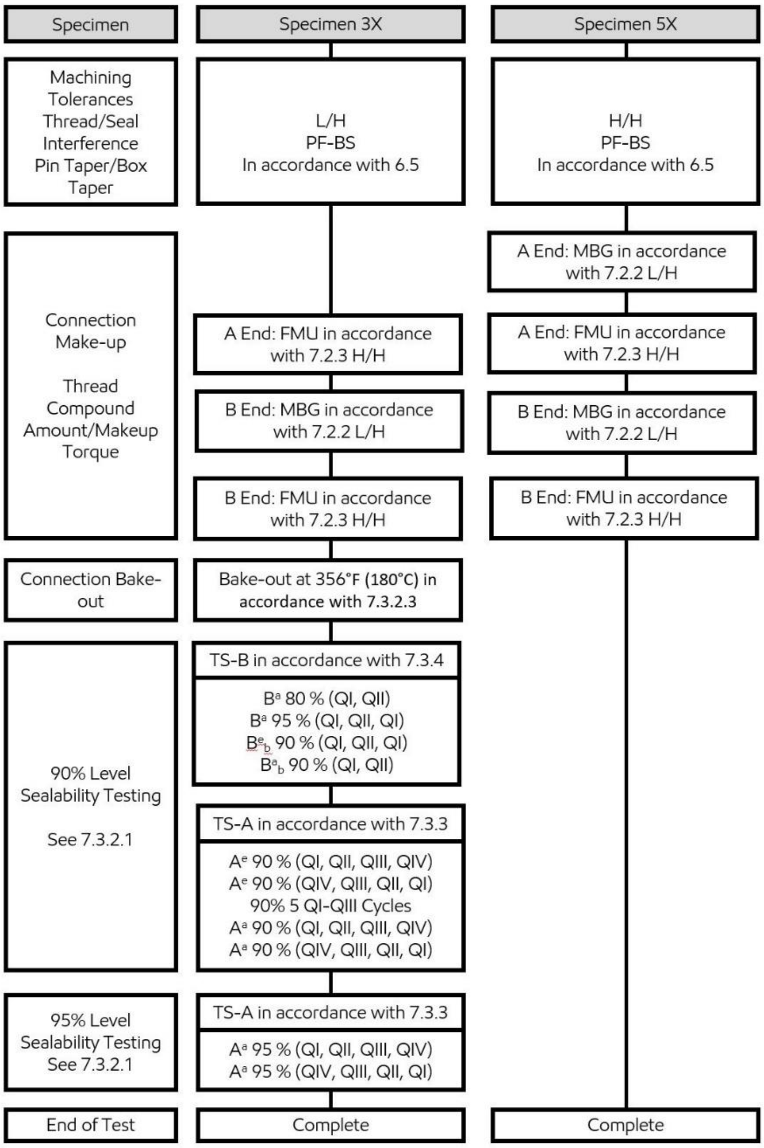


Figure 21 Incremental test for a tubing evaluation of a connection previously tested for casing

Variant 2: Incremental Evaluation of Connections Previously Evaluated with Other Alloy Contents

This section describes the incremental test program needed to evaluate a connection design on corrosion resistant alloys (CRA) that was previously evaluated for carbon steels or lower CRA; or the opposite. The previous evaluation must have been performed according to EMCEP, 1st Edition or later using the same connection design with the same size and weight. In addition, the yield strength of the proposed CRA connection must be with in the same grade band as the previously evaluated connection.

The incremental test program is shown in Figure 22 (Note that the terms and references used in the figure are taken from API RP 5C5, 4th Edition). The make and breaks for specimen 1C shall be performed on the A end of the connection. The Make and Breaks for the specimen 3C shall be performed on the B end of the connection. Series C shall not be performed on Specimen 1C.

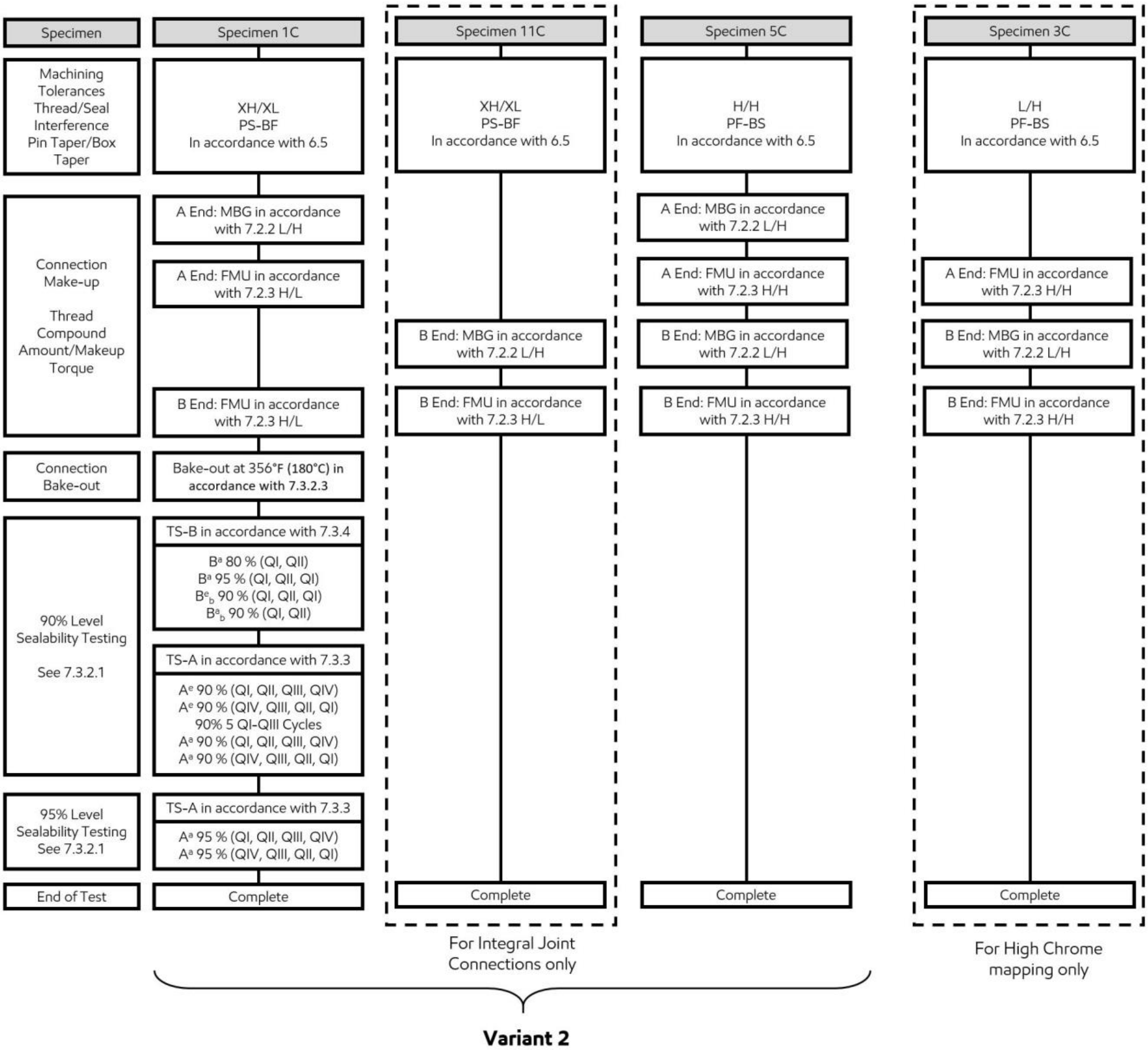


Figure 22 Incremental Evaluation of Connections Previously Evaluated with Other Alloy Contents

FEA may be required to extend the results across different materials as shown in Table 6.

Extension of Evaluation Results

See Table 6 for a summary of how the evaluation results may be extended across different materials.

Table 6 Extension of Results across different materials within the same grade band

Current Evaluation	Previous Evaluation[a]					
	Carbon steel	**13Cr80**	**S-13Cr, 15Cr, 17Cr**	**22Cr, 25Cr**	**SM2535, Alloy28**	**Other CRAs[c]**
Carbon steel		None	None	None[b]	None[b]	None[b]
13Cr80	VA 2		None (S13Cr-95 Only)			
S-13Cr, 15Cr, 17Cr	VA 2	None (S13Cr-95 Only)		None[b]	None[b]	None[b]
22Cr, 25Cr	VA 2 + FEA		VA 2 + FEA		None	None
SM2535, Alloy28	VA 2 + SP3C + FEA		VA 2 + SP3C + FEA	VA 2 + SP3C + FEA		None
Other CRAs[c]	VA 2 + SP3C + FEA		VA 2 + SP3C + FEA	VA 2 + SP3C + FEA	VA 2 + FEA	VA 2 + FEA

[a] Previous evaluation must have been for the same design, size and weight. The yield strengths must be within the same grade band. The previous evaluation must have been according to ExxonMobil Connection Evaluation Program (1st Edition or Later).

[b] FEA may be performed to capture the material behavior

[c] Examples: Alloys 825, 2550, G-3, C-276

Variant 3: Modified Thread Compound, Surface Finish, and Recommended Make-Up Torque: Incremental Tests for Previously Evaluated Connections

This section describes the requirements to extend the evaluation of a connection that has been previously evaluated with differences in one or more of the following features:

- Thread compound type
- Recommended thread compound amount
- Surface finish[d]
- Recommended maximum make-up torque
- Recommended minimum make-up torque

The previous evaluation must have been performed according to EMCEP, 1st Edition or later using the same connection design (drawing and revision numbers) with the same size and weight. Also, the yield strength of the proposed connection must be within the same grade band as the previously evaluated connection.

To extend the evaluation without additional testing, evidence must be presented to UIS that the proposed change(s) do not adversely affect connection performance. If no such evidence exists, or if the evidence is deemed insufficient by UIS, incremental testing must be performed.

The incremental test program is shown in Figure 23 (Note that the terms and references used in the figure are taken from API RP 5C5, 4th Edition). The make and breaks for specimen 1M shall be performed on the A end of the connection. Series C shall not be performed on Specimen 1M. No additional FEA is required for extending results to the new evaluation.

[d] Changes to surface finish do not require additional evaluation in cases where the previous evaluation was for a Corrosion Resistant Alloy (CRA) and the current evaluation is for Carbon Steel.

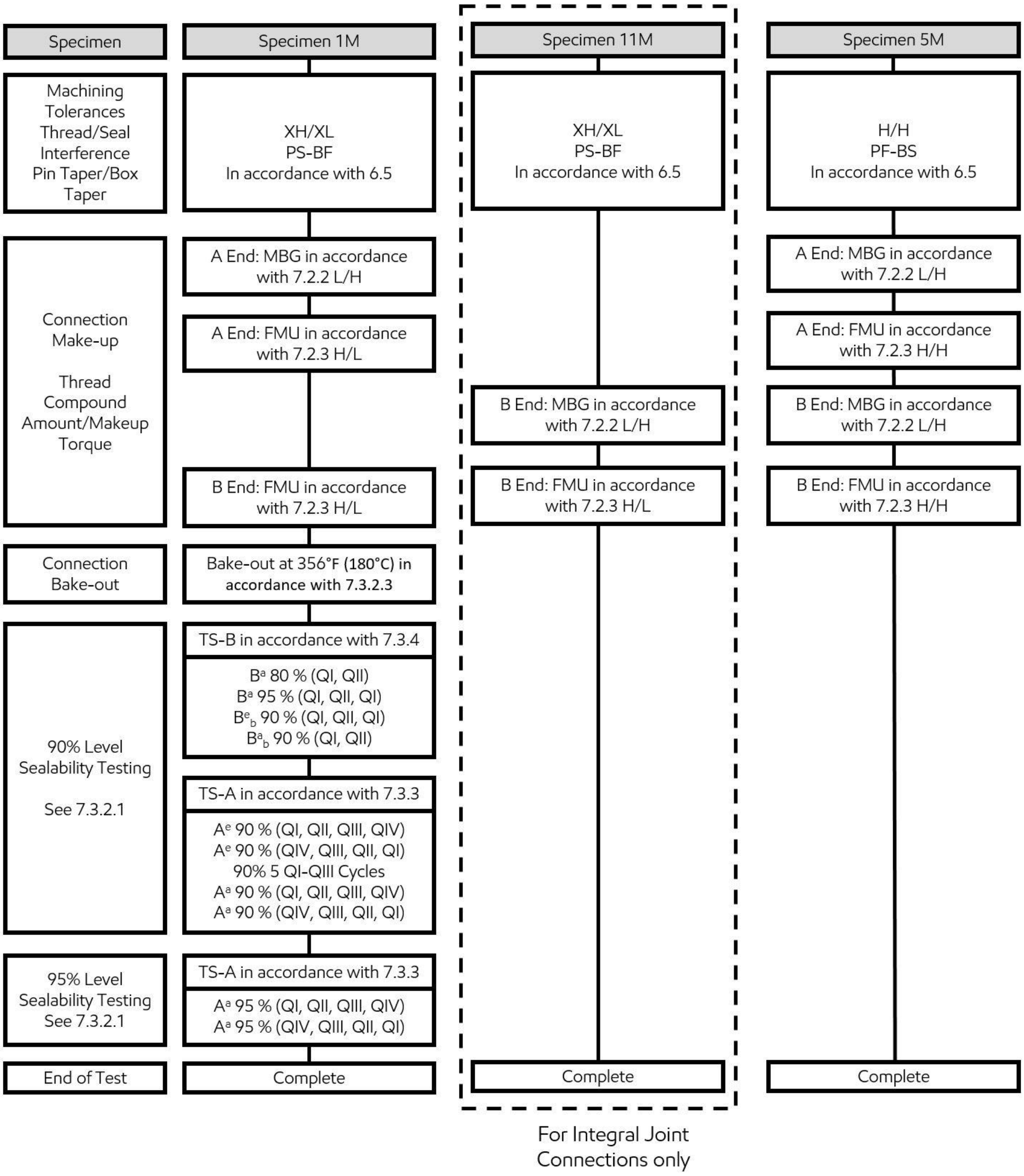


Figure 23 Incremental test to be performed on connection with modified thread compound, Surface finish and recommended make up torque

Variant 4: Testing for High Torque Applications

This section describes a test program that evaluates a connection design for high torque capacity, beyond published maximum makeup torque ratings. This test program may be incremental, building upon a previously completed full evaluation, or it may be incorporated into a planned full evaluation.

For a connection which has been already tested as per EMCEP, 1st Edition or later with the connection owner torque ratings, then an incremental test as per the flowchart shown in Figure 24 (Note that the terms and references used in the figure are taken from API RP 5C5, 4th Edition) shall be performed on Specimen 4T, 14T and **5T**.

If the connection is not tested as per EMCEP, 1st Edition or later, then a seven specimen test program shall be performed on Specimens 1, 2, 3, 4 (as per CAL IV of API RP 5C5, 4th Edition) and specimens 4T, 14T and **15T** as shown in Figure 24 (Note that the terms/references used in the figure is taken from API RP 5C5, 4th Edition).

Sealability testing may be fit-for-service for non-metal-to-metal seal connections. Specimens 4T and 14T shall be in "Minimum Material Configuration" (MMC) for non-metal-to-metal seal connections. Included in the MMC are minimum thread interference and minimum torsional load bearing areas. For a pin nose shouldered non-metal-to-metal sealed connection, the minimum torsional load bearing area would require that the pin nose thickness be machined to the minimum dimension allowed by the design tolerances. These MMC samples are designed to assess potential for structural deformation from high torque loading.

The drift tests performed after Make up to failure are for information purposes only.

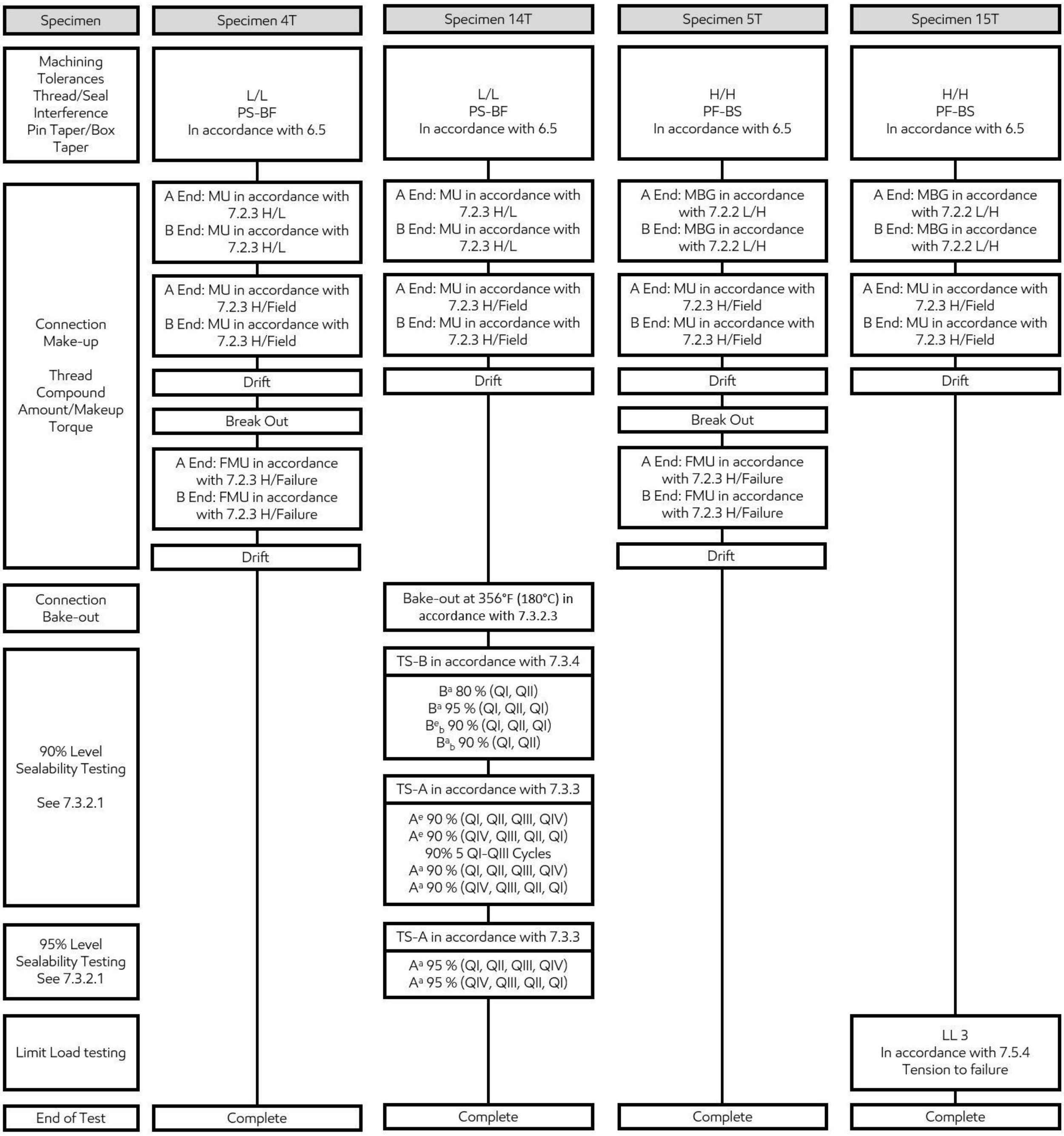


Figure 24 Testing Flowchart for High torque applications

Variant 5: Gas Sealability Testing for Large Diameter Connections

This section describes a test program that evaluates the gas sealing capacity of a large diameter connection used primarily as surface casing or conductor casing. Large OD is generally considered to be 16" and greater. Reduced test programs for larger OD connectors are based on intended service environment rather than an actual OD. This test program is intended to be stand-alone, however it may build upon a previously completed evaluation, or it may be incorporated with other test programs to assess a broader suite of performance.

For Large Diameter Connectors with metal-to-metal seal intended for use as surface casing with high liquid or any gas pressure, CAL II of API RP 5C5, 4th Edition shall be performed with the following guidance:

- The elevated bending step (B^e_b 90% (QI, QII, QI)) might be omitted.
- Using liquid as the internal pressurizing medium is acceptable, but will result in a liquid only evaluated envelope (no gas seal capacity assumed).

Large Diameter Connectors intended for use as surface casing with low liquid pressure shall require some demonstrated liquid sealability in the same size, weight, and grade. No specific protocol is required, CAL II of API RP 5C5, 4th Edition, ambient, liquid is suggested.

Large Diameter Connections intended for use as conductor casing with low liquid pressure shall require some demonstrated liquid sealability in the same size, weight, and grade. No specific protocol required, CAL I of API RP 5C5, 4th Edition is suggested.

Variant 6: Modified Drawing or Revision Number: Re-Evaluation of Previously Evaluated Connections

This section describes the requirements to re-evaluate connections that have previously been evaluated under a different drawing number or revision number. The previous evaluation must have been performed according to EMCEP, 1st Edition or later with the same size and weight. The yield strength of the proposed connection must be within the same grade band as that of the previously evaluated connection. In addition, the surface finish, thread compound type and amount, and recommended minimum and maximum make-up torques of the proposed connection must be the same as those of the previously evaluated connection.

Procedure

Depending on the nature of the modification, some or all of the following steps may be required to extend the evaluation to the new drawing or revision number. The Connection Owner should perform a review and propose a path forward to UIS:

1. Review of the old and new drawings to determine the nature of the change(s). If there is no material change in the connection, or the design is deemed similar to that of the previously evaluated connection, no further evaluation may be necessary.

2. The Connection Owner may choose to perform FEA on the connection model represented by the new drawing. If the mechanical response of the new connection to axial and pressure loads is deemed similar[e] to that of the previously evaluated connection, no further evaluation is necessary.

3. The new connection may be subjected to a customized test program drawn out of CAL IV of API RP 5C5, 4th Edition. The customized test program will focus on aspects of connection performance thought to be affected by the modifications that led to the change in drawing or revision number. If the connection successfully completes this test program, no further evaluation is necessary. The Connection Owner should propose a test program and confirm with UIS.

4. If none of the above conditions apply, the new connection must be fully evaluated as per CAL IV of API RP 5C5, 4th Edition.

[e] Similarity of mechanical response to axial and pressure loads will be determined by UIS, based on seal energization, connection shouldering, structural failure, etc.

Variant 7: Special Clearance and Special Drift Connections

This section describes the incremental test program needed for evaluating a Special Clearance (SC) or Special Drift (SD) connection if the standard version of the same size/weight/grade band/connection type and same torque values is previously evaluated as per EMCEP, 1st Edition or later.

Special Drift

An incremental test program on specimen 1 as per CAL IV of API RP 5C5, 4th Edition and FEA as per Part I – Finite Element Analysis of this document shall be performed in order to fully evaluate the connection.

Special Clearance

For SC80 and above, an incremental test program on specimens 1 and 5 as per CAL IV of API RP 5C5, 4th Edition and FEA as per Part I – Finite Element Analysis of this document shall be performed in order to fully evaluate the connection.

For anything below SC80, the connection owner needs to provide data to UIS marking the differences between Standard and Special Clearance (below SC80). In addition, the connection owner may propose an appropriate incremental test program to evaluate the special clearance (below SC80) connection.

For Special Clearance and Special Drift connections having different torques than their standard version follow Variant 3.

Appendix I

Adjusting Shoulder Interference in Connection Finite Element Models

This technical note describes the method used to adjust shoulder interference for finite element models of connections. The note describes the general approach used to modify the geometry. Examples are provided for two typical connection geometries. The general approach can be used to apply the method to other connection geometries.

General Approach

The following rules must be satisfied while adjusting the connection model to achieve the desired shoulder interference:

1. The shoulder interference is adjusted by adjusting the relative distance between cylindrical and helical features.
2. The relative distance between key cylindrical features must remain constant.
3. The relative arrangement within each group of key helical features must remain constant.

Key cylindrical features include the seal surfaces and shoulder surfaces. Key helical features are the threads.

As a convention, all modifications are made on the pin member.

This approach is not applicable to connections without shoulders (e.g., dovetail-thread products, API LTC and BTC connections).

Example

Pin Nose Shoulder with Pin Nose Seal

Shown in Figure 25, this geometry is typical of most threaded and coupled connections. The connection has a single seal that is located next to the shoulder at the pin nose.

The shoulder interference is controlled through the width of the section located between the seal region and the first thread. This is marked as section A in Figure 25. To increase the shoulder interference by a given amount, the width of section A is increased by the same amount. To decrease the interference, the width of section A is decreased.

Mid-Thread Seal with Pin Nose and Box Face Seals

Shown in Figure 26, this geometry is typical of integral joint connections. The connection has a two-step thread form and the shoulder is located between the two threads. There are two seal surfaces: the seal near the pin nose is for internal pressure resistance and the seal near the box face is for external pressure resistance.

In this geometry, the relative distances between the shoulder and the seals must be constant. Therefore, four sections are defined as A, B, C and D as shown in Figure 26. To increase the shoulder interference by a given amount, the widths of sections B and D are increased and the widths of sections A and C are decreased by the same amount. To decrease the shoulder interference, the widths of sections A and C are increased and the widths of sections B and D are decreased.

Some connection designs have seals located on either side of the mid-thread shoulder. This approach can still be used provided that the seals are located between sections B and C.

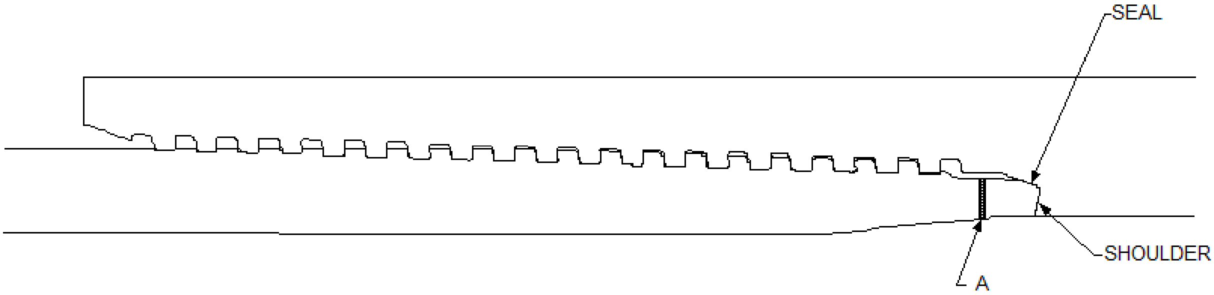


Figure 25 Pin Nose Shoulder and Seal

To increase shoulder interference, increase the width of section A. To decrease shoulder interference, decrease the width of section A.

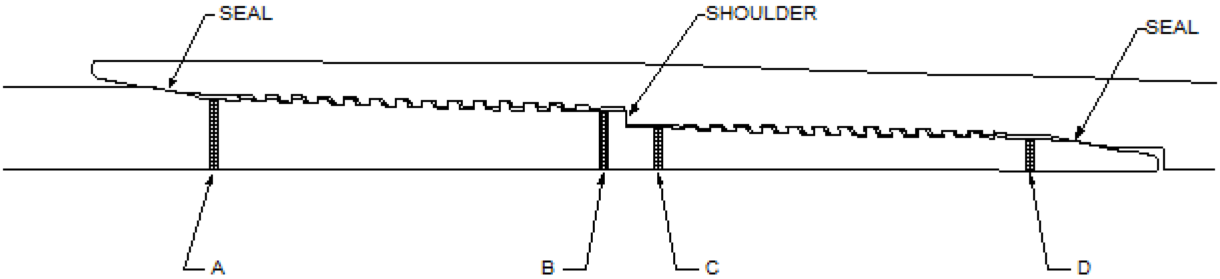


Figure 26 Mid-Thread Shoulder with Pin Nose and Box Face Seals

To increase shoulder interference, increase the widths of sections B and D and decrease the widths of sections A and C. To decrease shoulder interference, decrease the widths of sections B and D and increase the widths of sections A and C.

Appendix II

ExxonMobil Define Connector plug-in for Abaqus

A sequence of programs have been developed for Abaqus that automate a substantial portion of the pre-processing steps. The output from the Define Connector plug-in will be a set of input files for Abaqus solver. This section describes how to access, utilize and prepare generated Abaqus input files for solver.

The analyst shall verify the material properties, loads and boundary conditions in these input files before submitting them in the Abaqus solver.

Table 7 Abaqus Input files generated using Define Connector Plug-in

Input file name	Type
mmxmn	Make-up
apiseramxmn	Specimen
apiserbmxmn	Specimen
sp3mxmn	Specimen
sp5mxmn	Specimen
c25epmxmn	Standard
c25pmxmn	Standard
c50epmxmn	Standard
c50pmxmn	Standard
c75epmxmn	Standard
c75pmxmn	Standard
c90epmxmn	Standard
c90pmxmn	Standard
cepmxmn	Standard
cmxmn	Standard
cpmxmn	Standard
epmxmn	Standard

Input file name	Type
mtrpmxmn	Standard
pmxmn	Standard
ruppmxmn	Standard
t100pmxmn	Standard
t25epmxmn	Standard
t25pmxmn	Standard
t50epmxmn	Standard
t50pmxmn	Standard
t75epmxmn	Standard
t75pmxmn	Standard
tepmxmn	Standard
tmxmn	Standard
tpmxmn	Standard
vmcepmxmn	Standard
vmcpmxmn	Standard
vmepmxmn	Standard
vmpmxmn	Standard

Steps to Access Define Connector Plug-in

a) Define Connector plug-in and supporting scripts are clustered together in a single zip file "**DefineConnector_plugin_XOM**". Extract the zip file to a folder

b) Place the folder in Plug-in central directory (*e.g., for Abaqus 2017, C:\SIMULIA\CAE\plugins\2017*) or
Change the plug-in central directory address to the Define Connector plug-in folder address in the custom Abaqus environment file by editing the "**plugin_central_dir= <folder address>**" line.

Custom environment file (*.env) can be found in the following folder (*for 2017 version*):

C:\Program Files\Dassault Systemes\SimulationServices\V6R2017x\win_b64\SMA\site

c) **Define Connector** plug-in can be accessed from Abaqus CAE Assembly and Mesh Modules:

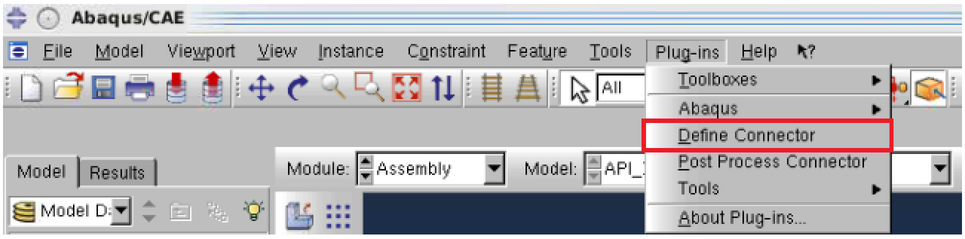


Figure 27 Define connector plug-in

Creating the Analysis Model using Define Connector

Define Connector plug-in needs the FE mesh of the connector transverse section (e.g., see Figure 28) to start the workflow. Mesh needs to be in axisymmetric elements (CAX4) and mesh quality has to be verified before the Defined Connector workflow.

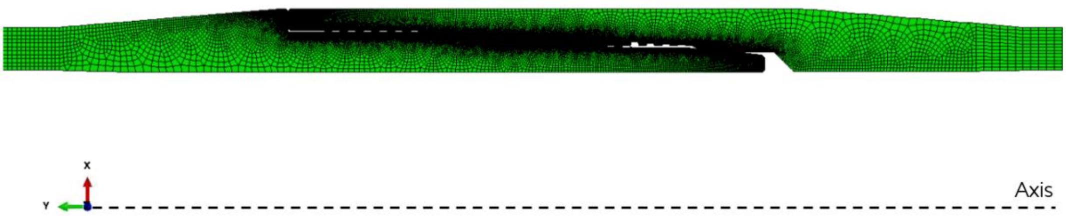


Figure 28 FE mesh of the connector transverse section

Once the mesh is imported or generated in Abaqus CAE, from the Mesh or Assembly module the plug-in can be accessed as shown in Figure 27. Step-by-step process of using the plug-in is shown below:

1. *Define Connector Properties*

The first dialogue box the plug-in shows is the “Define Connector Properties” dialogue box. In this the analyst needs to select OD and ID nodes from the viewport, and enter connector properties.

For connection evaluation with physical testing, the compression and tension efficiencies shall be the same as the efficiencies used to define the physical testing Connection Evaluation Envelope (CEE). For “FEA Only”, the efficiencies shall be as per the Connection Data Sheet.

An example is shown in Figure 29 below. After entering properties, and node selections, click on “Continue”.

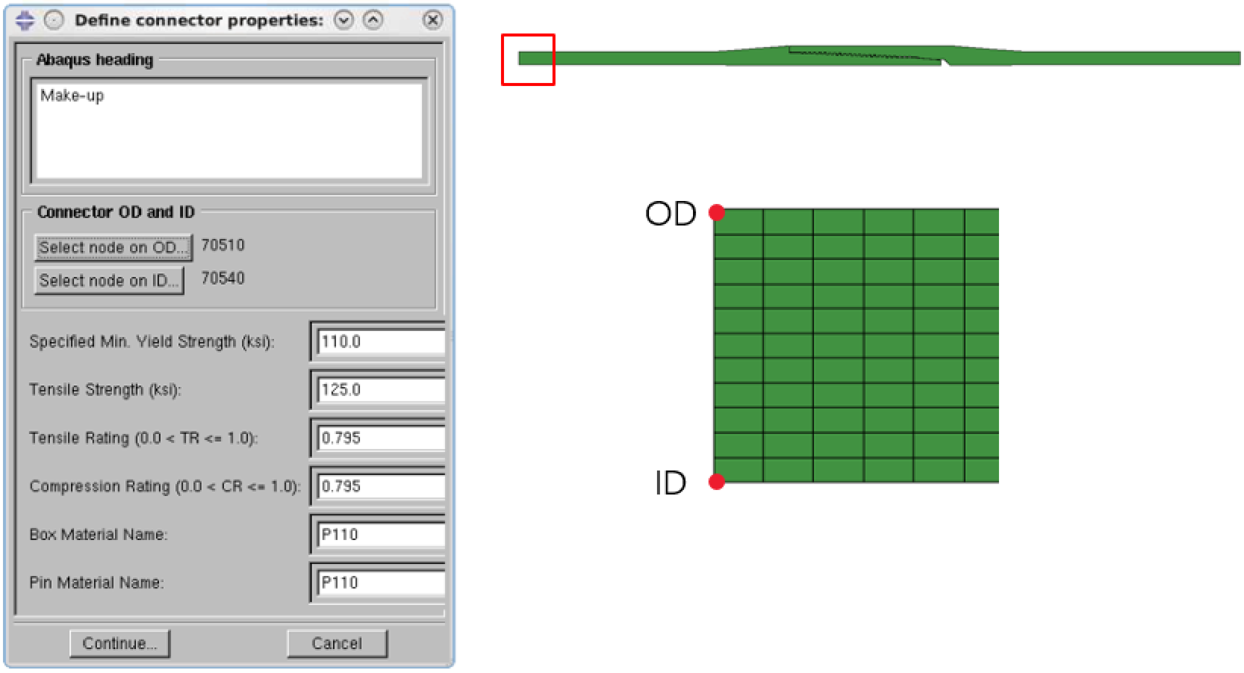


Figure 29 Define Connector Properties - Example

2. *Define Threads*

The second dialogue box the plug-in shows is the "Define Threads" dialogue box. In this the analyst needs to define Pin and Box thread surface pairs. The Box surfaces need to be defined as Slave and the Pin surfaces need to be defined as Master. Step-by step process of creating a thread surface pair is shown below:

i. Enter threads name (e.g., T09) and press **Enter** key.
ii. Enter slave surface name (e.g., TS09) and press **Enter** key. Click on "Select nodes" button, select first & second nodes on the Box thread, this will highlight the selected slave surface in yellow. Reverse the surface definition if required by clicking the "Reverse" button.
iii. Enter master surface name (e.g., TM09) and press **Enter** key. Click on "Select nodes" button, select first & second nodes on the Pin thread, this will highlight the selected master surface in red. Reverse the surface definition if required by clicking the "Reverse" button.
iv. In the interference option select "Shrink fit". Click on "Add" button to add the surface pair.
v. Repeat steps i to iv for all the individual threads.
vi. Use "Modify", "Delete" and "Delete All" buttons if required.
vii. Click on "Continue" to go to next step.

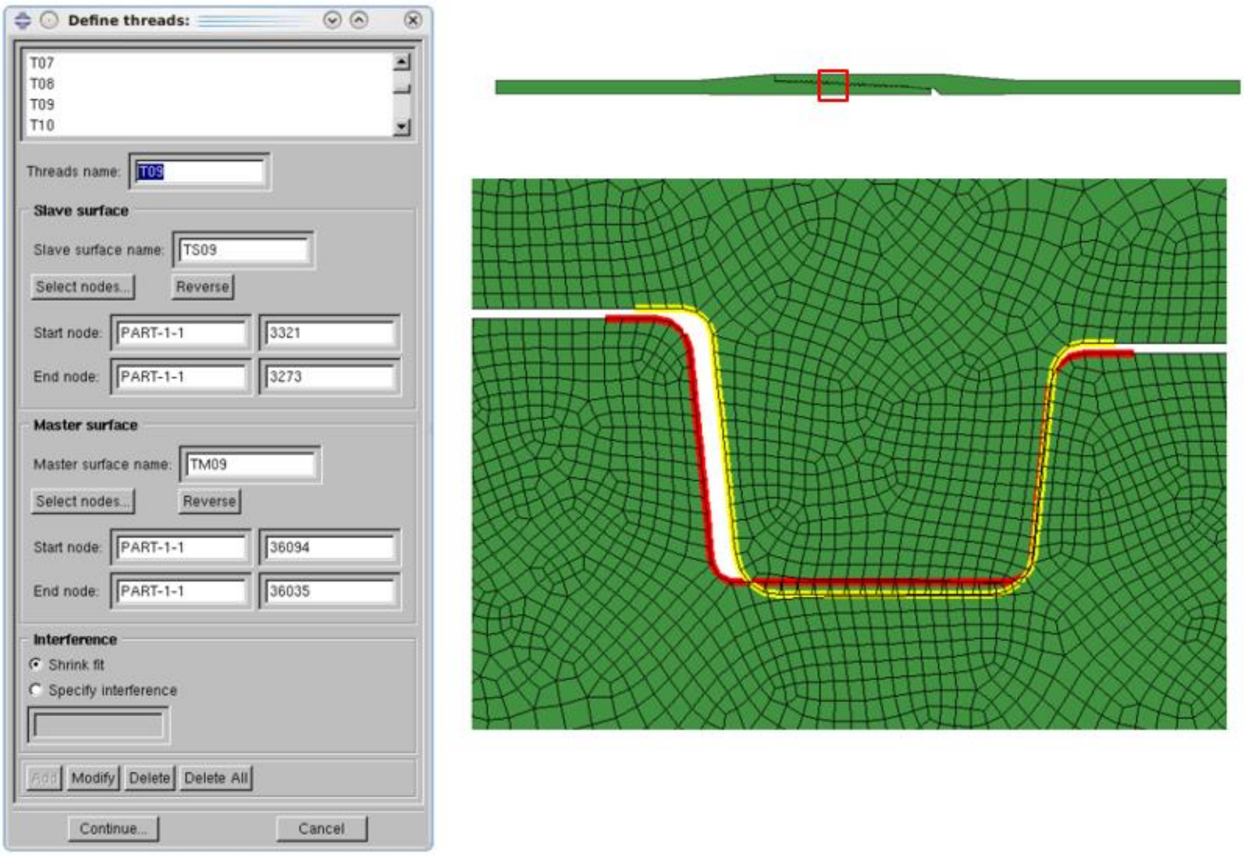


Figure 30 Define Threads - Example

3. *Define Seals*

The third dialogue box the plug-in shows is the “Define Seals” dialogue box. In this the analyst needs to define Pin and Box seal surface pair(s). The Box surfaces need to be defined as Slave and the Pin surfaces need to be defined as Master. Step-by step process of creating a seal surface pair is shown below:

i. Enter seals name (e.g., Seal01) and press **Enter** key.
ii. Enter slave surface name (e.g., SealS01) and press **Enter** key. Click on “Select nodes” button, select first & second nodes on the Box seal, this will highlight the selected slave surface in yellow. Reverse the surface definition if required by clicking the “Reverse” button.
iii. Enter master surface name (e.g., SealM01) and press **Enter** key. Click on “Select nodes” button, select first & second nodes on the Pin seal, this will highlight the selected master surface in red. Reverse the surface definition if required by clicking the “Reverse” button.
iv. In the interference option select “Automatically calculated”. Click on “Add” button to add surface pair.
v. Repeat steps i to iv to define remaining seal surface pairs, if the connection has more than one seal.
vi. Use “Modify”, “Delete” and “Delete All” buttons if required.
vii. Click on “Continue” to go to next step.

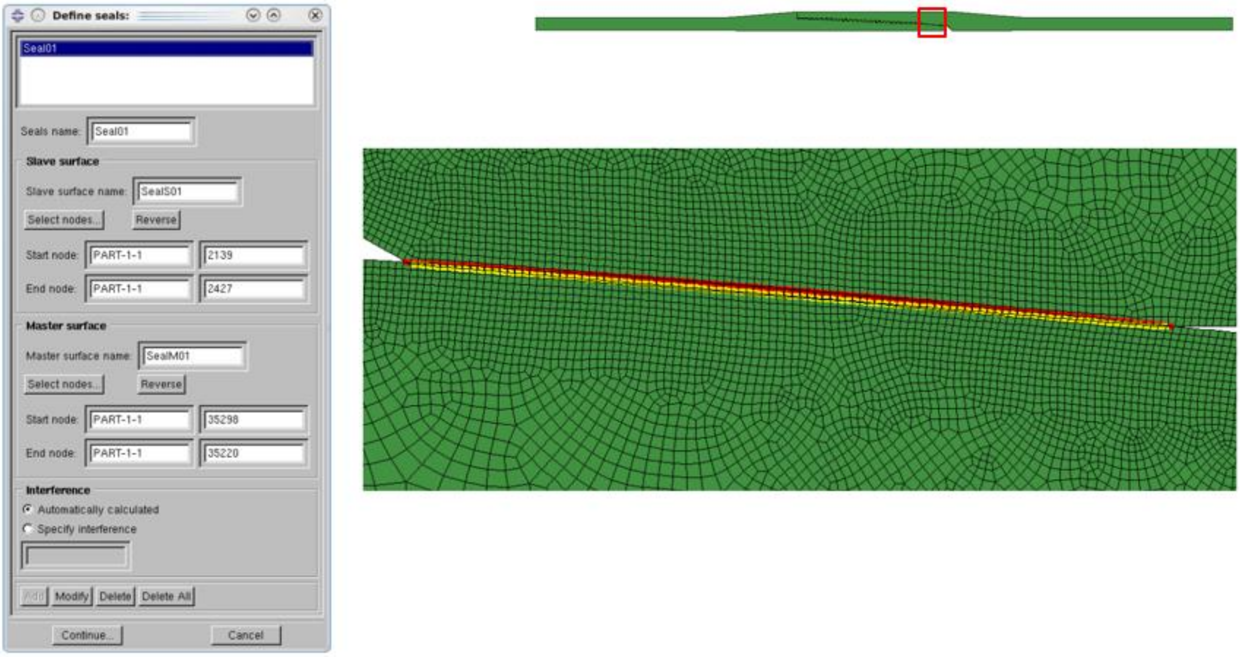


Figure 31 Define Seals - Example

4. *Define Torque Shoulders*

The fourth dialogue box the plug-in shows is the "Define Torque Shoulders" dialogue box. In this the analyst needs to define Pin and Box torque shoulder surface pair(s). The Box surfaces need to be defined as Slave and the Pin surfaces need to be defined as Master. Step-by step process of creating a torque shoulder surface pair is shown below, the analyst can skip this step by clicking "Continue" button if the connection does not have torque shoulder.

i. Enter torque shoulder name (e.g., TQ01) and press **Enter** key.
ii. Enter slave surface name (e.g., TQS01) and press **Enter** key. Click on "Select nodes" button, select first & second nodes on the Box torque shoulder, this will highlight the selected slave surface in yellow. Reverse the surface definition if required by clicking the "Reverse" button.
iii. Enter master surface name (e.g., TQM01) and press **Enter** key. Click on "Select nodes" button, select first & second nodes on the Pin torque shoulder, this will highlight the selected master surface in red. Reverse the surface definition if required by clicking the "Reverse" button.
iv. In the interference option select "Specify Interference" and enter the interference value to be 0.005. Click on "Add" button to add surface pair.
v. Repeat steps i to iv to define remaining torque shoulder surface pairs, if the connection has more than one torque shoulder.
vi. Use "Modify", "Delete" and "Delete All" buttons if required.
vii. Click on "Continue" to go to next step.

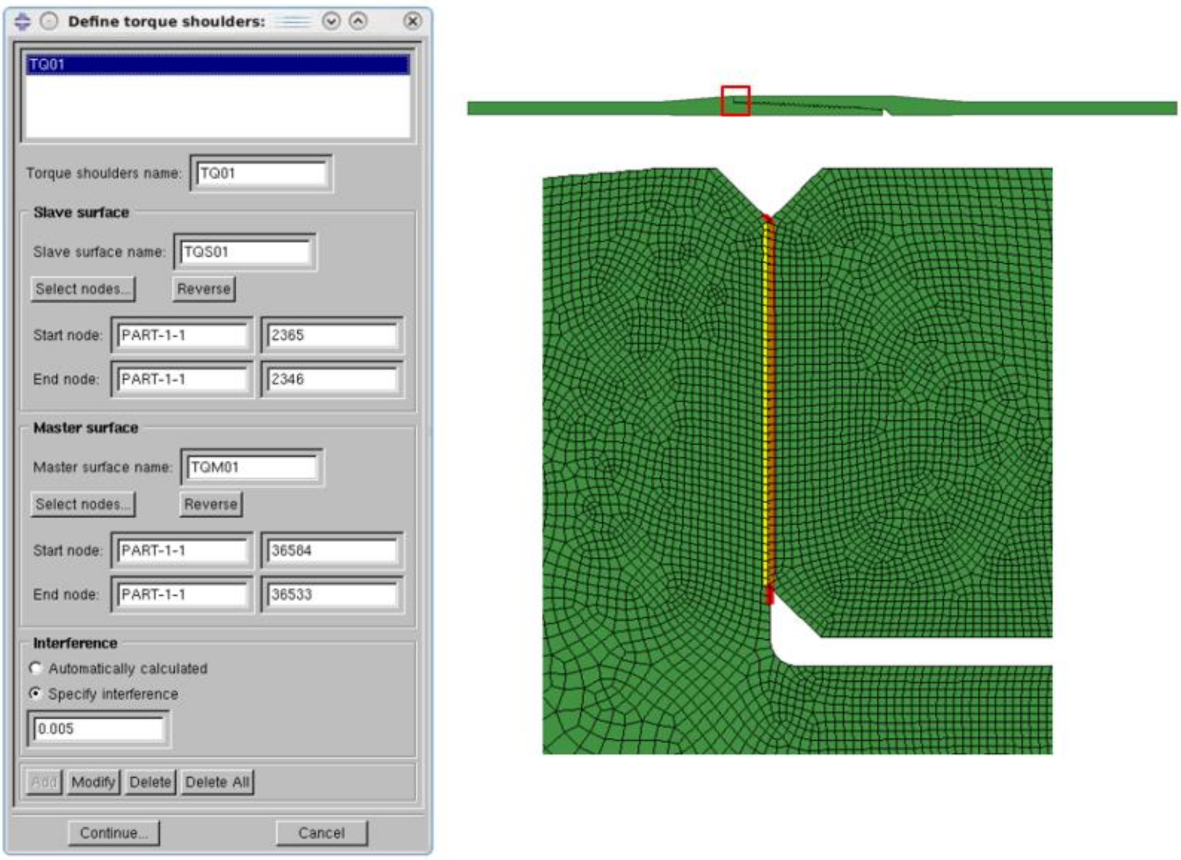


Figure 32 Define Torque Shoulders – Example

5. *Define External Pressure*

The fifth dialogue box the plug-in shows is the “Define External Pressure” dialogue box. In this the analyst needs to define Pin and Box surfaces where external pressure need to be applied. Step-by step process of creating external pressure surfaces on Pin and Box is shown below.

i. From the “Defined Seals” Box, select seal which is exposed to external pressure[f]. This will highlight the seal surface pair in the viewport.

ii. Enter external pressure surface name for Box (e.g., EP_Box). Click on “Select nodes” button, select first & second nodes on the Box external pressure surface[g], one end of the surface has to be the slave node of the selected seal which is exposed to external pressure. Other end has to be the last node on the Box OD. Reverse the surface definition if required by clicking the “Reverse” button. Click on “Add” button to add surface.

iii. Enter external pressure surface name for Pin (e.g., EP_Pin). Click on “Select nodes” button, select first & second nodes on the Pin external pressure surface, one end of the surface has to be the master node of the selected seal which is exposed to external pressure. Other end has to be the last node on the Pin OD. Reverse the surface definition if required by clicking the “Reverse” button. Click on “Add” button to add surface.

iv. Use “Modify”, “Delete” and “Delete All” buttons if required.

v. Click on “Continue” to go to next step.

Figure 33 shows an example of Pin and Box external pressure surfaces.

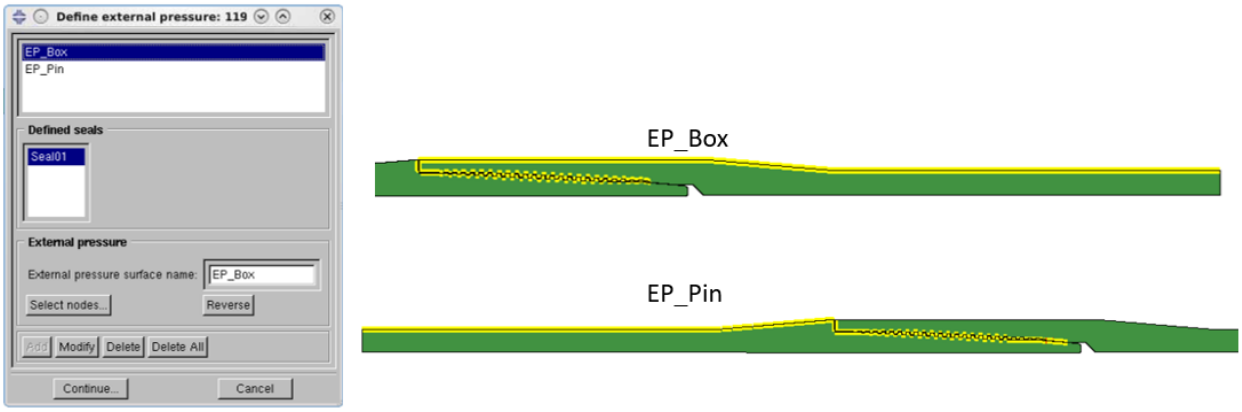


Figure 33 Define External Pressure - Example

[f] For connections with more than one seal, i) For standard analysis, select the seal which is exposed to external pressure ii) For “**Bleed Through**” analysis select the primary seal.

[g] Only metal-to-metal seal is considered as sealing element, i.e., threads and torque shoulders are treated as non-sealing elements.

6. *Define Internal Pressure*

The sixth dialogue box the plug-in shows is the "Define Internal Pressure" dialogue box. In this the analyst needs to define Pin and Box surfaces where internal pressure need to be applied. Step-by step process of creating internal pressure surfaces on Pin and Box is shown below.

i. From the "Defined Seals" Box, select seal which is exposed to internal pressure[h]. This will highlight the seal surface pair in the viewport.
ii. Enter internal pressure surface name for Box (e.g., IP_Box). Click on "Select nodes" button, select first & second nodes on the Box internal pressure surface, one end of the surface has to be the slave node of the selected seal which is exposed to internal pressure. Other end has to be the last node on the Box ID. Reverse the surface definition if required by clicking the "Reverse" button. Click on "Add" button to add surface.
iii. Enter internal pressure surface name for Pin (e.g., IP_Pin). Click on "Select nodes" button, select first & second nodes on the Pin internal pressure surface, one end of the surface has to be the master node of the selected seal which is exposed to internal pressure. Other end has to be the last node on the Pin ID. Reverse the surface definition if required by clicking the "Reverse" button. Click on "Add" button to add surface.
iv. Use "Modify", "Delete" and "Delete All" buttons if required.
v. Click on "Continue" to go to next step.

Figure 34 shows an example of Pin and Box internal pressure surfaces.

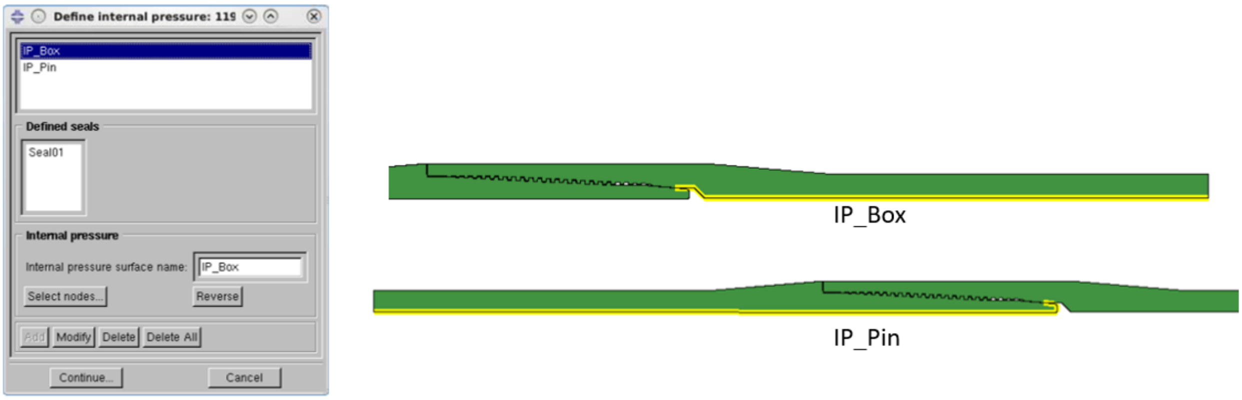


Figure 34 Define Internal Pressure - Example

[h] For connections with more than one seal, i) For standard analysis, select the seal which is exposed to internal pressure ii) For **Bleed Through** analysis select the primary seal.

7. Define Axial Pressure

The seventh dialogue box the plug-in shows is the “Define Axial Pressure” dialogue box. In this the analyst needs to define Pin surface where axial pressure (tension or compression) need to be applied. Step-by step process of creating axial pressure surface on Pin is shown below.

i. Enter axial pressure surface name (e.g., Axial_Load). Click on “Select nodes” button, select first & second nodes on the Pin axial pressure surface, one end of the surface has to be the OD node of the Pin. Other end has to be the ID node of the Pin. Reverse the surface definition if required by clicking the “Reverse” button. Click on “Add” button to add surface.

ii. Use “Modify”, “Delete” and “Delete All” buttons if required.

iii. Click on “Continue” to go to next step.

Figure 35 shows an example of defining axial pressure surface on the Pin.

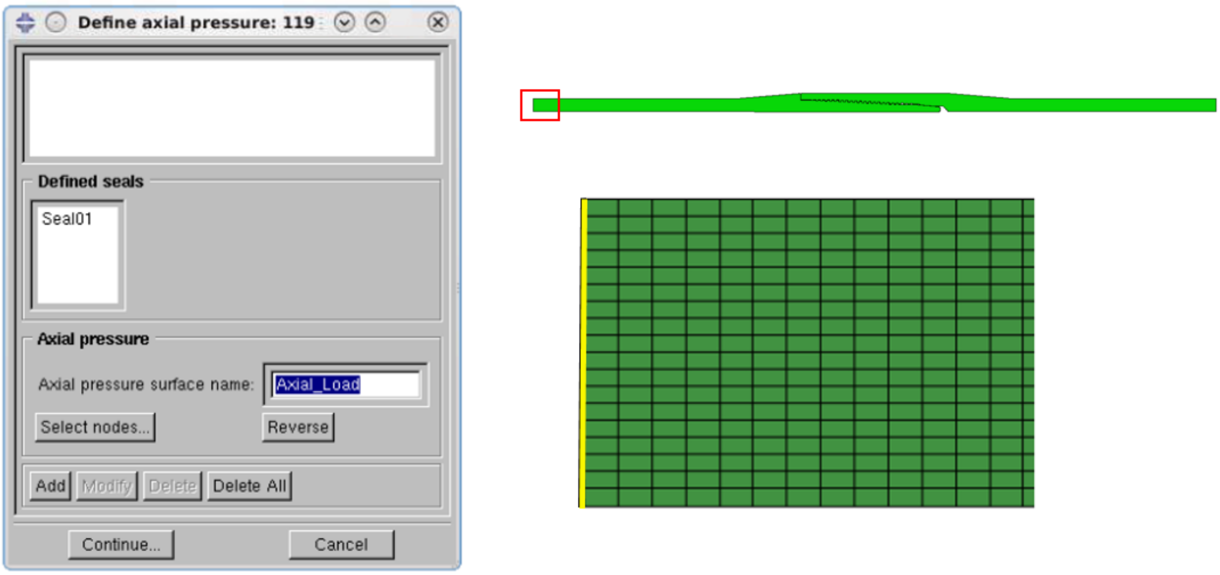


Figure 35 Define Axial Pressure - Example

8. *Define Pressure Penetration*

The eighth dialogue box the plug-in shows is the "Define Pressure Penetration" dialogue box. In this the analyst needs to define the pressure penetration definition(s).

To define pressure penetration, the exposure must be determined. If there is one seal, then the seal will be exposed to both internal and external pressure. If there are two seals, then the outer seal will be exposed only to external pressure and the inner seal exposed only to internal pressure[i]. Step-by step process of creating Pressure Penetration definition is shown below.

i. Enter Pressure Penetration definition name (e.g., PP01).
ii. Select the seal for which pressure penetration will be defined.
iii. Do one or both of the following
    a. Select the last seal slave node and the last seal master node for the internal pressure exposure (if applicable)
    b. Select the last seal slave node and the last seal master node for the external pressure exposure (if applicable)
iv. Click on "Add" button. Use "Modify" and "Delete" buttons if required.
v. Click on "Continue" to go to next step.

Figure 36 shows an example of Pressure Penetration definition.

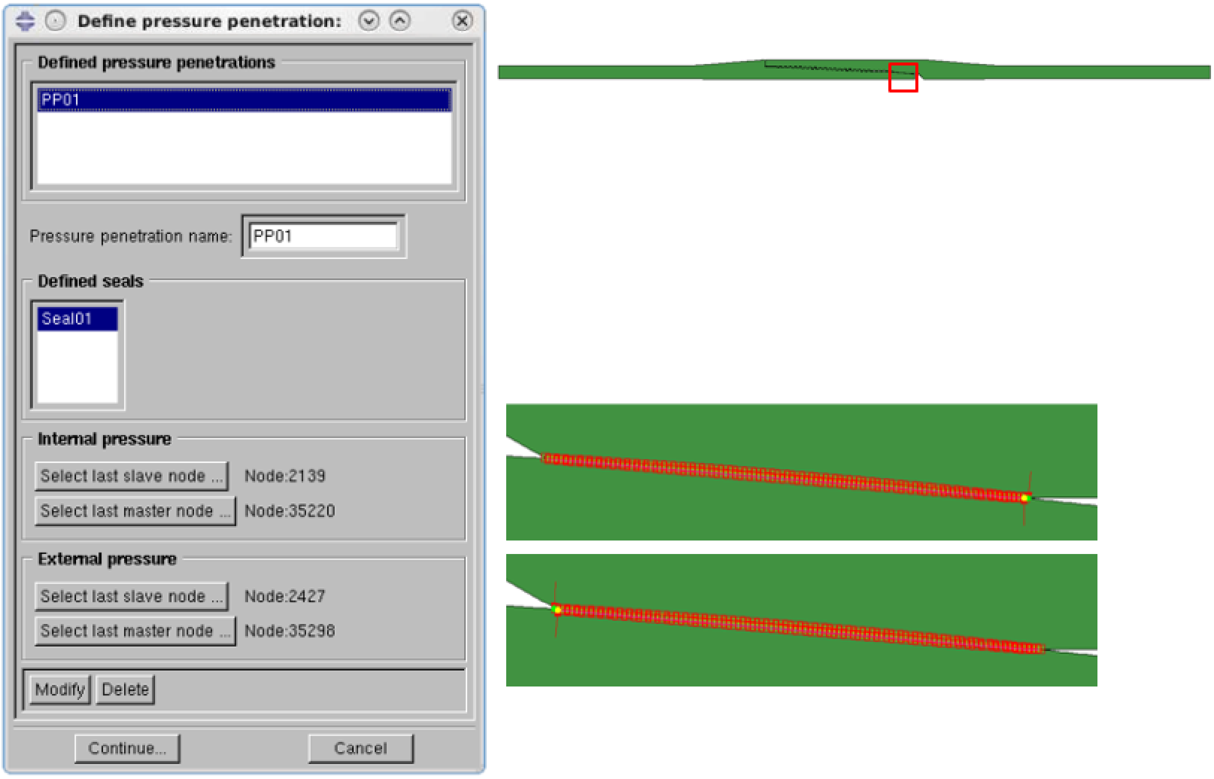


Figure 36 Define Pressure Penetration - Example

[i] For **Bleed Through** analysis define pressure penetration only on the primary seal

9. *Define Axial Constraint*

The ninth dialogue box the plug-in shows is the "Define Axial Constraint" dialogue box. In this the analyst needs to define Box surface which needs to be axially constrained. Step-by step process of creating axial constraint on the Box surface is shown below.

i. Enter axial constraint surface name (e.g., Ax_Constraint). Click on "Select nodes" button, select first & second nodes on the Box surface which needs to be axially constrained, one end of the surface has to be the OD node of the Box. Other end has to be the ID node of the Box. Reverse the surface definition if required by clicking the "Reverse" button.

ii. Click on "Add" button to add surface.

iii. Use "Modify", "Delete" and "Delete All" buttons if required.

iv. Click on "Continue" to go to next step.

Figure 37 shows an example of defining axial constraint on the Box.

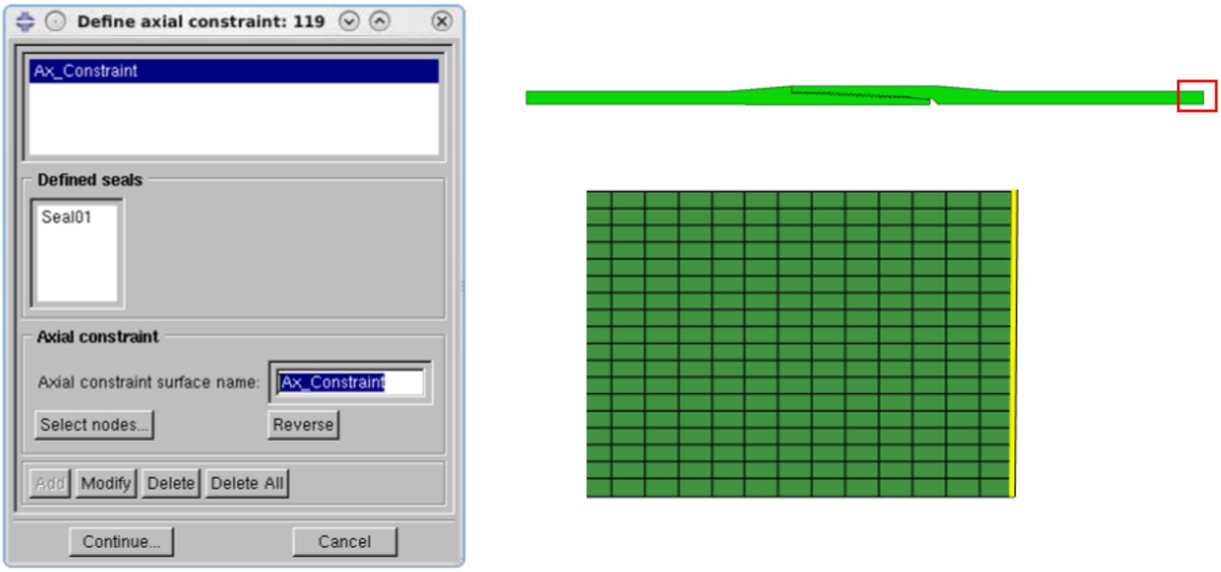


Figure 37 Define Axial Constraint - Example

10. Output Variables

The tenth dialogue box the plug-in shows is the "Output Variables" dialogue box. In this the analyst needs to select the element and node output variables required for the analysis. Contact Stress (CPRESS), contact displacement (CDISP), fluid pressure for pressure penetration analysis (PPRESS) and model energy output request are enabled by default in the plug-in. Step-by step process of selecting output variables is shown below.

i. In the "Element Variables" tab, Enter element output frequency (e.g., 999)
ii. Check the required element output variables. Following element output variables will be selected by default:
    a. Stress (S)
    b. Log Strain (LE)
    c. Plastic Strain (PE)
    d. Plastic Strain Magnitude (PEMAG)
iii. In the "Node Variables" tab, check the required node output variables. Following node output variables will be selected by default:
    a. Displacement (U)
    b. Reaction forces (RF)
iv. Click on "Continue" to go to next step.

Figure 38 shows an example of Output Variables tabs.

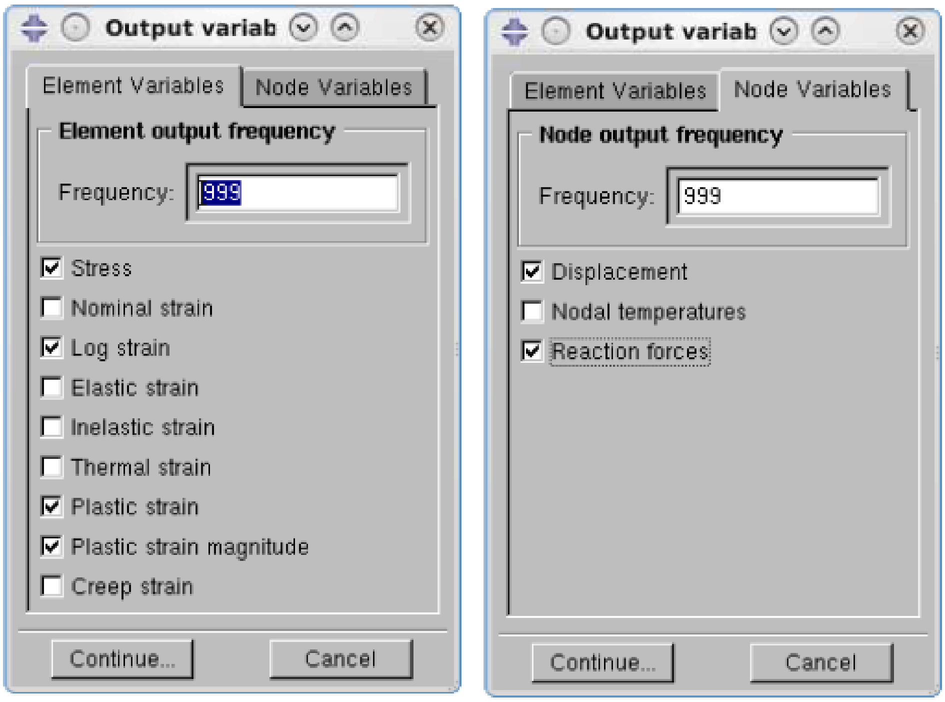


Figure 38 Output Variables – Example

11. Generate Input Files

The eleventh and final dialogue box the plug-in shows is the "Generate Input Files" dialogue box. In this the analyst needs to select the folder in which the Abaqus input files (*.inp) need to be generated. Step-by step process of selecting the folder is shown below.

i. Click on "Select directory for input files" button.
ii. Select the folder in which the Abaqus input files needs to be generated. Click "OK".
iii. Save the Abaqus CAE file.
iv. In the "Generate Input Files" dialogue box click on "Apply".

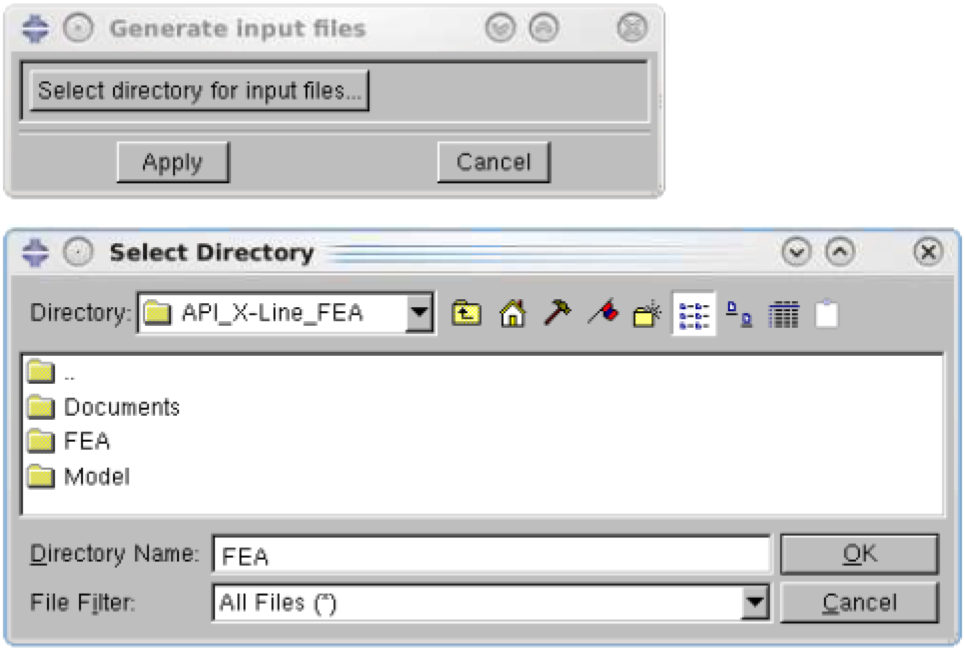


Figure 39 Generate Input Files - Example

After clicking on "Apply" button, input file generation progress bar will appear in the Abaqus CAE prompt area (Figure 40).

Figure 40 Input file generation progress bar – Example

The above 11 steps complete the Analysis Model generation process. This will generate Abaqus input files in the folder selected in step 11.

For the connection which has more than one seal, the analyst shall generate a set of input files for "**Standard analysis**" and another set of input files for "**Bleed through**" analysis, by making necessary modifications in steps 5, 6 and 8.

Preparing the Abaqus input files for Solver

“mmxmn.inp” amongst the input files corresponds to connection make-up simulation. The rest of the input files correspond to different load schedules (combination of tension, compression, external pressure and internal pressure loads) the connection is subjected to after make-up.

The plug-in assumes the collapse rating of the connection to be API Collapse. For high-collapse connections the external pressure shall be re-calculated using the collapse rating defined by the connection owner and equations described in Appendix III of ExxonMobil Connection Evaluation Program. The analyst shall modify the external pressure in the following load schedules:

1. c25epmxmn
2. c50epmxmn
3. c75epmxmn
4. c90epmxmn
5. cepmxmn
6. epmxmn
7. t25epmxmn
8. t50epmxmn
9. t75epmxmn
10. tepmxmn
11. vmcepmxmn

The analyst shall verify the material properties generated in the “**mmxmn**” input file and loads generated in the load schedule input files.

UIS recommends following material properties for API grades:

1. Young’s Modulus : $30 \times 10^6 psi$
2. Poisson’s ratio : 0.30
3. Plastic behavior : Bi-linear Stress – Strain curve (see Figure 41)

The points on the bi-linear curve should be calculated using specified minimum yield stress (SMYS) and specified minimum tensile strength (SMTS) tabulated in Table E.5 of API 5CT.

For proprietary grades, the specified minimum yield stress (SMYS) and specified minimum tensile strength (SMTS) shall be as defined by the connection owner. The elastic behavior should be only up to SMYS and strain at SMTS should be 2%.

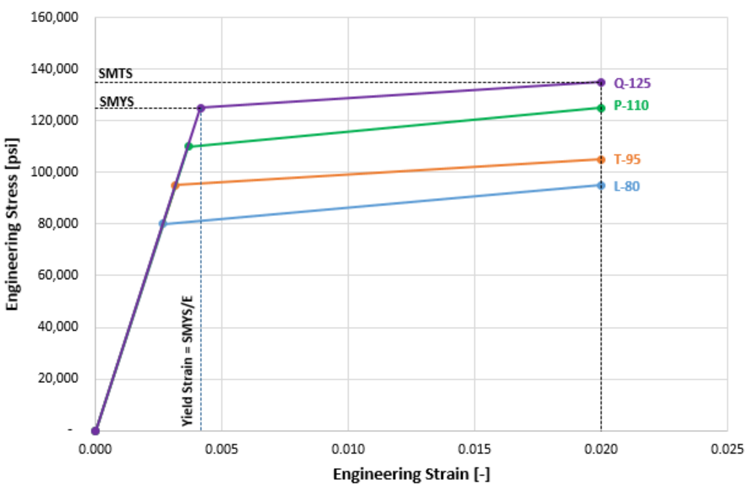


Figure 41 Engineering Stress-Strain curve for API grades

The engineering stress-strain data shall be converted to true stress - effective plastic strain data to use in Abaqus. The format in which the material properties should be defined is shown in Figure 42 below:

```
***
*MATERIAL, NAME=P110
*ELASTIC
30e6,0.3
*PLASTIC
110403.3,0.0000000
127500.0,0.0161427
***
```

Figure 42 Material Properties - Example

After verifying the material properties and loads, the analyst may run the Abaqus simulation. The first run shall be "**mmxmn**" which simulates make-up. Once the make-up simulation successfully completed the analyst should run the load schedule input files. The analyst can use a windows batch file to run the input files one after another. Figure 43 shows an example of a batch file.

Connector_FEA.bat

```
call abaqus cpus=24     job=mmxmn
call abaqus cpus=24     job=apiseramxmn     oldjob=mmxmn
call abaqus cpus=24     job=apiserbmxmn     oldjob=mmxmn
call abaqus cpus=24     job=c25epmxmn       oldjob=mmxmn
call abaqus cpus=24     job=c25pmxmn        oldjob=mmxmn
call abaqus cpus=24     job=c50epmxmn       oldjob=mmxmn
call abaqus cpus=24     job=c50pmxmn        oldjob=mmxmn
call abaqus cpus=24     job=c75epmxmn       oldjob=mmxmn
call abaqus cpus=24     job=c75pmxmn        oldjob=mmxmn
```

Figure 43 Connector FEA Batch file – Example

Appendix III

FEA Load Schedule Calculation

This section describes the FEA load schedule calculation. To calculate the loads the following connection properties are used:

D	Tubular outside diameter (tubular size), inches
Y_S	Specified Minimum yield strength, psi (e.g., for L-80, $Y_S = 80{,}000\ psi$)
t	Wall thickness (nominal), in
η_c	Compression efficiency of the connection ($\eta_c \leq 1$)
η_t	Tension efficiency of the connection ($\eta_t \leq 1$)

The rest of the connection properties are derived from the above properties. For connection evaluation with physical testing, the compression and tension efficiencies shall be the same as the efficiencies used to define the physical testing Connection Evaluation Envelope (CEE). For "FEA Only", the efficiencies shall be as per the Connection Data Sheet.

Load Schedules

See Formulas for Load Schedule Development section for explanation of symbols and functions. Both axial and pressure loads shall be calculated in "**psi**".

Positive axial load indicate tension; negative axial load indicates compression.

For all the load schedules, "**T**" is the axial load and "**P**" is the pressure load at the given step.

c25epmxmn

Description : **25% Compression Rating with External Pressure**

Number of load steps : **7**

Table 8 Load Schedule - c25epmxmn

Load Step	Axial Load	External Pressure
1	$0.25 \cdot compr$	$0.50 \cdot clps$
2	$0.25 \cdot compr$	$0.60 \cdot clps$
3	$0.25 \cdot compr$	$0.70 \cdot clps$
4	$0.25 \cdot compr$	$0.80 \cdot clps$
5	$0.25 \cdot compr$	$0.90 \cdot clps$
6	$0.25 \cdot compr$	$0.95 \cdot clps$
7	$0.25 \cdot compr$	$1.00 \cdot clps$

c25pmxmn

Description : **25% Compression Rating with Internal Pressure**

Number of load steps : **12**

Table 9 Load Schedule - c25pmxmn

Load Step	Axial Load	Internal Pressure
1	$0.25 \cdot compr$	$0.50 \cdot burst$
2	$0.25 \cdot compr$	$0.60 \cdot burst$
3	$0.25 \cdot compr$	$0.70 \cdot burst$
4	$0.25 \cdot compr$	$0.80 \cdot burst$
5	$0.25 \cdot compr$	$0.85 \cdot burst$
6	$0.25 \cdot compr$	$0.90 \cdot burst$
7	$0.25 \cdot compr$	$0.95 \cdot burst$
8	$0.25 \cdot compr$	$1.00 \cdot burst$
9	$0.25 \cdot compr$	$1.05 \cdot burst$
10	$0.25 \cdot compr$	$1.10 \cdot burst$
11	$0.25 \cdot compr$	$1.15 \cdot burst$
12	$0.25 \cdot compr$	$1.20 \cdot burst$

c50epmxmn

Description : **50% Compression Rating with External Pressure**

Number of load steps : **7**

Table 10 Load Schedule – c50epmxmn

Load Step	Axial Load	External Pressure
1	$0.50 \cdot compr$	$0.25 \cdot clps$
2	$0.50 \cdot compr$	$0.60 \cdot clps$
3	$0.50 \cdot compr$	$0.70 \cdot clps$
4	$0.50 \cdot compr$	$0.80 \cdot clps$
5	$0.50 \cdot compr$	$0.90 \cdot clps$
6	$0.50 \cdot compr$	$0.95 \cdot clps$
7	$0.50 \cdot compr$	$1.00 \cdot clps$

c50pmxmn

Description : **50% Compression Rating with Internal Pressure**

Number of load steps : **9**

Table 11 Load Schedule – c50pmxmn

Load Step	Axial Load	Internal Pressure
1	$0.50 \cdot compr$	$0.25 \cdot burst$
2	$0.50 \cdot compr$	$0.35 \cdot burst$
3	$0.50 \cdot compr$	$0.45 \cdot burst$
4	$0.50 \cdot compr$	$0.55 \cdot burst$
5	$0.50 \cdot compr$	$0.65 \cdot burst$
6	$0.50 \cdot compr$	$0.75 \cdot burst$
7	$0.50 \cdot compr$	$0.80 \cdot burst$
8	$0.50 \cdot compr$	$0.85 \cdot burst$
9	$0.50 \cdot compr$	$0.90 \cdot burst$

c75epmxmn

Description : **75% Compression Rating with External Pressure**

Number of load steps : **8**

Table 12 Load Schedule – c75epmxmn

Load Step	Axial Load	External Pressure
1	$0.75 \cdot compr$	$0.25 \cdot clps$
2	$0.75 \cdot compr$	$0.50 \cdot clps$
3	$0.75 \cdot compr$	$0.60 \cdot clps$
4	$0.75 \cdot compr$	$0.70 \cdot clps$
5	$0.75 \cdot compr$	$0.80 \cdot clps$
6	$0.75 \cdot compr$	$0.90 \cdot clps$
7	$0.75 \cdot compr$	$0.95 \cdot clps$
8	$0.75 \cdot compr$	$1.00 \cdot clps$

c75pmxmn

Description : **75% Compression Rating with Internal Pressure**

Number of load steps : **6**

Table 13 Load Schedule – c75pmxmn

Load Step	Axial Load	Internal Pressure
1	$0.75 \cdot compr$	$0.10 \cdot burst$
2	$0.75 \cdot compr$	$0.20 \cdot burst$
3	$0.75 \cdot compr$	$0.25 \cdot burst$
4	$0.75 \cdot compr$	$0.30 \cdot burst$
5	$0.75 \cdot compr$	$0.40 \cdot burst$
6	$0.75 \cdot compr$	$0.50 \cdot burst$

c90epmxmn

Description : **90% Compression Rating with External Pressure**

Number of load steps : **7**

Table 14 Load Schedule – c90epmxmn

Load Step	Axial Load	External Pressure
1	$0.90 \cdot compr$	$0.25 \cdot clps$
2	$0.90 \cdot compr$	$0.50 \cdot clps$
3	$0.90 \cdot compr$	$0.60 \cdot clps$
4	$0.90 \cdot compr$	$0.70 \cdot clps$
5	$0.90 \cdot compr$	$0.80 \cdot clps$
6	$0.90 \cdot compr$	$0.95 \cdot clps$
7	$0.90 \cdot compr$	$1.00 \cdot clps$

c90pmxmn

Description : **90% Compression Rating with Internal Pressure**

Number of load steps : **6**

Table 15 Load Schedule – c90pmxmn

Load Step	Axial Load	Internal Pressure
1	$0.90 \cdot compr$	$0.05 \cdot burst$
2	$0.90 \cdot compr$	$0.10 \cdot burst$
3	$0.90 \cdot compr$	$0.15 \cdot burst$
4	$0.90 \cdot compr$	$0.20 \cdot burst$
5	$0.90 \cdot compr$	$0.25 \cdot burst$
6	$0.90 \cdot compr$	$0.30 \cdot burst$

cepmxmn

Description : **Compression with External Pressure**

Number of load steps : **12**

Table 16 Load Schedule – cepmxmn

Load Step	Axial Load	External Pressure
1	$0.25 \cdot compr$	$0.25 \cdot clps$
2	$0.50 \cdot compr$	$0.50 \cdot clps$
3	$0.55 \cdot compr$	$0.55 \cdot clps$
4	$0.60 \cdot compr$	$0.60 \cdot clps$
5	$0.65 \cdot compr$	$0.65 \cdot clps$
6	$0.70 \cdot compr$	$0.70 \cdot clps$
7	$0.75 \cdot compr$	$0.75 \cdot clps$
8	$0.80 \cdot compr$	$0.80 \cdot clps$
9	$0.85 \cdot compr$	$0.85 \cdot clps$
10	$0.90 \cdot compr$	$0.90 \cdot clps$
11	$0.95 \cdot compr$	$0.95 \cdot clps$
12	$1.00 \cdot compr$	$1.00 \cdot clps$

cmxmn

Description : **Compression Only**

Number of load steps : **12**

Table 17 Load Schedule – cmxmn

Load Step	Axial Load	Internal Pressure
1	$0.25 \cdot compr$	0.00
2	$0.50 \cdot compr$	0.00
3	$0.55 \cdot compr$	0.00
4	$0.60 \cdot compr$	0.00
5	$0.65 \cdot compr$	0.00
6	$0.70 \cdot compr$	0.00
7	$0.75 \cdot compr$	0.00
8	$0.80 \cdot compr$	0.00
9	$0.85 \cdot compr$	0.00
10	$0.90 \cdot compr$	0.00
11	$0.95 \cdot compr$	0.00
12	$1.00 \cdot compr$	0.00

cpmxmn

Description : **Compression with Internal Pressure**

Number of load steps : **12**

Table 18 Load Schedule – cpmxmn

Load Step	Axial Load	Internal Pressure
1	$0.25 \cdot compr$	$0.25 \cdot burst$
2	$0.30 \cdot compr$	$0.30 \cdot burst$
3	$0.35 \cdot compr$	$0.35 \cdot burst$
4	$0.40 \cdot compr$	$0.40 \cdot burst$
5	$0.45 \cdot compr$	$0.45 \cdot burst$
6	$0.50 \cdot compr$	$0.50 \cdot burst$
7	$0.55 \cdot compr$	$0.55 \cdot burst$
8	$0.60 \cdot compr$	$0.60 \cdot burst$
9	$0.65 \cdot compr$	$0.65 \cdot burst$
10	$0.70 \cdot compr$	$0.70 \cdot burst$
11	$0.75 \cdot compr$	$0.75 \cdot burst$
12	$0.80 \cdot compr$	$0.80 \cdot burst$

epmxmn

Description : **External Pressure Only**

Number of load steps : **8**

Table 19 Load Schedule – epmxmn

Load Step	Axial Load	External Pressure
1	0.00	$0.25 \cdot clps$
2	0.00	$0.50 \cdot clps$
3	0.00	$0.60 \cdot clps$
4	0.00	$0.70 \cdot clps$
5	0.00	$0.80 \cdot clps$
6	0.00	$0.90 \cdot clps$
7	0.00	$0.95 \cdot clps$
8	0.00	$1.00 \cdot clps$

mtrpmxmn

Description : **100% Tension Rating with Internal Pressure**

Number of load steps : **16**

Table 20 Load Schedule – mtrpmxmn

Load Step	Axial Load	Internal Pressure
1	$tensr$	0.00
2	$tensr$	$0.25 \cdot P_I(T,1,+)$
3	$tensr$	$0.50 \cdot P_I(T,1,+)$
4	$tensr$	$0.60 \cdot P_I(T,1,+)$
5	$tensr$	$0.65 \cdot P_I(T,1,+)$
6	$tensr$	$0.70 \cdot P_I(T,1,+)$
7	$tensr$	$0.75 \cdot P_I(T,1,+)$
8	$tensr$	$0.80 \cdot P_I(T,1,+)$
9	$tensr$	$0.85 \cdot P_I(T,1,+)$
10	$tensr$	$0.90 \cdot P_I(T,1,+)$
11	$tensr$	$0.95 \cdot P_I(T,1,+)$
12	$tensr$	$1.00 \cdot P_I(T,1,+)$
13	$tensr$	$P_I(T,1,+) + 0.25 \cdot \left(P_R(T,+) - P_I(T,1,+)\right)$
14	$tensr$	$P_I(T,1,+) + 0.50 \cdot \left(P_R(T,+) - P_I(T,1,+)\right)$
15	$tensr$	$P_I(T,1,+) + 0.75 \cdot \left(P_R(T\ ,+) - P_I(T,1,+)\right)$
16	$tensr$	$P_R(T,+)$

ruppmxmn

Description : **Tension with Internal Pressure on Rupture**

Number of load steps : **5**

Table 21 Load Schedule – ruppmxmn

Load Step	Axial Load	Internal Pressure
1	$T_R(P,+)$	$0.50 \cdot burst$
2	$T_R(P,+)$	$0.625 \cdot burst$
3	$T_R(P,+)$	$0.75 \cdot burst$
4	$T_R(P,+)$	$0.875 \cdot burst$
5	$T_R(P,+)$	$1.00 \cdot burst$

pmxmn

Description : **Internal Pressure Only**

Number of load steps : **14**

Table 22 Load Schedule – pmxmn

Load Step	Axial Load	Internal Pressure
1	0.00	$0.25 \cdot burst$
2	0.00	$0.50 \cdot burst$
3	0.00	$0.60 \cdot burst$
4	0.00	$0.70 \cdot burst$
5	0.00	$0.75 \cdot burst$
6	0.00	$0.80 \cdot burst$
7	0.00	$0.85 \cdot burst$
8	0.00	$0.90 \cdot burst$
9	0.00	$0.95 \cdot burst$
10	0.00	$1.00 \cdot burst$
11	0.00	$burst + 0.25 \cdot (P_R(0.0,+) - burst)$
12	0.00	$burst + 0.50 \cdot (P_R(0.0,+) - burst)$
13	0.00	$burst + 0.75 \cdot (P_R(0.0,+) - burst)$
14	0.00	$P_R(0.0,+)$

t25epmxmn

Description : **25% Pipebody Tension Rating + External Pressure**

Number of load steps : **7**

Table 23 Load Schedule – t25epmxmn

Load Step	Axial Load	External Pressure
1	$0.25 \cdot tensr$	$0.50 \cdot clps$
2	$0.25 \cdot tensr$	$0.60 \cdot clps$
3	$0.25 \cdot tensr$	$0.70 \cdot clps$
4	$0.25 \cdot tensr$	$0.80 \cdot clps$
5	$0.25 \cdot tensr$	$0.90 \cdot clps$
6	$0.25 \cdot tensr$	$0.95 \cdot clps$
7	$0.25 \cdot tensr$	$1.00 \cdot clps$

t25pmxmn

Description : **25% Pipebody Tension Rating + Internal Pressure**

Number of load steps : **14**

Table 24 Load Schedule – t25pmxmn

Load Step	Axial Load	Internal Pressure
1	$0.25 \cdot tensr$	$0.50 \cdot burst$
2	$0.25 \cdot tensr$	$0.60 \cdot burst$
3	$0.25 \cdot tensr$	$0.70 \cdot burst$
4	$0.25 \cdot tensr$	$0.75 \cdot burst$
5	$0.25 \cdot tensr$	$0.80 \cdot burst$
6	$0.25 \cdot tensr$	$0.85 \cdot burst$
7	$0.25 \cdot tensr$	$0.90 \cdot burst$
8	$0.25 \cdot tensr$	$0.95 \cdot burst$
9	$0.25 \cdot tensr$	$0.95 \cdot P_I(T,1,+)$
10	$0.25 \cdot tensr$	$P_I(T,1,+)$
11	$0.25 \cdot tensr$	$P_I(T,1,+) + 0.25 \cdot \big(P_R(T,+) - P_I(T,1,+)\big)$
12	$0.25 \cdot tensr$	$P_I(T,1,+) + 0.50 \cdot \big(P_R(T,+) - P_I(T,1,+)\big)$
13	$0.25 \cdot tensr$	$P_I(T,1,+) + 0.75 \cdot \big(P_R(T,+) - P_I(T,1,+)\big)$
14	$0.25 \cdot tensr$	$P_R(T,+)$

t50epmxmn

Description : **50% Pipebody Tension Rating + External Pressure**

Number of load steps : **7**

Table 25 Load Schedule – t50epmxmn

Load Step	Axial Load	External Pressure
1	$0.50 \cdot tensr$	$0.25 \cdot clps$
2	$0.50 \cdot tensr$	$0.30 \cdot clps$
3	$0.50 \cdot tensr$	$0.40 \cdot clps$
4	$0.50 \cdot tensr$	$0.50 \cdot clps$
5	$0.50 \cdot tensr$	$0.60 \cdot clps$
6	$0.50 \cdot tensr$	$0.70 \cdot clps$
7	$0.50 \cdot tensr$	$0.80 \cdot clps$

t50pmxmn

Description : **50% Pipebody Tension Rating + Internal Pressure**

Number of load steps : **14**

Table 26 Load Schedule – t50pmxmn

Load Step	Axial Load	Internal Pressure
1	$0.50 \cdot tensr$	$0.25 \cdot burst$
2	$0.50 \cdot tensr$	$0.60 \cdot burst$
3	$0.50 \cdot tensr$	$0.70 \cdot burst$
4	$0.50 \cdot tensr$	$0.75 \cdot burst$
5	$0.50 \cdot tensr$	$0.80 \cdot burst$
6	$0.50 \cdot tensr$	$0.85 \cdot burst$
7	$0.50 \cdot tensr$	$0.90 \cdot burst$
8	$0.50 \cdot tensr$	$0.95 \cdot burst$
9	$0.50 \cdot tensr$	$0.95 \cdot P_I(T, 1, +)$
10	$0.50 \cdot tensr$	$P_I(T, 1, +)$
11	$0.50 \cdot tensr$	$P_I(T, 1, +) + 0.25 \cdot \left(P_R(T, +) - P_I(T, 1, +)\right)$
12	$0.50 \cdot tensr$	$P_I(T, 1, +) + 0.50 \cdot \left(P_R(T, +) - P_I(T, 1, +)\right)$
13	$0.50 \cdot tensr$	$P_I(T, 1, +) + 0.75 \cdot \left(P_R(T, +) - P_I(T, 1, +)\right)$
14	$0.50 \cdot tensr$	$P_R(T, +)$

t75epmxmn

Description : 7**5% Pipebody Tension Rating + External Pressure**

Number of load steps : **6**

Table 27 Load Schedule – t75epmxmn

Load Step	Axial Load	External Pressure
1	$0.75 \cdot tensr$	$0.30 \cdot clps$
2	$0.75 \cdot tensr$	$0.40 \cdot clps$
3	$0.75 \cdot tensr$	$0.50 \cdot clps$
4	$0.75 \cdot tensr$	$0.60 \cdot clps$
5	$0.75 \cdot tensr$	$0.70 \cdot clps$
6	$0.75 \cdot tensr$	$0.80 \cdot clps$

t75pmxmn

Description : **75% Pipebody Tension Rating + Internal Pressure**

Number of load steps : **14**

Table 28 Load Schedule – t75pmxmn

Load Step	Axial Load	Internal Pressure
1	$0.75 \cdot tensr$	$0.25 \cdot burst$
2	$0.75 \cdot tensr$	$0.60 \cdot burst$
3	$0.75 \cdot tensr$	$0.70 \cdot burst$
4	$0.75 \cdot tensr$	$0.75 \cdot burst$
5	$0.75 \cdot tensr$	$0.80 \cdot burst$
6	$0.75 \cdot tensr$	$0.85 \cdot burst$
7	$0.75 \cdot tensr$	$0.90 \cdot burst$
8	$0.75 \cdot tensr$	$0.95 \cdot burst$
9	$0.75 \cdot tensr$	$0.95 \cdot P_I(T, 1, +)$
10	$0.75 \cdot tensr$	$P_I(T, 1, +)$
11	$0.75 \cdot tensr$	$P_I(T, 1, +) + 0.25 \cdot \left(P_R(T, +) - P_I(T, 1, +)\right)$
12	$0.75 \cdot tensr$	$P_I(T, 1, +) + 0.50 \cdot \left(P_R(T, +) - P_I(T, 1, +)\right)$
13	$0.75 \cdot tensr$	$P_I(T, 1, +) + 0.75 \cdot \left(P_R(T, +) - P_I(T, 1, +)\right)$
14	$0.75 \cdot tensr$	$P_R(T, +)$

t100pmxmn

Description : **100% Pipebody Tension Rating + Internal Pressure**

Number of load steps : **10**

Table 29 Load Schedule – t100pmxmn

Load Step	Axial Load	Internal Pressure
1	$tensr$	$0.25 \cdot burst$
2	$tensr$	$0.50 \cdot burst$
3	$tensr$	$0.60 \cdot burst$
4	$tensr$	$0.70 \cdot burst$
5	$tensr$	$0.80 \cdot burst$
6	$tensr$	$0.85 \cdot burst$
7	$tensr$	$0.90 \cdot burst$
8	$tensr$	$0.95 \cdot burst$
9	$tensr$	$1.00 \cdot burst$
10	$tensr$	$P_R(T, +)$

tepmxmn

Description : **Tension + External Pressure**

Number of load steps : **12**

Table 30 Load Schedule – tepmxmn

Load Step	Axial Load	External Pressure
1	$0.25 \cdot tensr$	$0.25 \cdot clps$
2	$0.30 \cdot tensr$	$0.30 \cdot clps$
3	$0.35 \cdot tensr$	$0.35 \cdot clps$
4	$0.40 \cdot tensr$	$0.40 \cdot clps$
5	$0.45 \cdot tensr$	$0.45 \cdot clps$
6	$0.50 \cdot tensr$	$0.50 \cdot clps$
7	$0.55 \cdot tensr$	$0.55 \cdot clps$
8	$0.60 \cdot tensr$	$0.60 \cdot clps$
9	$0.65 \cdot tensr$	$0.65 \cdot clps$
10	$0.70 \cdot tensr$	$0.70 \cdot clps$
11	$0.75 \cdot tensr$	$0.75 \cdot clps$
12	$0.80 \cdot tensr$	$0.80 \cdot clps$

tmxmn

Description : **Tension Only**

Number of load steps : **12**

Table 31 Load Schedule – tmxmn

Load Step	Axial Load	Internal Pressure
1	$0.25 \cdot tensr$	0.00
2	$0.50 \cdot tensr$	0.00
3	$0.55 \cdot tensr$	0.00
4	$0.60 \cdot tensr$	0.00
5	$0.65 \cdot tensr$	0.00
6	$0.70 \cdot tensr$	0.00
7	$0.75 \cdot tensr$	0.00
8	$0.80 \cdot tensr$	0.00
9	$0.85 \cdot tensr$	0.00
10	$0.90 \cdot tensr$	0.00
11	$0.95 \cdot tensr$	0.00
12	$1.00 \cdot tensr$	0.00

tpmxmn

Description : **Tension + Internal Pressure**

Number of load steps : **12**

Table 32 Load Schedule – tpmxmn

Load Step	Axial Load	Internal Pressure
1	$0.25 \cdot tensr$	$0.25 \cdot burst$
2	$0.50 \cdot tensr$	$0.50 \cdot burst$
3	$0.55 \cdot tensr$	$0.55 \cdot burst$
4	$0.60 \cdot tensr$	$0.60 \cdot burst$
5	$0.65 \cdot tensr$	$0.65 \cdot burst$
6	$0.70 \cdot tensr$	$0.70 \cdot burst$
7	$0.75 \cdot tensr$	$0.75 \cdot burst$
8	$0.80 \cdot tensr$	$0.80 \cdot burst$
9	$0.85 \cdot tensr$	$0.85 \cdot burst$
10	$0.90 \cdot tensr$	$0.90 \cdot burst$
11	$0.95 \cdot tensr$	$0.95 \cdot burst$
12	$1.00 \cdot tensr$	$1.00 \cdot burst$

vmcepmxmn

Description : **Compression + External Pressure on VonMises curve**

Number of load steps : **2**

Table 33 Load Schedule – vmcepmxmn

Load Step	Axial Load	External Pressure
1	$0.75 \cdot compr$	$1.00 \cdot clps$
2	$1.00 \cdot compr$	$0.75 \cdot clps$

vmcpmxmn

Description : **Compression + Internal Pressure on VonMises curve**

Number of load steps : **2**

Table 34 Load Schedule – vmcpmxmn

Load Step	Axial Load	Internal Pressure
1	$T_{II}(P, 1)$	$0.50 \cdot burst$
2	$T_{II}(P, 1)$	$0.75 \cdot burst$

vmepmxmn

Description : **Tension + External Pressure on VonMises curve**

Number of load steps : **2**

Table 35 Load Schedule – vmepmxmn

Load Step	Axial Load	External Pressure
1	$T_{IV}(-P, 1)$	$0.50 \cdot clps$
2	$T_{IV}(-P, 1)$	$0.75 \cdot clps$

vmpmxmn

Description : **Tension + Internal Pressure on VonMises curve**

Number of load steps : **3**

Table 36 Load Schedule – vmpmxmn

Load Step	Axial Load	Internal Pressure
1	$T_I(P, 1, +)$	$0.25 \cdot burst$
2	$T_I(P, 1, +)$	$0.50 \cdot burst$
3	$T_I(P, 1, +)$	$0.75 \cdot burst$

apiseramxmn

Description : **Axial Load + Internal Pressure as per API 5C5 Series-A**

Number of load steps : **11**

Table 37 Load Schedule – apiseramxmn

Load Step	Axial Load	Internal Pressure
1	$0.90 \cdot tensr$	0.00
2	$0.90 \cdot tensr$	$0.50 \cdot P_I(T, 0.95, +)$
3	$0.90 \cdot tensr$	$P_I(T, 0.95, +)$
4	$0.80 \cdot tensr$	$P_I(T, 0.95, +)$
5	$ECT(P)$	$0.95 \cdot P_{15a}$
6	0.00	$0.95 \cdot burst$
7	$0.25 \cdot compr$	$P_{II}(T, 0.95)$
8	$0.50 \cdot compr$	$P_{II}(T, 0.95)$
9	$0.75 \cdot compr$	$P_{II}(T, 0.95)$
10	$0.90 \cdot compr$	$P_{II}(T, 0.95)$
11	$0.90 \cdot compr$	0.00

apiserbmxmn

Description : **Axial Load + Internal Pressure as per API 5C5 Series-B**

Number of load steps : **12**

Table 38 Load Schedule – apiserbmxmn

Load Step	Axial Load	Internal Pressure
1	$0.90 \cdot tensr$	0.00
2	$0.90 \cdot tensr$	$0.25 \cdot P_I(T, 0.95, +)$
3	$0.90 \cdot tensr$	$0.50 \cdot P_I(T, 0.95, +)$
4	$0.90 \cdot tensr$	$P_I(T, 0.95, +)$
5	$0.80 \cdot tensr$	$P_I(T, 0.95, +)$
6	$ECT(P)$	$0.95 \cdot P_{15a}$
7	0.00	$0.95 \cdot burst$
8	$0.25 \cdot compr$	$P_{II}(T, 0.95)$
9	$0.50 \cdot compr$	$P_{II}(T, 0.95)$
10	$0.75 \cdot compr$	$P_{II}(T, 0.95)$
11	$0.90 \cdot compr$	$P_{II}(T, 0.95)$
12	$0.90 \cdot compr$	0.00

sp3mxmn

Description : **Axial Load + Internal Pressure as per EMCEP 1st Ed. SP3**

Number of load steps : **33**

Table 39 Load Schedule – sp3mxmn

Load Step	Axial Load	Internal Pressure
1	$0.25 \cdot tensr$	0.00
2	$0.50 \cdot tensr$	0.00
3	$0.67 \cdot tensr$	0.00
4	$0.67 \cdot tensr$	$0.25 \cdot burst$
5	$0.67 \cdot tensr$	$0.50 \cdot burst$
6	$0.67 \cdot tensr$	$0.80 \cdot burst$
7	$0.67 \cdot \left(tensr + T_{BIT}(5)\right)$	$0.80 \cdot burst$
8	$0.67 \cdot \left(tensr + T_{BIT}(10)\right)$	$0.80 \cdot burst$
9	$ECT(P)$	$0.80 \cdot burst$
10	$0.50 \cdot compr$	0.00
11	$0.50 \cdot compr$	$0.25 \cdot burst$
12	$0.50 \cdot compr$	$0.50 \cdot burst$
13	$0.50 \cdot compr$	$P_{II}(T, 1.0)$
14	$0.50 \cdot compr$	$P_{II}(T, 1.0)$
15	$0.50 \cdot compr$	$P_{II}(T, 1.0)$
16	0.00	$0.80 \cdot burst$
17	$0.67 \cdot tensr$	$0.80 \cdot burst$
18	$0.75 \cdot tensr$	$0.80 \cdot burst$
19	$0.75 \cdot \left(tensr + T_{BIT}(5)\right)$	$0.80 \cdot burst$
20	$0.75 \cdot \left(tensr + T_{BIT}(10)\right)$	$0.80 \cdot burst$
21	$T_I(P, 0.80, +)$	0.00
22	$T_I(P, 0.80, +)$	$0.30 \cdot burst$
23	$T_I(P, 0.80, +)$	$0.50 \cdot burst$
24	$T_I(P, 0.80, +)$	$0.70 \cdot burst$
25	$T_I(P, 0.80, +)$	$0.80 \cdot burst$
26	$0.67 \cdot tensr$	$0.80 \cdot burst$
27	$0.90 \cdot tensr$	$0.80 \cdot burst$
28	$0.75 \cdot tensr$	$0.90 \cdot burst$
29	$0.75 \cdot tensr$	$0.95 \cdot burst$
30	$0.75 \cdot tensr$	$1.00 \cdot burst$
31	0.00	$1.00 \cdot burst$
32	$ECT(P)$	$1.00 \cdot burst$
33	0.00	0.00

sp5mxmn

Description : **Axial Load + Internal Pressure as per EMCEP 1st Ed. SP5**

Number of load steps : **46**

Table 40 Load Schedule – sp5mxmn

Load Step	Axial Load	Internal Pressure
1	$0.50 \cdot tensr$	0.00
2	$0.67 \cdot tensr$	$0.80 \cdot burst$
3	$0.67 \cdot \left(tensr + T_{BIT}(5)\right)$	$0.80 \cdot burst$
4	$0.67 \cdot \left(tensr + T_{BIT}(10)\right)$	$0.80 \cdot burst$
5	$1.00 \cdot tensr$	0.00
6	$tensr + \ 0.50 \cdot (T_B - tensr)$	$P_I(T, 1, -)$
7	T_B	$P_I(T, 1, +)$
8	$tensr + \ 0.50 \cdot (T_B - tensr)$	$P_I(T, 1, +)$
9	$tensr$	$P_I(T, 1, +)$
10	$T_C + \ 0.67 \cdot (tensr - T_C)$	$P_I(T, 1, +)$
11	$T_C + \ 0.33 \cdot (tensr - T_C)$	$P_I(T, 1, +)$
12	T_C	$P_I(T, 1, +)$
13	$ECT(P)$	$1.00 \cdot burst$
14	$ECT(P)$	$0.80 \cdot burst$
15	0.00	$0.80 \cdot burst$
16	$ECT(P)$	$0.80 \cdot burst$
17	$0.67 \cdot T_C$	$P_I(T, 1, +)$
18	$0.33 \cdot T_C$	$P_I(T, 1, +)$
19	0.00	$1.00 \cdot burst$
20	$0.17 \cdot compr$	$P_{II}(T, 1.0)$
21	$0.33 \cdot compr$	$P_{II}(T, 1.0)$
22	$0.50 \cdot compr$	$P_{II}(T, 1.0)$
23	$0.50 \cdot compr$	$0.50 \cdot P_{II}(T, 1.0)$
24	$0.50 \cdot compr$	0.00
25	0.00	0.00
26	$0.50 \cdot compr$	$P_{II}(T, 1.0)$
27	$0.50 \cdot compr$	$P_{II}(T, 1.0)$
28	0.00	$1.00 \cdot burst$
29	T_C	$P_I(T, 1, +)$
30	$1.00 \cdot \left(T_C + T_{BIT}(10)\right)$	$P_I(T, 1, +)$

continued on the next page

Table 41 Load Schedule – sp5mxmn (continued)

Load Step	Axial Load	Internal Pressure
Continued from the previous page		
31	$tensr$	$P_I(T,1,+)$
32	T_B	$P_I(T,1,+)$
33	$tensr$	0.00
34	$0.67 \cdot \left(tensr + T_{BIT}(10)\right)$	$0.80 \cdot burst$
35	0.00	0.00
36	T_C	$0.5 \cdot \left(P_I(T,1,+) + P_R(T,+)\right)$
37	$1.00 \cdot \left(T_C + T_{BIT}(10)\right)$	$0.5 \cdot \left(P_I(T,1,+) + P_R(T,+)\right)$
38	0.00	$0.5 \cdot \left(P_I(0.0,1,+) + P_R(0.0,+)\right)$
39	$T_{BIT}(5)$	$0.5 \cdot \left(P_I(0.0,1,+) + P_R(0.0,+)\right)$
40	$T_C + 0.33 \cdot (tensr - T_C)$	$0.5 \cdot \left(P_I(T,1,+) + P_R(T,+)\right)$
41	$T_C + 0.67 \cdot (tensr - T_C)$	$0.5 \cdot \left(P_I(T,1,+) + P_R(T,+)\right)$
42	$tensr$	$0.5 \cdot \left(P_I(T,1,+) + P_R(T,+)\right)$
43	$tensr + 0.50 \cdot (T_B - tensr)$	$0.5 \cdot \left(P_I(T,1,+) + P_R(T,+)\right)$
44	T_B	$P_I(T,1,+)$
45	$tensr$	0.00
46	0.00	0.00

Formulas for Load Schedule Development

Nomenclature

Symbols and Functions Used in Load Schedules

$tensr$	Tension rating of the connection in psi. Use equation (9) to compute.
$compr$	Compression rating of the connection in psi. Use equation (10) to compute.
$clps$	Collapse rating of the connection. For high-collapse connections, value of $\boldsymbol{clps}$ shall be the collapse rating defined by the connection owner. For other connections, use equation (11) to compute.
$burst$	Burst rating of the connection. For connections with hoop anisotropy, value of $\boldsymbol{burst}$ shall be the burst rating defined by the connection owner. For other connections, use equation (12) to compute.
P_{15a95}	Internal pressure at yield for a capped-end thick tube. Use equation (13) to compute.
$P_I(T,g,\pm)$	Pressure at a given axial load on a specified von Mises ellipse in quadrant I (tension with internal pressure). Use equation (16) to compute. The second argument (g) specifies the stress level as fraction of Y_S. The third argument $(+$ or $-)$ indicates the root to select.
$P_{II}(T,g)$	Pressure at a given axial load on a specified von Mises ellipse in quadrant II (compression with internal pressure). Use equation (19) to compute. The second argument (g) specifies the stress level as fraction of Y_S.
$P_R(T,\pm)$	Pressure at a given axial load on the rupture envelope in quadrant I (tension with internal pressure). Use equation (28) to compute. The second argument $(+$ or $-)$ indicates the root to select.
$P_{EXT}(T)$	External test pressure at a given axial load. The external pressure is the lesser (in absolute value) of (i) the tension-de-rated API collapse pressure and (ii) the pressure at given axial load on the von Mises ellipse at Y_S (i.e., $g=1$). To compute (i), see API Bulletin 5C3, *Bulletin on Formulas and Calculations for Casing, Tubing, Drill Pipe, and Line Pipe Properties*, Sixth Edition (October 1994), Section 2. To compute (ii), use equation (22) for quadrant III (compression with external pressure) or equation (25) for quadrant IV (tension with external pressure). API collapse pressure is not adjusted for compression.
$T_I(P,g,\pm)$	Axial load at a given pressure on a specified von Mises ellipse in quadrant I (tension with internal pressure). Use equation (15) to compute. The second argument (g) specifies the stress level as fraction of Y_S. The third argument $(+$ or $-)$ indicates the root to select.
$T_{II}(P,g)$	Axial load at a given pressure on a specified von Mises ellipse in quadrant II (compression with internal pressure). Use equation (18) to compute. The second argument (g) specifies the stress level as fraction of Y_S.
$T_{IV}(P,g)$	Axial load at a given pressure on a specified von Mises ellipse in quadrant IV (tension with external pressure). Use equation (15) to compute. The second argument (g) specifies the stress level as fraction of Y_S. The third argument $(+$ or $-)$ indicates the root to select.

$T_R(P, \pm)$	Axial Load at a given pressure on the rupture envelope in quadrant I (tension with internal pressure). Use equation (27) to compute. The second argument ($+$ or $-$) indicates the root to select.
$T_B,\ T_C$	Axial load at points B and C. See Figure 44. Use equations (29b) and (29c) to compute.
$T_{BIT}\left(\frac{\Delta\theta}{\Delta x}\right)$	Bending-induced tension. Use equation (30) to compute.
$ECT(P)$	End cap tension due to internal pressure (no applied machine load). Use equation (31) to compute.

Other Symbols

A_w	Wall cross sectional area (based on nominal wall thickness), in
D	Tubular outside diameter (tubular size), in
P	Pressure, psi
T	Tension, psi
Y_s	Minimum yield strength, lb/in^2 (e.g., for L-80, $Y_s = 80{,}000\ psi$)
d	Pipe inside diameter (based on minimum wall thickness), in
d_n	Pipe inside diameter (based on nominal wall thickness), in
f	Minimum allowable wall thickness as a fraction of nominal wall thickness (for API pipe, $f = 0.875$)
t	Wall thickness (nominal), in
β	Non-dimensional factor (see equation (4) for definition)
δ	Non-dimensional factor (see equation (5) for definition)
η	Non-dimensional factor (see equation (6) for definition)
γ	Non-dimensional factor (see equation (7) for definition)
κ	Non-dimensional factor (see equation (8) for definition)
$\frac{\Delta\theta}{\Delta x}$	Bending curvature, deg/100ft
η_c	Compression efficiency of the connection ($\eta_c \leq 1$)
η_t	Tension efficiency of the connection ($\eta_t \leq 1$)

Tubular Quantities

$$d_n = D - 2 \cdot t \tag{1}$$

$$d = D - 2 \cdot f \cdot t \text{ where } f = 0.875 \text{ for API pipe} \tag{2}$$

$$A_w = \frac{\pi}{4}\left(D^2 - {d_n}^2\right) \tag{3}$$

$$\beta \equiv \frac{D^2 + d^2}{D^2 - d^2} \tag{4}$$

$$\delta \equiv \frac{2D^2}{D^2 - d^2} \tag{5}$$

$$\eta \equiv \frac{d^2}{D^2 - {d_n}^2} \tag{6}$$

$$\gamma \equiv \frac{2}{\sqrt{3}} \ln\left(\frac{D}{d_n}\right) \tag{7}$$

$$\kappa \equiv \frac{\pi}{4} \cdot \frac{{d_n}^2}{A_w} \tag{8}$$

$$tensr = \eta_t \cdot Y_S \tag{9}$$

$$compr = -\eta_c \cdot Y_S \tag{10}$$

$$clps = P_{EXT}(0) \tag{11}$$

$$burst = 2 \cdot Y_S \cdot f \cdot \frac{t}{D} \tag{12}$$

$$P_{15a95} = Y_S \Bigg/ \sqrt{\frac{3 \cdot D^4 + d^4}{(D^2 - d^2)^2} + \frac{{d_n}^4}{\left((D^2 - {d_n}^2) \cdot \eta_t\right)^2} - 2 \cdot {d_n}^2 \cdot \frac{d^2}{(D^2 - {d_n}^2) \cdot \eta_t \cdot (D^2 - d^2)}} \tag{13}$$

Connection Von Mises Envelope

See Figure 44. The conventions are as follows:

- T is axial load in psi. Positive axial load indicate tension; negative axial load indicates compression.
- P is pressure in psi. Positive pressure indicates internal pressure; negative pressure indicates external pressure. If there are both internal pressure (P_i) and external pressure (P_o), $P \equiv P_i - P_o$.
- g specifies the stress level for the von Mises ellipse. It is expressed as a fraction of Y_s (e.g. $g = 0.8$).
- The equations do not change for reduced efficiency connections.

Quadrant I – Tension and Internal Pressure

The following equation defines a von Mises ellipse at a specified stress level:

$$\left(\frac{T}{\eta_t Y_s}\right)^2 - (\beta - 1)\left(\frac{P}{Y_s}\right)\left(\frac{T}{\eta_t Y_s}\right) + (\beta^2 + \beta + 1)\left(\frac{P}{Y_s}\right)^2 = g^2 \tag{14}$$

The following formula may be used to solve for tension when pressure is specified:

$$\frac{T}{\eta_t Y_s} = \frac{\beta - 1}{2}\left(\frac{P}{Y_s}\right) \pm \frac{1}{2}\sqrt{(\beta - 1)^2\left(\frac{P}{Y_s}\right)^2 - 4\left[(\beta^2 + \beta + 1)\left(\frac{P}{Y_s}\right)^2 - g^2\right]} \tag{15}$$

Positive root is used between point A and point C. Negative root is used between point C and point D.

The following formula may be used to solve for pressure when tension is specified:

$$\frac{P}{Y_s} = \frac{\beta - 1}{2(\beta^2 + \beta + 1)}\left(\frac{T}{\eta_t Y_s}\right) \pm \frac{1}{2(\beta^2 + \beta + 1)}\sqrt{(\beta - 1)^2\left(\frac{T}{\eta_t Y_s}\right)^2 - 4(\beta^2 + \beta + 1)\left[\left(\frac{T}{\eta_t Y_s}\right)^2 - g^2\right]} \tag{16}$$

Positive root is used between point D and point B. Negative root is used between point B and point A.

Quadrant II – Compression and Internal Pressure

The following equation defines a von Mises ellipse at a specified stress level:

$$\left(\frac{T}{\eta_c Y_s}\right)^2 - (\beta - 1)\left(\frac{P}{Y_s}\right)\left(\frac{T}{\eta_c Y_s}\right) + (\beta^2 + \beta + 1)\left(\frac{P}{Y_s}\right)^2 = g^2 \tag{17}$$

The following formula may be used to solve for tension when pressure is specified:

$$\frac{T}{\eta_c Y_s} = \frac{\beta-1}{2}\left(\frac{P}{Y_s}\right) - \frac{1}{2}\sqrt{(\beta - 1)^2\left(\frac{P}{Y_s}\right)^2 - 4\left[(\beta^2 + \beta + 1)\left(\frac{P}{Y_s}\right)^2 - g^2\right]} \tag{18}$$

The following formula may be used to solve for pressure when tension is specified:

$$\frac{P}{Y_s} = \frac{\beta-1}{2(\beta^2+\beta+1)}\left(\frac{T}{\eta_c Y_s}\right) + \frac{1}{2(\beta^2+\beta+1)}\sqrt{(\beta - 1)^2\left(\frac{T}{\eta_c Y_s}\right)^2 - 4(\beta^2 + \beta + 1)\left[\left(\frac{T}{\eta_c Y_s}\right)^2 - g^2\right]} \tag{19}$$

Quadrant III – Compression and External Pressure

The following equation defines a von Mises ellipse at a specified stress level:

$$\left(\frac{T}{\eta_c Y_s}\right)^2 - \delta\left(\frac{P}{Y_s}\right)\left(\frac{T}{\eta_c Y_s}\right) + \delta^2\left(\frac{P}{Y_s}\right)^2 = g^2 \tag{20}$$

The following formula may be used to solve for tension when pressure is specified:

$$\frac{T}{\eta_c Y_s} = \frac{\delta}{2}\left(\frac{P}{Y_s}\right) \pm \frac{1}{2}\sqrt{\delta^2\left(\frac{P}{Y_s}\right)^2 - 4\left[\delta^2\left(\frac{P}{Y_s}\right)^2 - g^2\right]} \tag{21}$$

Positive root is used between point H and point G. Negative root is used between point G and point E.

The following formula may be used to solve for pressure when tension is specified:

$$\frac{P}{Y_s} = \frac{1}{2\delta}\left(\frac{T}{\eta_c Y_s}\right) \pm \frac{1}{2\delta}\sqrt{\left(\frac{T}{\eta_c Y_s}\right)^2 - 4\left[\left(\frac{T}{\eta_c Y_s}\right)^2 - g^2\right]} \tag{22}$$

Positive root is used between point E and point F. Negative root is used between point F and point H.

Quadrant IV – Tension and External Pressure

The following equation defines a von Mises ellipse at a specified stress level:

$$\left(\frac{T}{\eta_t Y_s}\right)^2 - \delta\left(\frac{P}{Y_s}\right)\left(\frac{T}{\eta_t Y_s}\right) + \delta^2\left(\frac{P}{Y_s}\right)^2 = g^2 \tag{23}$$

The following formula may be used to solve for tension when pressure is specified:

$$\frac{T}{\eta_t Y_s} = \frac{\delta}{2}\left(\frac{P}{Y_s}\right) + \frac{1}{2}\sqrt{\delta^2\left(\frac{P}{Y_s}\right)^2 - 4\left[\delta^2\left(\frac{P}{Y_s}\right)^2 - g^2\right]} \tag{24}$$

The following formula may be used to solve for pressure when tension is specified:

$$\frac{P}{Y_s} = \frac{1}{2\delta}\left(\frac{T}{\eta_t Y_s}\right) - \frac{1}{2\delta}\sqrt{\left(\frac{T}{\eta_t Y_s}\right)^2 - 4\left[\left(\frac{T}{\eta_t Y_s}\right)^2 - g^2\right]} \tag{25}$$

Connection Rupture Envelope

The rupture envelope is given by the following equation in the first quadrant (tension with internal pressure):

$$\left(\kappa^2+\frac{1}{\gamma^2}\right)\left(\frac{P}{Y_s}\right)^2-2\kappa\left(\frac{T}{\eta_t Y_s}\right)\left(\frac{P}{Y_s}\right)+\left(\frac{T}{\eta_t Y_s}\right)^2=1 \qquad (26)$$

The following formula may be used to solve for tension when pressure is specified:

$$\frac{T}{\eta_t Y_s}=\kappa\left(\frac{P}{Y_s}\right)\pm\sqrt{\kappa^2\left(\frac{P}{Y_s}\right)^2+\left[1-\left(\kappa^2+\frac{1}{\gamma^2}\right)\left(\frac{P}{Y_s}\right)^2\right]} \qquad (27)$$

Positive root is used between point A and point C'. Negative root is used between point C' and point D'.

The following formula may be used to solve for pressure when tension is specified:

$$\frac{P}{Y_s}=\frac{\kappa\left(\frac{T}{\eta_t Y_s}\right)\pm\sqrt{\kappa^2\left(\frac{T}{\eta_t Y_s}\right)^2+\left(\kappa^2+\frac{1}{\gamma^2}\right)\left[1-\left(\frac{T}{\eta_t Y_s}\right)^2\right]}}{\left(\kappa^2+\frac{1}{\gamma^2}\right)} \qquad (28)$$

Positive root is used between point D' and point B'. Negative root is used between point B' and point A.

Axial load and Pressure load Calculations at Points A-H

The following formulas may be used to calculate the tension and pressure at the points marked on Figure 44. All are calculated on the yield envelope (e.g. $g = 1$).

Point	Tension	Pressure	
A	$=\eta_t Y_s$	$=0$	(29a)
B	$=\sqrt{\frac{4(\beta^2+\beta+1)}{3(\beta^2+2\beta+1)}}\,\eta_t Y_s$	$=\frac{\beta-1}{\sqrt{3(\beta^2+\beta+1)(\beta^2+2\beta+1)}}$	(29b)
C	$=\frac{\beta-1}{\sqrt{3(\beta^2+2\beta+1)}}\,\eta_t Y_s$	$=\sqrt{\frac{4}{3(\beta^2+2\beta+1)}}\,Y_s$	(29c)
D	$=0$	$=\frac{Y_S}{\sqrt{\beta^2+\beta+1}}$	(29d)
E	$=-\eta_c Y_s$	$=0$	(29e)
F	$=-\frac{2}{\sqrt{3}}\,\eta_c Y_s$	$=-\frac{1}{\delta\sqrt{3}}\,Y_s$	(29f)
G	$=-\frac{1}{\sqrt{3}}\,\eta_c Y_s$	$=-\frac{2}{\delta\sqrt{3}}\,Y_s$	(29g)
H	$=0$	$=-\frac{Y_S}{\delta}\,Y_s$	(29h)

Substitute $g \cdot Y_s$ for Y_s to calculate values at equivalent points on other von Mises ellipses (e.g., 80% yield envelope).

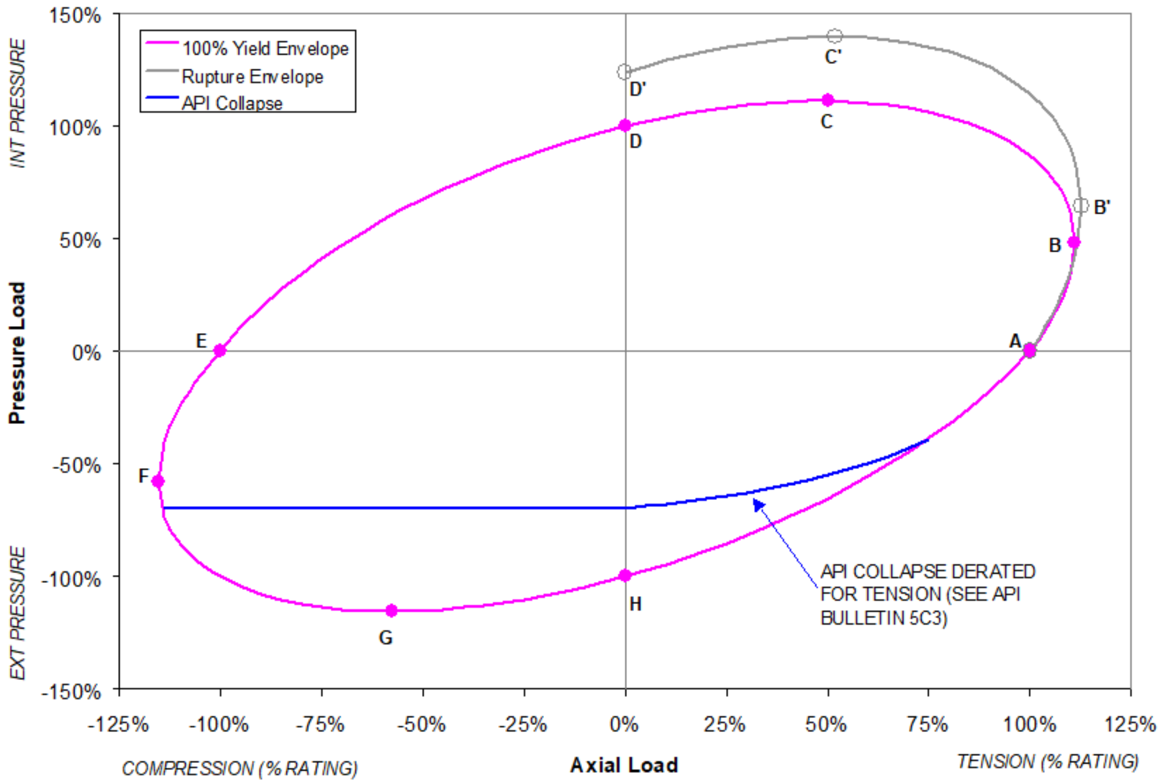


Figure 44 Symbols and Functions for Defining Connection Rupture and Connection Yield Envelopes

Bending-Induced Tension

T_{BIT} is the bending-induced tension and is calculated using the following:

$$T_{BIT} = 218.166 \cdot D \cdot \frac{\Delta\theta}{\Delta x} \tag{30}$$

End Cap Tension

ECT is the end cap tension due to internal pressure and is calculated using the following:

$$ECT = P \cdot \kappa \tag{31}$$

Appendix IV

ExxonMobil Post-Process Connector plug-in for Abaqus

This technical note describes the steps required to access and use the ExxonMobil Post-Process Connector plug-in for Abaqus.

The purpose of this plug-in is to extract results from Abaqus output files (*.odb). The output from the Post-Process Connector plug-in will be XOM files (*.xom), which will be used for further post-processing.

Steps to Access Post-Process Connector Plug-in

a) Post-Process Connector plug-in and supporting scripts are clustered together in a single zip file "**PostProcessConnector_plugin_XOM**". Extract the zip file to a folder

b) Place the folder in the Plug-in central directory (*E.g., for Abaqus 2017, C:\SIMULIA\CAE\plugins\2017*)

   or

   Change the plug-in central directory address to the Define Connector plug-in folder address in the custom Abaqus environment file by editing the "**plugin_central_dir= <folder address>"** line.

   Custom environment file (*.env) can be found in the following folder (*for 2017 version*):

   C:\Program Files\Dassault Systemes\SimulationServices\V6R2017x\win_b64\SMA\site

c) **Post-Process Connector** plug-in can be accessed from Abaqus Visualization Module:

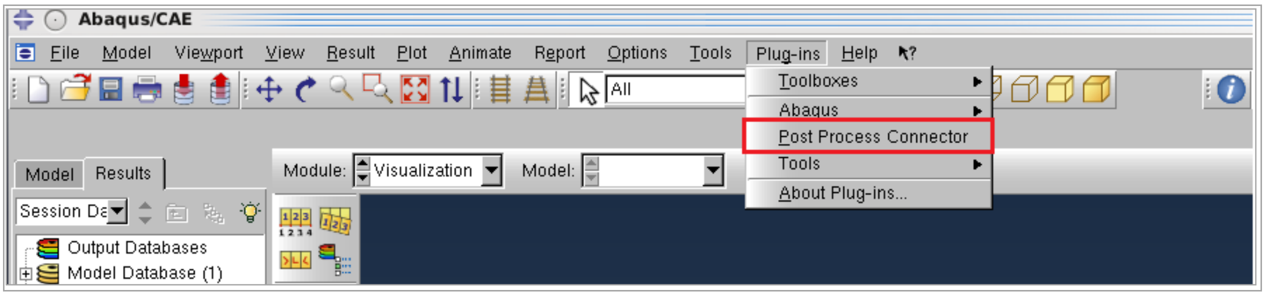


Figure 45 Post-Process connector plug-in

Post-Process Connector is a single dialogue box plug-in. Before accessing the plug-in the analyst needs to complete Reviewing FEA Run status and Preparing XOM Files for Result Extraction steps described below:

Reviewing FEA Run status

1. The analyst needs to generate Abaqus input files using **ExxonMobil Define Connector plug-in**
2. The analyst needs to complete FEA runs of all 35 input files generated in above step.
3. The analyst needs to make sure the following 5 FEA runs are fully converged:

   i. *SP3MXMN* *ii.* *SP5MXMN*

   iii. *APISERAMXMN* *iv.* *APISERBMXMN*

   v. *CTN12MXMN*

Preparing XOM Files for Result Extraction

1. Create 4 new text documents and save them with "**.xom**" extension.
2. Rename the XOM files in the following format:

   i. <Connection name>_API.xom

   ii. <Connection name>_CTN12.xom

   iii. <Connection name>_SP3SP5.xom

   iv. <Connection name>_Standard.xom

An example of XOM files with preferred nomenclature is shown in
Figure 46 below:

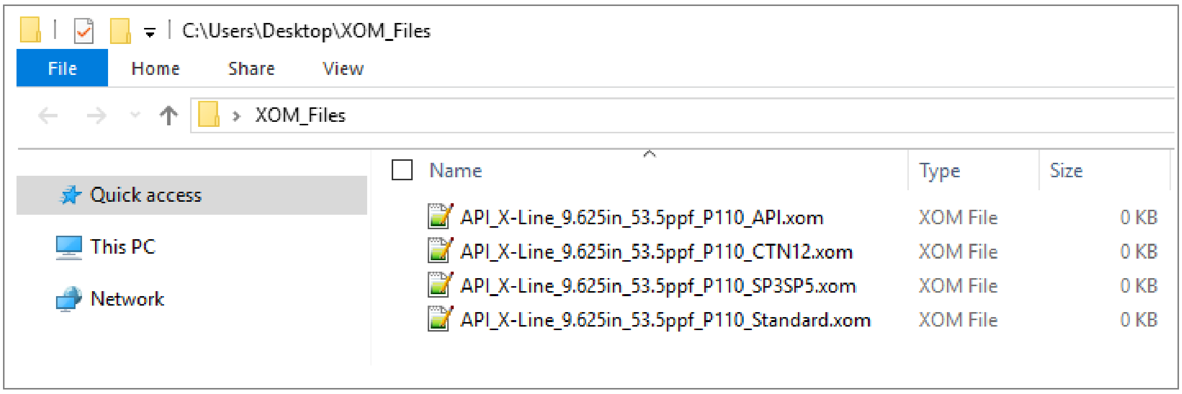


Figure 46 XOM File Nomenclature - Example

Extracting FEA Results using Post-Process Connector

1. Open Abaqus CAE or Abaqus Viewer, and go to Visualization module.
2. Go to Plug-ins menu and click on “Post-Process Connector” plug-in.
3. Click on “Select make-up odb” and select make-up odb file (*mmxmn.odb*)
4. In “Slave surface” dropdown box select the connection seal slave surface[j].
5. Click on “Select make-up input file” and select make-up input file (*mmxmn.inp*)
6. Click on “Select xom file” and select XOM file corresponds to Combination-1 shown in Table 42.
7. Close “Select files” dialogue box and verify the XOM file address populated in “Connector Post-Processor” dialogue box is correct.
8. Click on “Select odb files” and select ODB file(s) corresponds to Combination-1 shown in Table 42.
9. Click “OK” (Figure 48)

   A progress bar as shown in Figure 47 will appear after clicking “OK”. In this step all the required results are extracted from the ODB files and will be written to selected XOM file.

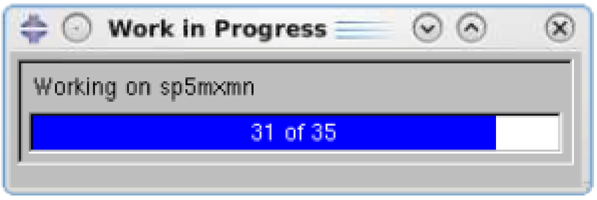


Figure 47 XOM File Generation – Example

10. Repeat steps- 1 to step-9, but select XOM file and ODB file(s) corresponds to Combination-2 in step-6 and step-8 respectively (Figure 49).
11. Repeat steps- 1 to step-9, but select XOM file and ODB file(s) corresponds to Combination-3 in step-6 and step-8 respectively (Figure 50).
12. Repeat steps- 1 to step-9, but select XOM file and ODB file(s) corresponds to Combination-4 in step-6 and step-8 respectively (Figure 51).

[j] If the connection has more than one seal then create new set of XOM files and repeat the entire **Extracting FEA Results using Post-Process Connector plug-in** step for each of the remaining seal surfaces.

Table 42 XOM File and ODB File(s) Combinations

Combination No.	XOM File	ODB File(s)
Combination-1	<Connection Name>_Standard.xom	c25epmxmn c25pmxmn c50epmxmn c50pmxmn c75epmxmn c75pmxmn c90epmxmn c90pmxmn cepmxmn cmxmn cpmxmn epmxmn ruppmxmn mtrpmxmn pmxmn t100pmxmn t25epmxmn t25pmxmn t50epmxmn t50pmxmn t75epmxmn t75pmxmn tepmxmn tmxmn tpmxmn vmpmxmn vmcpmxmn vmepmxmn vmcepmxmn
Combination-2	<Connection Name>_SP3SP5.xom	sp3mxmn sp5mxmn
Combination-3	<Connection Name>_API.xom	apiseramxmn apiserbmxmn
Combination-4	<Connection Name>_CTN12.xom	ctn12mxmn

Figure 48, Figure 49, Figure 50 and Figure 51 show the Post-Process Connector plug-in dialogue box with XOM files and ODB files correspond to Combination-1, Combination-2, Combination-3 and Combination-4 respectively.

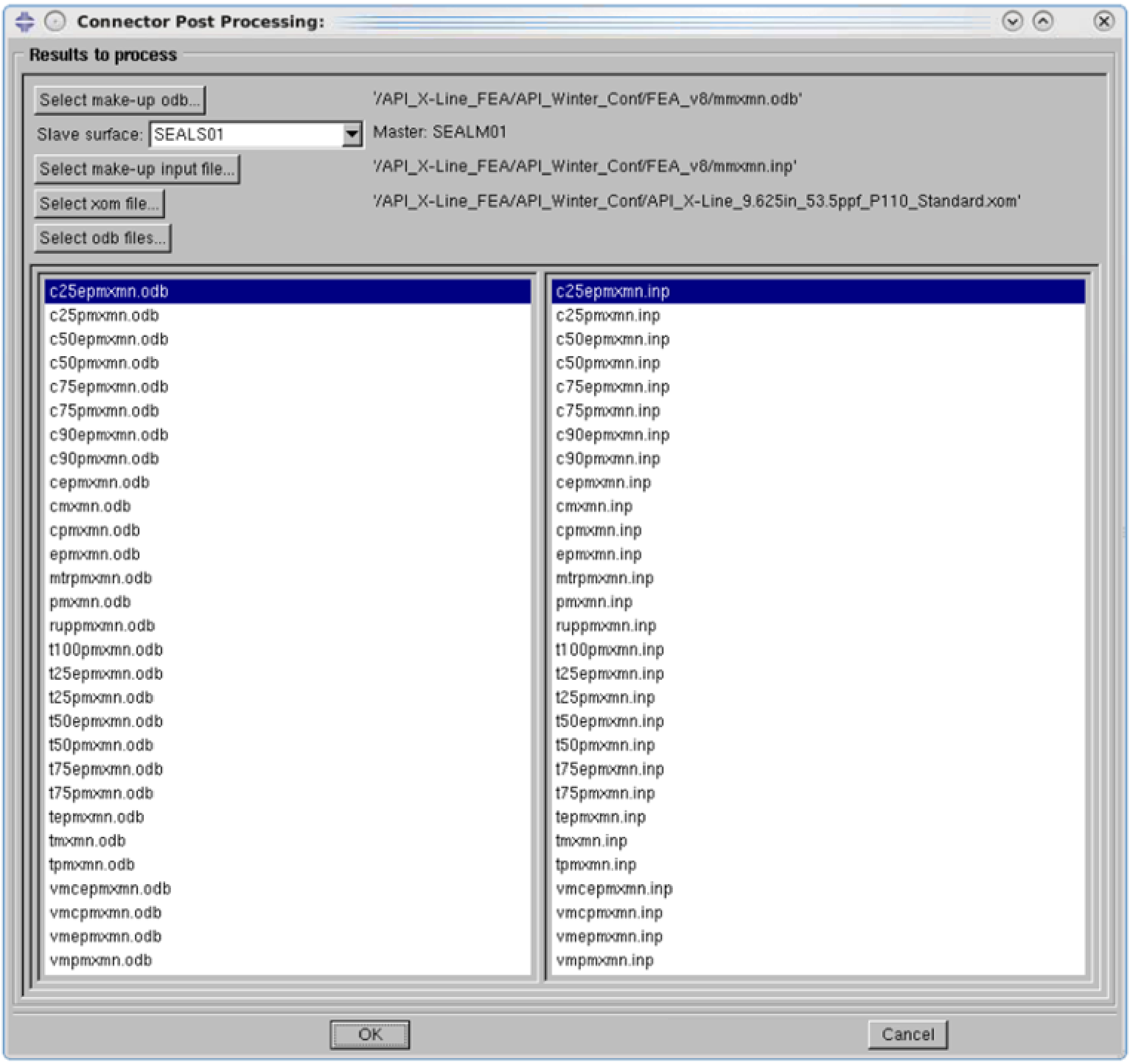


Figure 48 Post-Process Connector dialogue box – Combination-1

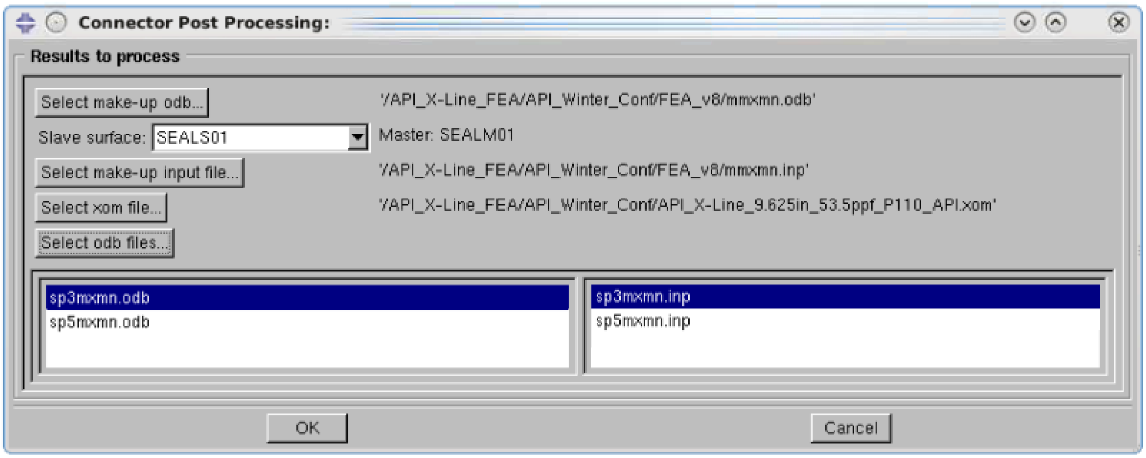


Figure 49 Post-Process Connector dialogue box – Combination-2

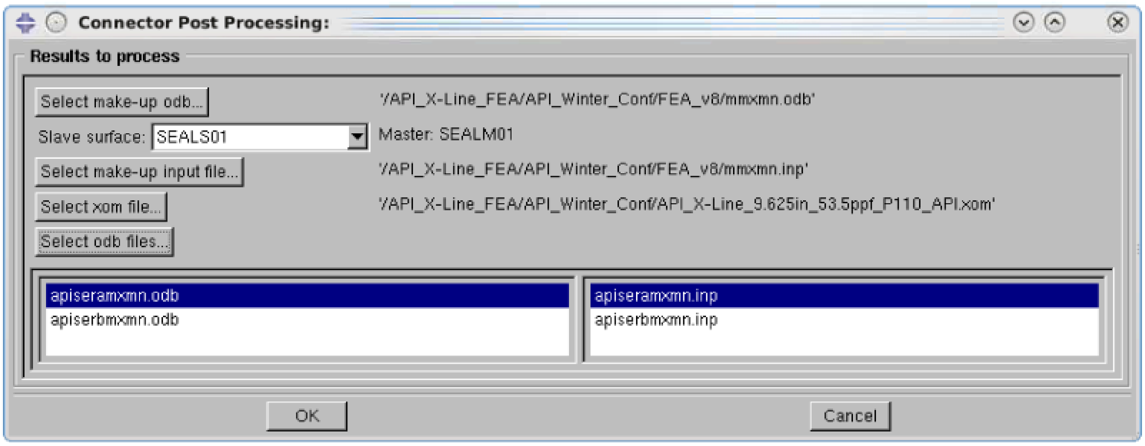


Figure 50 Post-Process Connector dialogue box – Combination-3

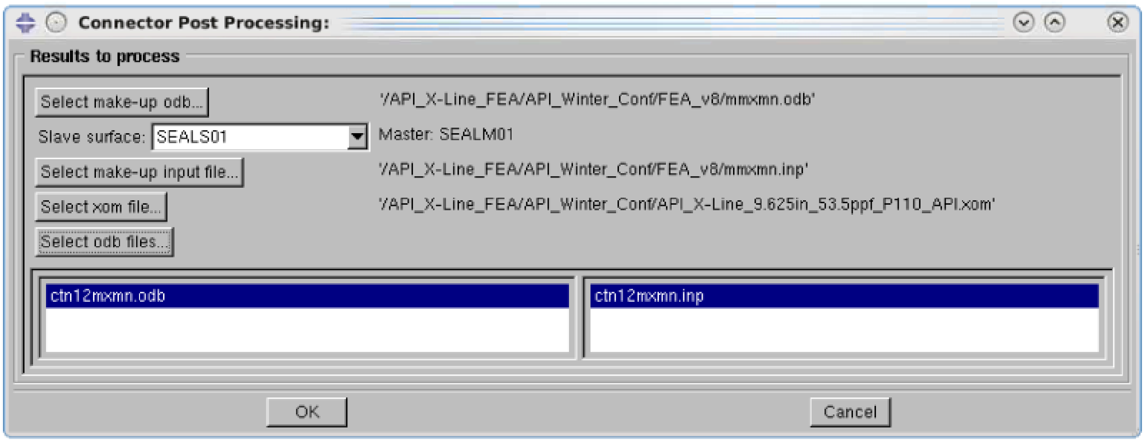


Figure 51 Post-Process Connector dialogue box – Combination-4

Appendix V

TPI Instructions

Some of the activities to be carried out by third party inspectors are tabulated in Table 43 below. All documents generated out of physical testing activities which are witnessed shall be endorsed by TPI.

Third Party Inspection Activity Description

Designation	Inspection Activity	Description
N	No TPI Needed	Activity does not receive TPI witness
V	Visual Inspection	Look for damage / repair / trace of water after external pressure test.
E	Examination of Records	Compliant with operating procedures.
W	Witness	100% attendance (or equivalent) for compliance with operating procedures.
SW	Spot Witness	At any time without announcing (at least once)
R	Review	Compliance with the specification and test carried out (reports).

Third Party Inspection Program

S. No	All Specimens	Inspection Designation
1	Pipe Material Selection	R
2	Forging Material Selection (Variant 4)	R
3	Welding (Variant 4)	E
4	Mapping	E, V
5	Saw cutting	N
6	End Forming	N
7	Coupon Testing	N
8	Machining	N
9	Re-gauging	V, W, E
10	Non-Destructive Testing (if applicable)	N
11	Surface Finish	E
12	Initial Parts Condition	V
13	Walls and OD’s	E
14	Make and Breaks testing	V, W, E
15	Bake out	E
16	Test System Setup	N
17	Sensitivity Check (anytime)	SW, E
18	Series B (all parts)	W
19	Series C	SW, E
20	Series A_e (90%)	SW, E
21	Series A (Q Cycles)	SW, E
22	Series A_a (90%)	W
23	Series A_a (95%)	W
24	Limit Load Tests	W
25	Test Report	N
26	TPI Summary/Exceptions/Daily Reports	**

**TPI will list all exceptions taken and witness points in *physical testing summary form*

Table 43 TPI Instructions